Scuola Superiore Sant'Anna
Bionics Engineering
Istituto di BioRobotica

Lecture Notes

Soft Robotics Technologies

– Ansys Part –

Professor:
Matteo Cianchetti

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Finite Element Method (FEM)

The Finite Element Method can be applied basically to everything that is physics. We will use Ansys, and its workbench, as Multiphysics tool, to study mechanical structural simulations. Mechanics is everywhere, so we need to find a solution to study and develop system that can support our simulation, used for designing and optimizing things.

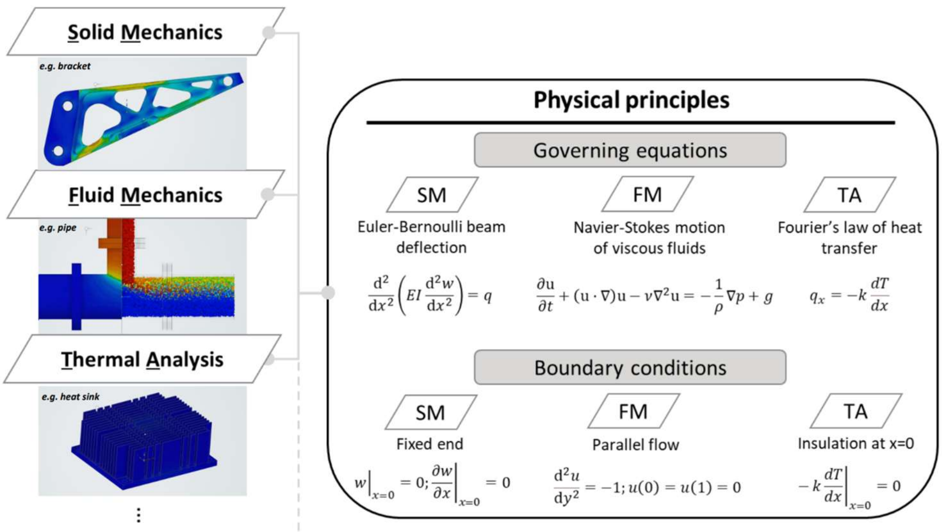
Understanding Mechanics in FEM

Mechanics is the study of how things work. We want to understand how to pass from physics (material properties, system of reference, etc.) into a virtual simulation.

In mechanics, a body can be treated as a rigid body or as a flexible body. When we deal with rigid bodies, the system could move, with some constraints, in a rigid way without considering any deformation. Continuum mechanics deals with material deformation, so with flexible bodies (flexible means deformable, not soft: A flexible body can be of metal). During the interaction, a flexible body can have mutual deformation.

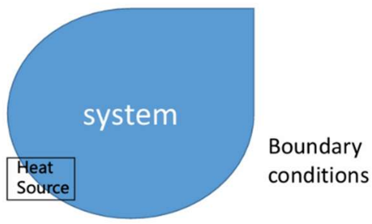
Mechanics can be divided into two main groups: continuum mechanic, in which materials are treated as continuum mass instead of discrete particles/atoms; while molecular, or quantum, mechanics studies the microstructure. Continuum is divided into fluid mechanics (study of gas and liquids) and solid mechanics.

In the following image there are governing equations and boundary conditions used in FEM and by Ansys to define the simulations:



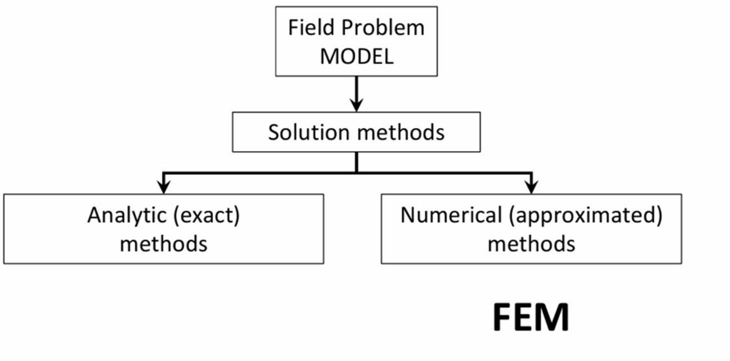
Field Problems and Solution methodologies

Field problem: Given a certain continuum system, with defined volume and boundary conditions, we are interested in knowing what happens inside it in terms of spatial distribution of one or more dependent variables. For example, find the distribution of temperature in the volume when a certain source of heat is applied in a specific point of the boundary of the volume. Other examples are displacement, stress, strain, electrical charge or magnetic field.

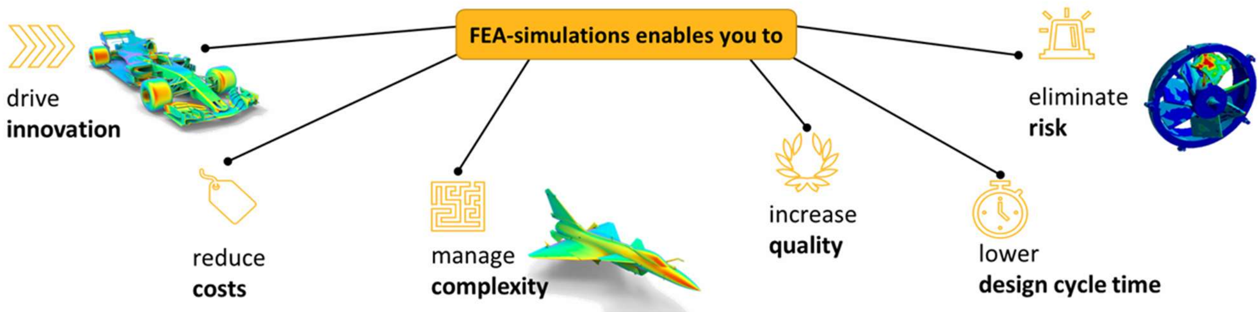


A field problem could be solved in two ways:

- Analytical Methods: give the exact solution
- Numerical methods: approximated solutions (for example Finite Element Method)



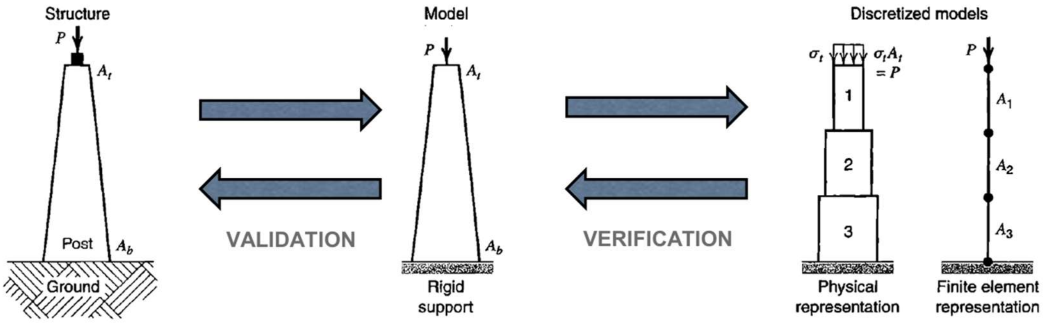
FEA and FEM are sometimes used as synonymous but are two different but related concepts. FEM stands for Finite Element Method and is the mathematical method in which partial differential equations are solved to describe physical phenomena. FEM is used to conduct Finite Element Analysis (FEAs), which are implemented in computer-aided engineering (CAE) tools, providing a solution for virtual prototyping. FEA helps to solve structural, heat transfer, fluid flow or electromagnetic problems quickly by exploiting computational power and make design changes early product development stage. FEA is powerful and widely used in many industries. It is fast (faster than prototyping) because it allows to do multiple designs iterations quickly since are virtual, simulating also interactions between bodies, and allowing to avoid the physical fabrication, test evaluations, modifications and redesigning usually needed to develop a new system/object. This allows to reduce the costs, in terms of time, human resources and costs of materials. FEA enables to manage complexity, entering into fine details of the simulation. It increases quality and eliminates risks because you can also simulate conditions that cannot be covered by real test validations (for example, stability of structures, in particular the effect of buckling when a structure is subject to mechanical instability that can quickly evolve into a complete failure). Furthermore, FEA can enable innovation that would be tedious or impossible with any other method.



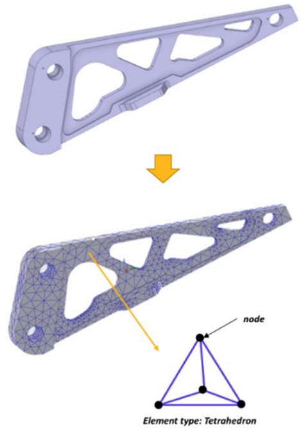
FEA can be used in a lot of engineering sectors, due to its multi-physics approach.

Discretization Process in FEM

FEM is an approximated model, we can understand that using an example. We consider a sort of column, that is grounded, with a block (weight) on the top. We want to know what happens to the pillar when the force is applied. We are not interested on what happens on the ground, so it is considered as a rigid support. The first model (first level of abstraction) keeps variable cross-sectional areas and same pressure applied. On this first simplified model, analytical methods can applied. However, it can be simplified more through discretization achieved by FEM or other numerical methods. Discretization divides the problem, which is continuous, into smaller pieces. For example, the trapezoidal shape of the pillar model can be modelled with several rectangles, one on the top of the previous, keeping the top and bottom cross-sectional areas and the same pressure (P). This discretized model creates a simplified physical representation. By increasing the number of the rectangles, we will be more precise in describing the model itself, better approximating the pilar structure. We are describing a continuous structure with smaller parts, but the number of the parts is Finite. We can even further simplify the model, from bi-dimensional into monodimensional, by using segments, still maintaining physical meaning (we are allowed to do that only because the load is vertical).

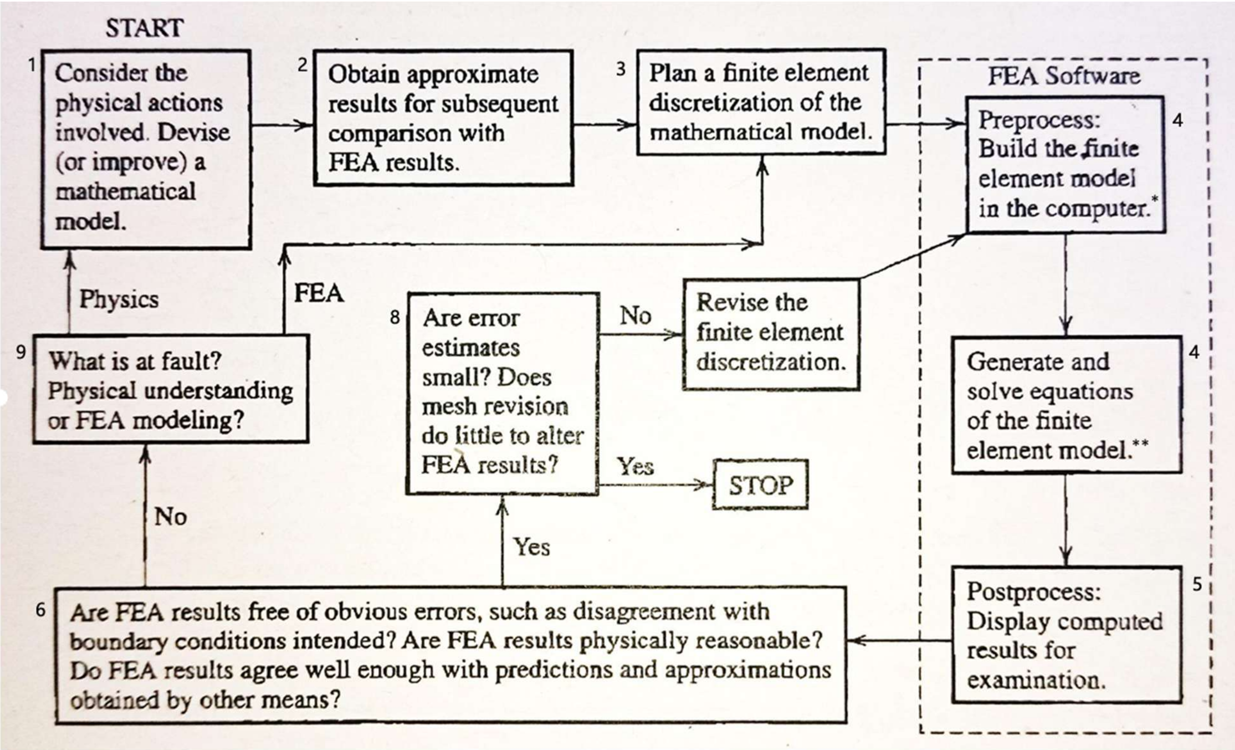


However, we need some Verification and validation steps. The *verification* confirms that FEM calculations correctly represent mathematical model/equations on which it is based. Usually, the verification is made by the software itself. The *validation* confirms that the mathematical model you have solved with FEM is an accurate representation of the real physical system. Validation is usually done by comparison of FEM results to experiments or simple hand calculations. The validation MUST be done by the user. The user must define what is relevant or not in the virtual representation of the system (in the example we choose that to not consider deformation on the ground). The essence of FEA is approximation by piecewise interpolation of a field quantity. The validation can be done only once, even if you do small changes: for example, we can start with a very simplified version of the structure, do the validation and then add complexity in geometry or material (without doing again the validation).



We can conclude that FEM is a numerical method that exploits computational power to calculate solutions for a range of different engineering problems, offering good approximate solution to complex problems in several fields (e.g. solid mechanics, statics and dynamics). At the core of FEM lies the discretization, which means a model is divided into a finite number of smaller bodies or *elements* that are connected at common points, the so-called *nodes*. The entire model is called *body*. All elements share nodes with other elements. The shared nodes are formally and mathematically part of several elements. Each node has a limited number of degrees of freedom (DoFs), so e.g. a continuum solid becomes discretized into a finite number of DoFs, based on the number and type of elements. For each of those elements a set of algebraic equations are solved in an iterative way. The sum of all elements in which the body is divided is called *mesh*. The meshing task consists of generate a grid by subdividing the body into elements. Each of those elements represents the unit of the mesh. Instead, the nodes are the solution points of the element.

FEA Workflow



* Implementing simplifications (e.g. symmetries) if possible

** Defining boundary conditions

1. The first step is to consider the physical actions involved studying the real model. We have to identify the important 'characteristics' that must be created in the model, translating the real system into a first physical representation. The first model is usually extremely simplified but is useful to check if the FEM works well.
2. The second step consists into obtain the simplified approximated model
3. the third consists into plan the discretization (deciding shape and number of elements). Meshing is fundamental since accuracy of the model depends on that. For example, a fewer number of elements leads to a non-accurate solution (low computational power) but is fast, while a high number of elements allows to obtain an accurate solution but is slow; we need a trade-off. (The discretization can also be done by the software).
4. The fourth step is performed by the FEA software. It consists into preprocessing (build the FE model in the computer) and generate and solve equations of the FEM.
5. The software displayed results and solutions of the field problems obtained for examination and to calculate further parameters.
6. Checking for obvious errors, such as disagreement with boundary conditions or results that are physically reasonable.
7. Mutual validation: is a decision step. Are FEA results free of obvious errors?
    - If no obvious errors are present: proceed to Step 8.
    - If significant errors are present: proceed to Step 9.

8. Error sensitivity and accuracy assessment: Evaluate whether the estimated errors are sufficiently small and whether further refinement would have a negligible effect on the results.
    - If YES (errors are small): accept the solution and STOP.
    - If NO (errors are not negligible): refine the finite element discretization and return to Step 4.
9. Model Revision: Identify the source of the discrepancy:
    - If due to modelling assumptions and misunderstanding on the underline physics: return to Step 1 and revise the physical/mathematical laws.
    - If due to discretization (discretization modifies a lot the results), return to Step 3 and recreate the discrete modelling.

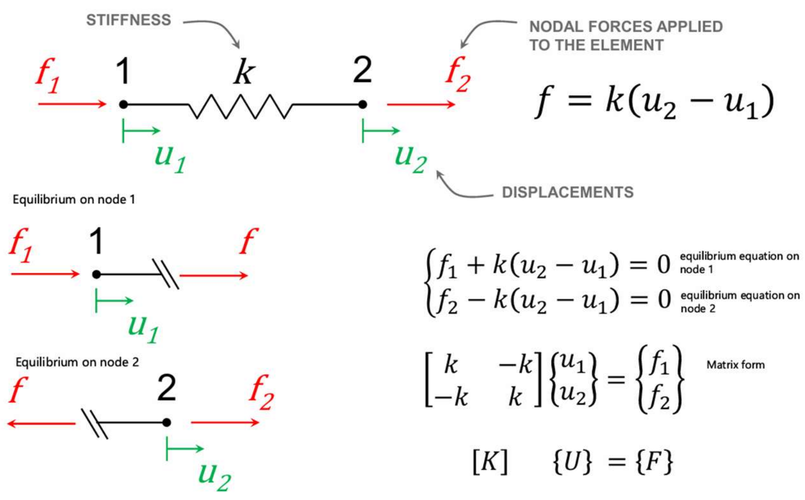
IMPORTANT: To be sure about accuracy of the model and number of meshes → We should increase the number of elements until the last two recent simulations give us the same results, or slightly neglectable differences.

Mathematical equations in FEM

Understanding the mathematical equations implemented within the software is essential, as they define how the simulation operates. A solid grasp of these equations enables the user to make informed decisions about which simulation settings to adjust and how to modify them effectively.

1D Bar Element Example

In this example the 1D body is divided into 1 unique element. The element describes the entire body and has 2 nodes. The body stiffness is described by k and two forces, f_1 and f_2, which are applied in the nodes. The displacements caused by the forces are u_1 and u_2 . The internal force inside the element is f. Now we want to see the equilibrium of the system node by node:



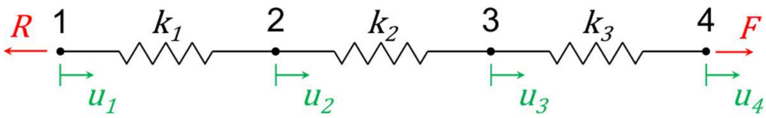
The system of equilibrium equations can be expressed in matrix form: where [K] is the stiffness matrix, {U} is the displacement vector and {F} is the external force vector. By using this simple example, we identify one matrix and two vectors that describe the stiffness, the displacement and the force. K, U and F are used to describe the problem, and by solving these equations, we can solve the field problem.

Another example with 3 elements: the 3 elements are in series, for a total of 4 nodes.

The stiffness matrix

Assembly

$$\begin{cases} -R + k_1(u_2 - u_1) = 0 \\ -k_1(u_2 - u_1) + k_2(u_3 - u_2) = 0 \\ -k_2(u_3 - u_2) + k_3(u_4 - u_3) = 0 \\ -k_3(u_4 - u_3) + F = 0 \end{cases}$$



$$\begin{bmatrix} k_1 & -k_1 & 0 & 0 \\ -k_1 & k_1 + k_2 & -k_2 & 0 \\ 0 & -k_2 & k_2 + k_3 & -k_3 \\ 0 & 0 & -k_3 & k_3 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{Bmatrix} = \begin{Bmatrix} -R \\ 0 \\ 0 \\ F \end{Bmatrix}$$

GLOBAL STIFFNESS MATRIX $[K] \quad \{U\} = \{F\}$

R is the reaction force, F is the applied force, u_i is the displacement of node i. Following the same approach, we can consider the equilibrium at each node and then transform the system of equilibrium equations into matrix form (k_1 is stiffness of node 1, k_2 is the stiffness of node 2, etc.). The matrix form is derived by the software, and it Is one of the most computationally expensive part of the simulation. In technical terms, the generation of the matrix is called Assembly, because the matrix is composed by small parts. The global stiffness matrix can be assembled and interpreted in two equivalent ways:

- As blocks (element-wise view): The matrix can be seen as the assembly of smaller submatrices, each corresponding to a finite element. For example, the top-left 2×2block represents the local stiffness matrix of element 1, which connects nodes 1 and 2. In this view, each element contributes its local stiffness matrix to the global matrix according to its connectivity.
- As columns/rows (degree-of-freedom view): The matrix can also be interpreted column by column (or row by row), where each column corresponds to a degree of freedom (DOF) of a node. For instance, column 1 represents the response of the system when a unit displacement is applied at the DOF of node 1. In this sense, the matrix describes how each DOF interacts with all the others.

FEA process

It consists of:

1. Set model and elements
2. Develop element stiffness matrix, [k]
3. Assembly to form global stiffness matrix, [K]
4. Apply loads to nodes

5. Apply boundary conditions to supports (in ansys application of loads and boundary conditions is basically the same)

6. Solve for {U} and {F}

7. Compute gradients (for example, $\varepsilon = u_{,x}$) and related quantities (by constitutive equation: $\sigma = E\varepsilon$)

The description of Finite Element Method (FEM) involves understanding and calculating stresses, which are related to body and surface forces, and strains, which are associated with displacements. However, not all solutions provided by the software are guaranteed to be feasible. While the software performs certain checks, others still require verification by the user. For example, the equilibrium criteria, ensuring the equilibrium of internal stresses, must be validated at every point to confirm the solution's physical accuracy.

The three checks automatically done by the software are:

- Equilibrium:
- Constitutive laws
- Compatibility

Equilibrium: The primary goal in FEM is to ensure the equilibrium of the body in terms of the position, arrangement, and shape of all the elements after the load conditions are applied. Since the body is usually divided into many elements, the challenge lies in finding a combination of deformations and forces that are distributed across the body in a way that ensures overall equilibrium. Forces are applied to individual elements, and the reaction forces are applied to adjacent elements, with each force contributing to the overall force balance. Only if the system achieves overall equilibrium, the solution can be considered acceptable. While dividing the body into elements simplifies the analysis for each individual element, the entire mesh must be consistent in terms of force propagation. The sum of all forces must be zero for the solution to be valid.

Constitutive laws: This refers to the application of material behaviour in the model. The software uses the defined constitutive laws to calculate how the material responds to the applied loads. These laws are automatically applied during the simulation and ensure that the stress-strain relationships are consistent with the material properties.

Compatibility: Compatibility is especially important in simulations involving cracks or discontinuities in the model. In most simulations, the displacement and strain fields are assumed to be homogeneous and continuous throughout the entire body. However, when the model involves breaking (e.g., cracking, material failure), compatibility checks become critical. The software ensures that the displacement and strain fields are continuous and compatible. These checks are automatically performed if the model includes failure criteria or fracture mechanics.

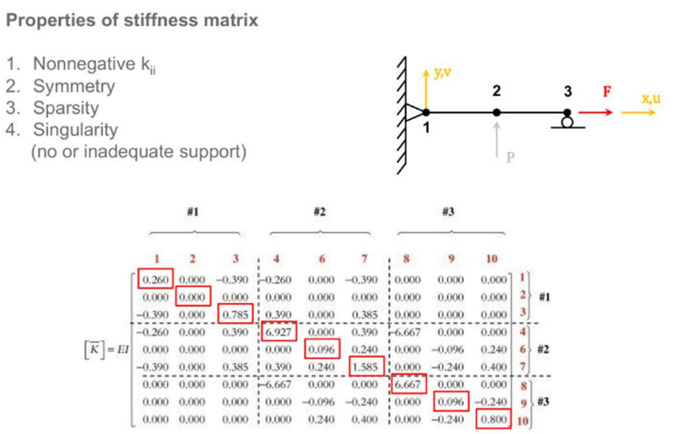
By understanding these checks, the user can better interpret the results and ensure that the solution provided by the software is both physically accurate and feasible. However, while these checks are done automatically, the user should still verify that the model setup, mesh, and material properties are appropriate for the problem

Properties of the Stiffness Matrix

The properties of the stiffness matrix are fundamental for understanding and interpreting warnings or errors provided by FEM software. The main properties are:

1. Non-negative diagonal terms (k_{ii})
2. Symmetry
3. Sparsity
4. Singularity (due to missing or inadequate constraints)

These properties can be clearly understood by referring to the same 2D model shown in the figure, composed of three nodes and three elements.



Reference Model

In this system:

- Node 1 is fully fixed (no displacement in x or y).
- Node 2 is free to move.
- Node 3 is constrained vertically but free in the horizontal direction (roller constraint).
- A force is applied at Node 3 in the horizontal direction.

The stiffness matrix shown below corresponds to this discretized system.

Interpretation of Matrix Properties

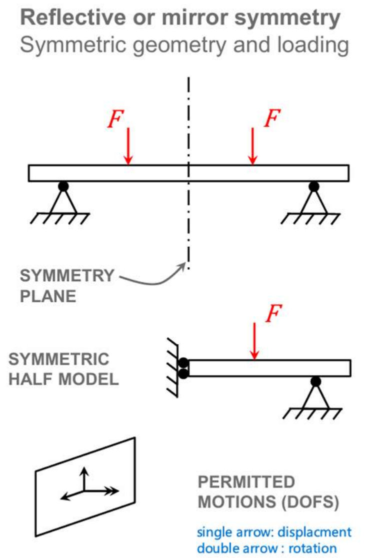
- Non-negative diagonal terms: The diagonal entries k_{ii}, highlighted in the matrix, are all non-negative. This reflects a physical requirement: it is not physically meaningful and possible for a force to produce displacement in the opposite direction. These terms represent the stiffness associated with each degree of freedom.
- Symmetry: The matrix is symmetric, meaning that the interaction between two degrees of freedom is reciprocal. Physically, this implies that the influence of node i on node j is equal to the influence of node j on node i.
- Sparsity: Most of the entries in the matrix are zero. This is because each node is only connected to adjacent elements, and therefore only interacts with nearby nodes. As seen in the matrix, non-zero terms appear only in localized blocks corresponding to connected elements, while the rest are zeros. This sparsity is typical in FEM and allows efficient storage and computation.
- Singularity: Singularity arises when the stiffness matrix cannot be inverted, meaning that the system does not have a unique solution. In this example, despite the presence of constraints, the system may still allow a rigid body motion, such as rotation around Node 1. This happens

when some degrees of freedom are not sufficiently constrained. In fact, when the force is applied at Node 3, one might expect a well-defined deformation. However, if the constraints are not sufficient to prevent all rigid body motions, the structure can still rotate. Even if this motion is not explicitly visible, the solver detects that the system lacks enough information to fully constrain all degrees of freedom. As a result, the stiffness matrix includes unconstrained degrees of freedom, becomes singular, and cannot be inverted. This is why the solver reports a singularity error, even though the model may appear correctly defined at first glance.

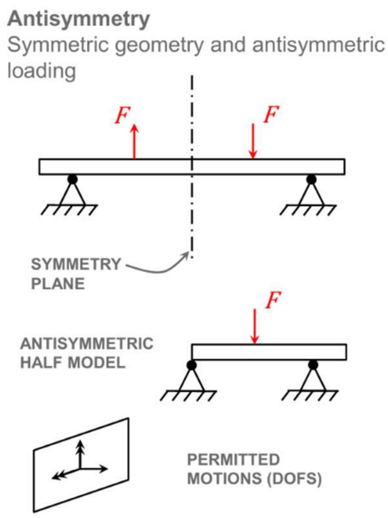
Symmetry Of a Model

A lot of models represent symmetry which allow to simplify the system itself, by considering only a part of it, fully maintaining the complexity. There exist two symmetry types:

Reflective or mirror symmetry: the system is perfectly symmetric, both in geometry and load distribution, with respect to an plane called symmetry plane. In this case, we can remove a part of the model, by applying some specific conditions on the symmetry plane, and study only the symmetric half model. In this way, by reducing the number of elements considered, also the computational time. We are allowed to neglect half of the model but we must take care of the nodes that are on the symmetry plane: every node that is on the symmetry plane must be associated with its permitted degree of freedom. In the example, displacement IN plane and rotation on the axis perpendicular to the symmetry plane are permitted, all the rest rotations and displacement are not permitted.



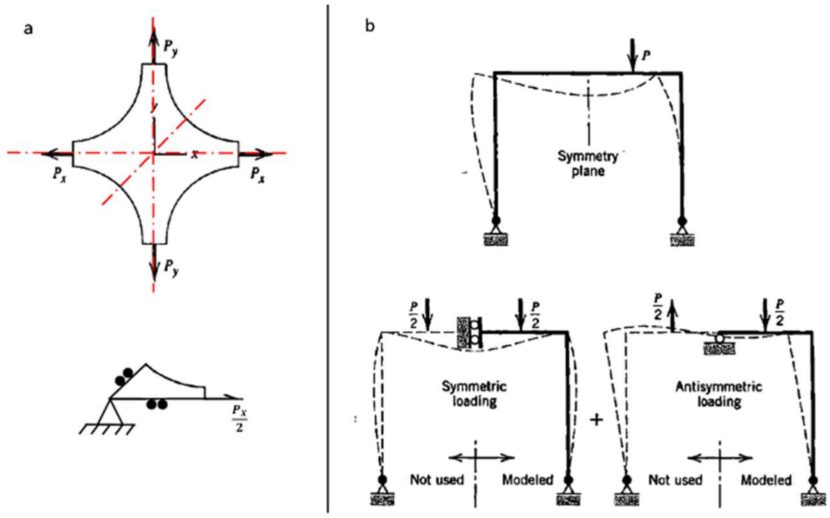
Antisymmetry: antisymmetric condition consist into geometrical symmetry and load conditions that are opposite with respect to the symmetry plane. Also in this case we are still allowed to consider and study only the antisymmetric half model. However in this case the permitted dofs are displacement out of the symmetry plane and rotations in the plane (the opposite).



⚠ It is Antisymmetry, not Asymmetry

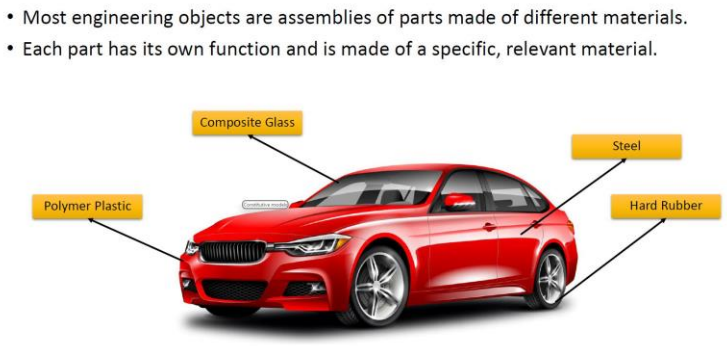
In the following example there are two cases of simplified model due to symmetry:

a) the model is completely symmetric. In particular, it has 4 symmetric axes, which allow to study only 1/8 of the system, reducing a lot the complexity.
b) The model is symmetric in the geometrical features, but the load condition is not symmetric not antisymmetric. But we can we can 'decompose' the system into two system, a symmetric and a antisymmetric loading systems, which sum give the original system. The two systems are with the same geometrical structure of the original one, but on the symmetric there are applied half of the force P in a symmetric way, and on the antisymmetric are applied two antisymmetric forces of value P/2.



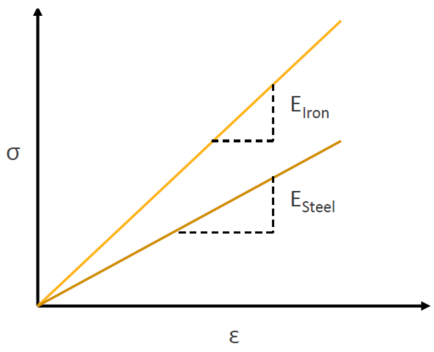
1.1 FUNDAMENTALS ON LINEAR ELASTICITY

What is the importance of material? What is the difference between rubber and steel? The teacher assumes that we know it, at least from a mechanical point of view. So, why choosing a certain material for a certain part of the car should be something quite trivial. Why we are going for rubber for tires and structural steel for the rest of the car. What is the material for engine?

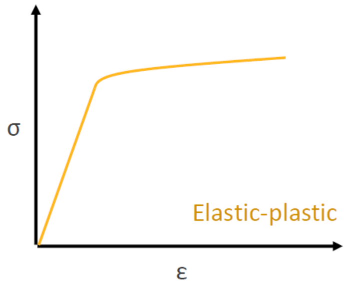


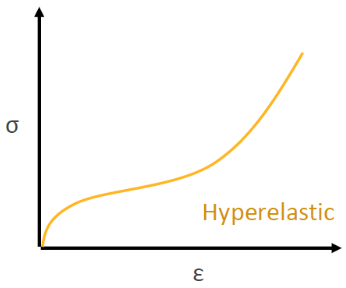
Therefore, we need CONSTITUTIVE MODEL.

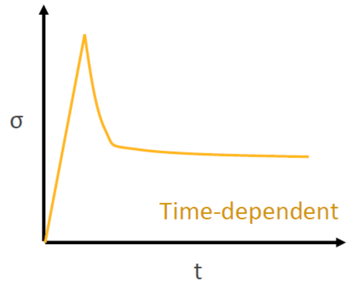
CONSTITUTIVE MODELS give us essential information about mechanical properties of materials. We know that trough the relationship between stress and strain we know the stiffer material and then, in this case, make our choice.



This function, that is relating stress and strain, usually is linear for most of materials, especially for small deformations. For metals is smaller. For elastic materials, is more extended. But, in any case the very first part of this relationship is pretty much linear, so we can also use a constat that is called Youngs Modulus.







But not all the materials are linear. We can linearize the description, the relationship. But, in reality, we usually have other non-linear behaviour like in elastic plastic, where you consider plasticity in the material behaviour.

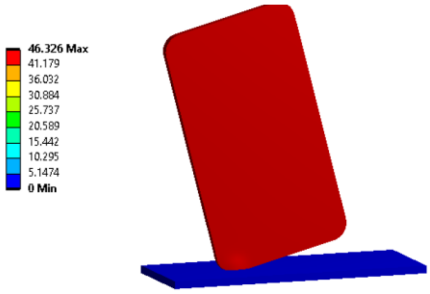
Hyper elasticity → elastomers and rubbers, rubber-like materials.

There are also materials time dependent. So, the stress is also dependent on the time you apply, for example, a certain deformation. When you go into non-liner behaviours, most of the time, unless you are considering hyper-elastic materials, also in the first part of the curve you get some plasticity. This means that there are plastic deformations, *permanent deformations that you cannot recover.* So, when you release normal condition, you have permanent deformations.

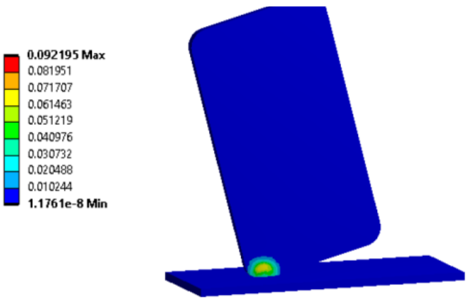
Again, stress is dependent on the modulus. There are some terms that, in colloquial languages, have more or less the same mining, but not for fem!!!

1. <u>*Distortion and strain*</u> actually are the same. Distortion is basically never used. You will always see strain.
2. <u>*Strain ≠ deformation.*</u>
3. <u>*Deformation = strain + rigid motion.*</u> Deformation is strain, so the real distortion of our body, plus the rigid motion.

   What is the implication of this? If you are simulating the impact of a mobile phone on a flat plane. You have the mobile phone that first moves down, and then there is the impact with the distortion of the edge. If you see the result of the simulation, and you ask for deformation, you will see that the mobile phone has more or less the same deformation everywhere.

   This happens because deformation includes also the rigid body motion. So, the red image includes also the entire motion of the mobile phone before the impact. And this covers what is happening locally in the impact, because the distortion of the edge is very small.

   Instead, if you go for strain, you are able to see what happens locally, and probably what you are searching for. In this scale you have another scale that is only related to what happens locally.

4. <u>*Poisson ratio*</u> = it's a ratio usually between 0 and 0.5. Elastomers and natural rubber are very closed to 0.5. The poison ration can be applied in more dimension.

$$\epsilon_{YY} = \epsilon_{ZZ} = -\nu\epsilon_{XX}$$

5. <u>*COMPLIANCE TENSOR*</u>

   in this case there the strain in function of the compliance tensor and the stress. $\epsilon = C\sigma$

$$\begin{bmatrix} \epsilon_{XX} \\ \epsilon_{YY} \\ \epsilon_{ZZ} \\ 2\epsilon_{YZ} \\ 2\epsilon_{XZ} \\ 2\epsilon_{XY} \end{bmatrix} = \frac{1}{E} \begin{bmatrix} 1 & -\nu & -\nu & 0 & 0 & 0 \\ -\nu & 1 & -\nu & 0 & 0 & 0 \\ -\nu & -\nu & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2(1+\nu) & 0 & 0 \\ 0 & 0 & 0 & 0 & 2(1+\nu) & 0 \\ 0 & 0 & 0 & 0 & 0 & 2(1+\nu) \end{bmatrix} \begin{bmatrix} \sigma_{XX} \\ \sigma_{YY} \\ \sigma_{ZZ} \\ \sigma_{YZ} \\ \sigma_{XZ} \\ \sigma_{XY} \end{bmatrix}$$

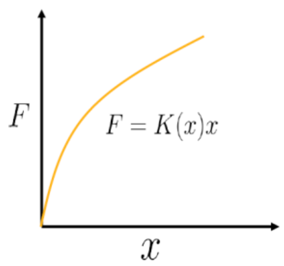
6. *STIFFENESS MATRIX* is the same thing above but expressed in form of stress vector.

$$\begin{bmatrix}\sigma_{XX}\\ \sigma_{YY}\\ \sigma_{ZZ}\\ \sigma_{YZ}\\ \sigma_{XZ}\\ \sigma_{XY}\end{bmatrix} = \frac{E}{(1+\nu)(1-2\nu)}\begin{bmatrix}1-\nu & \nu & \nu & 0 & 0 & 0\\ \nu & 1-\nu & \nu & 0 & 0 & 0\\ \nu & \nu & 1-\nu & 0 & 0 & 0\\ 0 & 0 & 0 & \frac{(1-2\nu)}{2} & 0 & 0\\ 0 & 0 & 0 & 0 & \frac{(1-2\nu)}{2} & 0\\ 0 & 0 & 0 & 0 & 0 & \frac{(1-2\nu)}{2}\end{bmatrix}\begin{bmatrix}\epsilon_{XX}\\ \epsilon_{YY}\\ \epsilon_{ZZ}\\ \epsilon_{YZ}\\ \epsilon_{XZ}\\ \epsilon_{XY}\end{bmatrix}$$

This followed the Hooke's Law, and this is very simple to applied when you have isotropic materials, but it's also applied when you have anisotropic materials, like natural materials. But you must know it the direction that is relevant for that relationship and the value of confidence for that direction.

1.2. INTRODUCTION TO STRUCTURAL NONLINEARITIES

Now we can start the second part. This is what we know, via linear theory → a spring-weight system, in this case force is directly proportional to the stiffness of the spring and deformation (F=Kx). This lazy simplification is very simple and useful, but when you start getting serious in analysing systems, this relation is by far the simplest.



In nature nothing is weaker. This is the reality. In this case you can start to deal with a simple system spring-mass system and in reality, the spring is not linear at all, and the stiffness (K) depends on the deformation itself. This means that the relationship between the block Force and the displacement is not linear anymore.

When you want to get serious in analysing the real word system, so crash and impacts → If you do not consider plasticity, you cannot analyse the impact of any kind of materials that are together. If you don't consider the viscoelasticity in biological tissue, you cannot produce simulation about that. But what is the sources of nonlinearities? Why we have these nonlinearities? There are 3 main sources that affect most of the simulations:

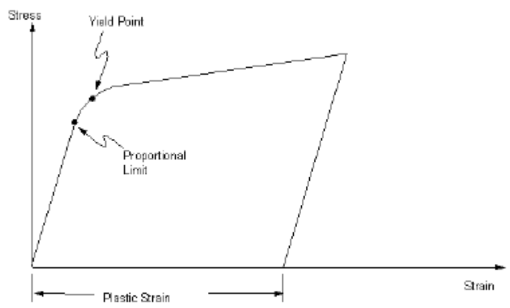
1. **Large deformations.**
2. **Material nonlinearities.**
3. **Contacts.**

Whenever you have one of those, you have a nonlinear system.

- Large deformations:
  1. one example is when you have a change of **wall thickness** (when you have a material in the tube and you start stretching, the thickness will decrease)
  2. change of cross-section areas. (if you have a cylinder and you start stretching it, there is a decrease in the cross-section area)
  3. changes in direction of the loading (in the example of the fish = the relative angle that is connecting the road to the fishing line is changing, and this orientation is introducing a nonlinear effect)

- Material Nonlinearity:

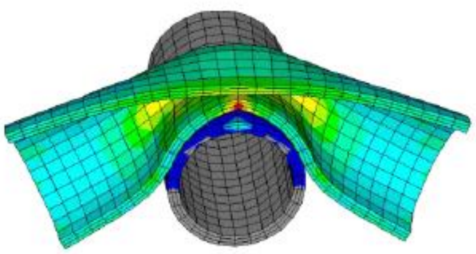
They are nonlinearity that are already intrinsic in the material. For example, → **PLASTICITY** = (usually in metals) after a linear behaviour, you have proportional limit and Yield point, when materials start fading in terms of elastic properties, and then the sloop is changed and also the material properties. When the sloop changes, you are out of elastic domain, and you start to deform the material plastically. This means that when you remove loading conditions, and stress returns to zero, you still have some permanent stain. This strain cannot be recovered because you enter the plastic domain. This is a very common nonlinearity, and you must be able to consider in a simulation. Another kind of nonlinearity related to materials **is HYPERELASTICITY** = in this case you have a displacement-force ratio that is not linear. The plot on slide is a little bit limited. Usually, you have intrinsic nonlinearities, and you have to consider them in your simulations.



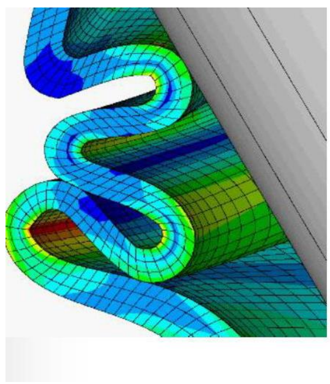
- Contacts:

It's a particular source of nonlinearities because **it's a changing status to nonlinearity**. This introduces another change in stiffness due to the bodies that are your simulations. The bodies that are in your simulation enter in contact or go out of contact. Of course, this only when you are simulating multi-bodies simulations, usually excluding self-contacts.

For example:



these are 2 bodies that are moving. In this case there is a very fast change on relationship between force and displacement due to the sudden loss of stiffness.



This is another example of nonlinearity and self-contact. There is a large deformation.

Nonlinear solution using linear solvers

Nonlinearities are essential for replicating realistic scenarios without losing the crucial aspects of real material behavior, structural interactions, etc.
We cannot ignore nonlinearity. The question is: *Are we still able to use FEA*, which is based on linear approximation, for highly nonlinear problems?
Here, we will explain how to handle nonlinearity using a linear solver.
The **Answer**: We use an iterative series of linear approximations and corrections to solve highly nonlinear problems.

Newton-Raphson Method

There are several methods available, but this is one of the most widely used and robust.

Let's consider a single-variable scenario with a nonlinear equation. Assuming the function is continuous, a Taylor series expansion can be performed:

$$f(x^{(i+1)}) \approx f(x^{(i)}) + \frac{\partial f(x^{(i)})}{\partial x}(x^{(i+1)} - x^{(i)}) + \cdots$$

We neglect the higher-order terms.

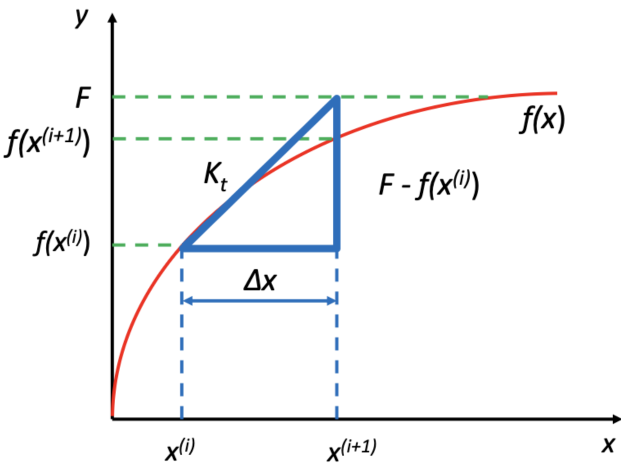
Rearranging the terms gives:

$$(x^{(i+1)} - x^{(i)}) = \frac{f(x^{(i+1)}) - f(x^{(i)})}{f'(x^{(i)})} \qquad \mathbf{\Delta x} = (\mathbf{x^{(i+1)}} - \mathbf{x^{(i)}})$$

$$\Delta x = \frac{F - f(x^{(i)})}{K_t} = K_t^{-1}(F - f(x^{(i)})) \qquad \mathbf{K_t} = \mathbf{f}'(\mathbf{x^i})$$

Let's define the different variables:

- F = desired value (external force).
- $x^{(i)}$ = initial value of x.
- $f(x^{(i)})$ = initial value of the function.
- K_t = slope of the tangent line.
- Δx = increment in x.



In FEA:

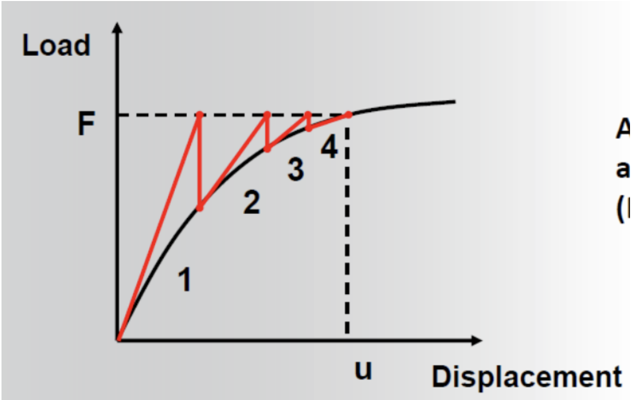
- $F - f(x^{(i)})$ = difference between the applied force (F) and the internal forces ($f(x^{(i)})$).
- Δx = displacement increment.

The problem is that we do not know the $x^{(i)}$ that generates the force a priori. The purpose of the iterative procedure is to find this value.

NOTE: L'idea alla base è semplice, non avendo linearità e quindi non potendo invertire la funzione, non riusciamo a trovare il valore che stiamo cercando, tuttavia riusciamo a misurarne l'effetto. Per trovarlo dobbiamo usare questo metodo iterativo.

How does it work?

1. We choose an initial arbitrary value for $x^{(i)}$.
2. We compute $f(x^{(i)})$, $K_t = f'(x^{(i)})$, and the difference between F and the initial internal force $f(x^{(i)})$.
3. Using these values, we determine a new $x^{(i+1)}$. This is no longer arbitrary but is defined as $x^{(i+1)} = x^{(i)} + \Delta x$, where Δx is computed using the values found in the second step.
4. We iterate until the residual $F - f(x^{(i)})$ is sufficiently small. Each step is called an **equilibrium iteration**.



The **Problem in FEA**: When applying an external force, we must compute the material deformation that generates a resisting internal force equal and opposite to the applied force. Since this deformation is not known a priori, we aim to minimize the difference between these two forces until it is close to zero. A difference of zero means we have found the exact deformation generating the correct resisting force. Because the system is nonlinear and not directly invertible, we cannot use the force to calculate the deformation directly; instead, we must guess and iteratively correct the deformation.

NOTE: Fa sto esempio con Forza e deformazione che è essenzialmente il $f(x^{(i)})$ e il $x^{(i)}$.

In summary, we are using a linear solver to solve a nonlinear problem:

- The difference between external forces $\{F^a\}$ and internal forces $\{F^{nr}\}$ is called the **residual**. It is a measure of the force imbalance in the structure (these are just different terms for the concepts discussed above).
- The goal is to iterate until the residual becomes acceptably small, meaning the solution has **converged**.
- When convergence is achieved, the system is in **equilibrium** within an acceptable tolerance.

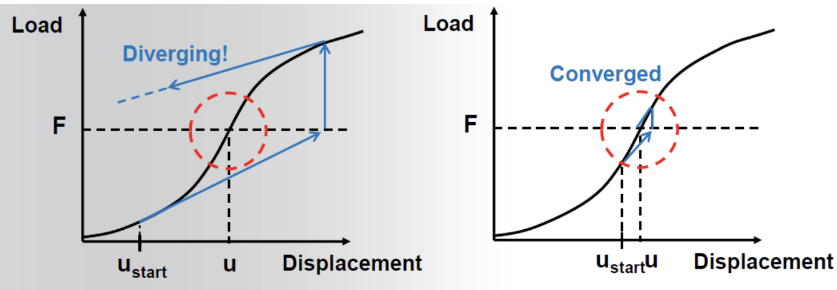
NOTE: When iterating, it is impossible to reach exactly 0; in FEA, a perfect 0 does not exist. Therefore, we must define a tolerance that is ”small enough.”

Convergence in the Newton-Raphson method

The choice of initial displacement can cause the system to either converge or diverge. Therefore, the convergence of the system depends heavily on the arbitrary initial guess.

There are two main points to consider:

- This method does not guarantee convergence in all cases.
- To achieve *convergence*, the initial guess must fall within the **radius of convergence**.



Starting outside the radius of convergence

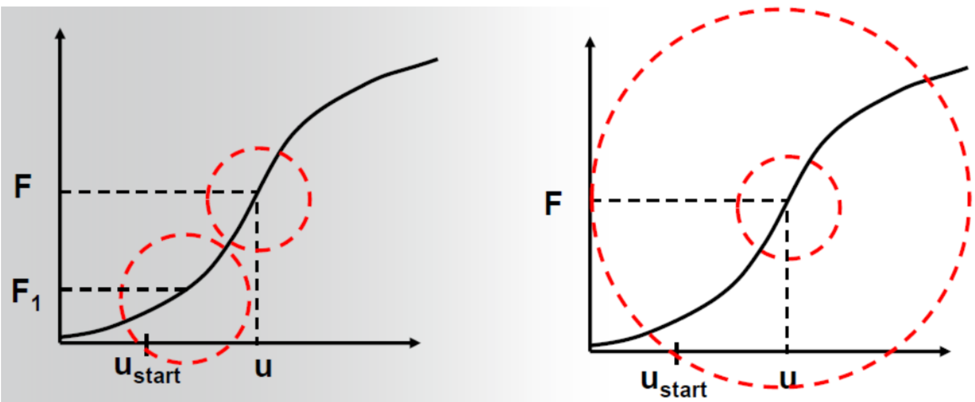
Starting inside the radius of convergence

NOTE: È stata fatta na domanda, dove si è chiesto se è possibile convergere a soluzioni diverse partendo da punti diversi, il prof non l'ho sentito sicuro, ma ha risposto di sì, senza approfondire. Poi ha anche detto su ad ansys dicendo che a volte da soluzioni strane.

How do we limit the dependency on the initial guess?

There are two main approaches:

- We apply the load incrementally. Instead of applying the entire loading condition at once, we apply only a fraction of it. By applying a smaller load, we increase the probability that our initial guess falls within the radius of convergence. Essentially, we solve the main problem by breaking it down into a series of smaller problems. Consequently, our guess and the radius of convergence will evolve alongside the load increments.
- We use convergence-enhancement tools to artificially enlarge the radius of convergence.

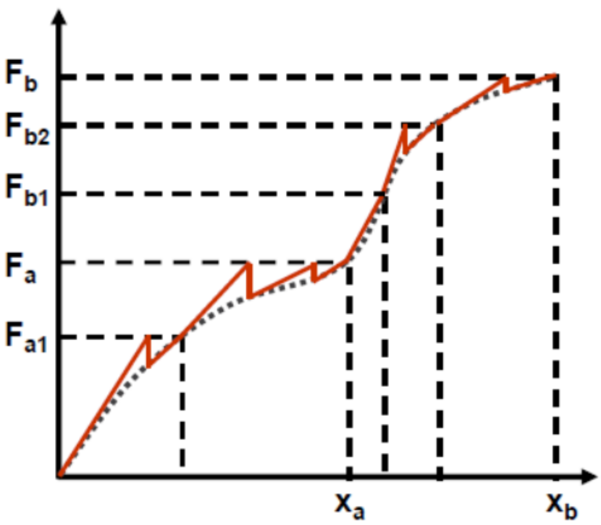


Notation used in Ansys

NOTE: I seguenti concetti sono parametri modificabili nel solver di ansys e ti permettono di risolvere i vari warning di convergenza che a volte saltano fuori.

Let's explore how loads are managed:

- **Load steps**: These represent the different types of forces being applied. For example, if you apply a normal force and gravity, you have two load steps. In the figure, F_a and F_b are load steps. Thus, a load step is a way to differentiate and sequence loading conditions.
- **Substeps**: These are used to apply the loads incrementally. Every load step is divided into several smaller steps called substeps. This allows us to solve smaller intermediate problems, gradually increasing the percentage of the total load step applied.
- **Equilibrium iterations**: These are the iterative corrections performed to achieve a converged solution for each substep.

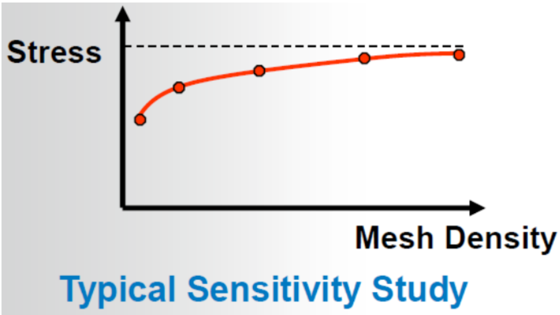


Nonlinear FEA issues

Typically, three main issues arise when performing a nonlinear finite element analysis:

- Obtaining convergence. This depends heavily on the *radius of convergence*. While there are tools to aid convergence, even experienced users often rely on trial-and-error to achieve it.
- Balancing computational expense versus accuracy. The trade-off between expense and accuracy is largely tied to the **number of load increments**. Increasing the number of iterations enhances accuracy but also significantly increases computational demand, so finding the right balance is essential.
- Verification.

A final interesting aspect is the mesh density. Obviously, increasing the mesh density increases both computational demand and accuracy. However, this is only true up to a certain point; beyond that threshold, further increasing the density will not significantly alter the results.



GENERAL PROCEDURE

Building a Nonlinear Model

We are going to use the Newton-Raphson method to solve these types of models.

Differences Between Building a Nonlinear and a Linear Model

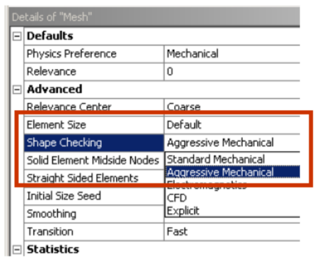
In some cases, there will be no difference:

- A model undergoing mildly nonlinear behavior due to large deflections and stress stiffening effects might need no modifications with regard to geometry setup and meshing.

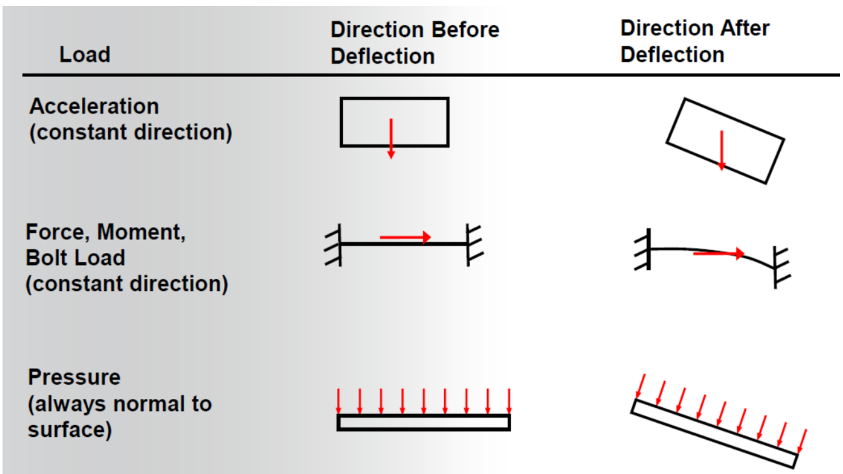
In other cases, specific features must be included:

- **Elements with special properties**: *Contact elements* can be a source of nonlinearity. If contact between elements is anticipated, appropriate connections must be defined in the software to ensure it recognizes the interacting surfaces.
- **Material properties**: Particularly if they exhibit nonlinear behavior (e.g., plasticity and creep data).
- **Geometric features**: Another common source of nonlinearity.
- **Mesh controls**: Necessary when experiencing *large deflections* (significant deformation) or when large displacements are expected.
- **Element technology**.
- **Loads and boundary conditions**: These can introduce complexities under large deflections.

Regarding meshing, if *large strains* and nonlinearities are expected, set the shape checking to "Aggressive Mechanical".
PS: The professor mentioned he is unsure if the default in newer versions is "Aggressive"; previously, it was definitely "Standard".



Let's focus on loads and boundary conditions, particularly when large deformations or displacements are involved. The application of loads and boundary conditions may vary depending on the current state or deformation of the model.



NOTE: Depending on the type of load, deflections can cause changes relative to the local or global reference system. For example:

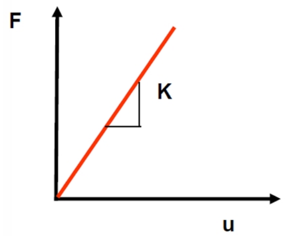
1. Changes relative to the local reference system, but not the global one.
2. Changes relative to both local and global reference systems.
3. Pressure is always normal to the surface, meaning it changes relative to the global reference system but remains constant relative to the local one.

What is different about obtaining a nonlinear solution?

Let's focus on two different cases:

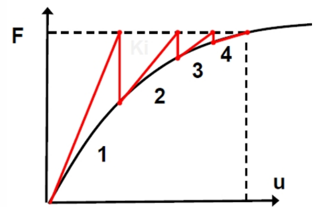
- Linear static analysis requires only one pass through the matrix equation solver:

$$F = Ku$$



- Nonlinear analysis performs a new solution for every iteration:

$$F_i = K_i u_i$$



Analysis Settings

Generally, you will need to manage many of these parameters, though some are automatically handled by the software so you don't have to deal with them.

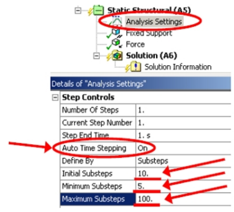
- **Step controls**: Load steps (defined completely by the user), Substeps (creation, generation, and sizing can be managed by the software), and Auto Time Stepping. This refers to the last part of the overview.
- **Solver controls**: Choosing the appropriate type of solver.
- **Nonlinear controls**: Newton-Raphson convergence criteria.
- **Restart controls**: Resuming a previous analysis.
- **Output controls**: Determining what data is saved.
- **Analysis data management**: Managing the retention or deletion of files.

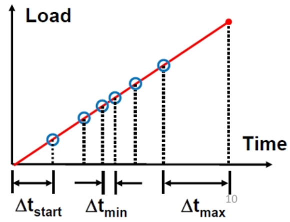
NOTE: The concept of **time** in Ansys represents the fraction of the loading condition applied in the simulation. Time ranges from 0 to 1 (i.e., 0% to 100%). This is useful for identifying which loading condition (or load step) is causing convergence issues, making debugging easier. If there are multiple load steps, the first ranges from 0 to 1, the second from 1 to 2, the third from 2 to 3, and so on.

Step Controls

Remember: A substep is a percentage of a load step. By applying only a fraction of the load, it is more likely that the solution remains within the radius of convergence.

Let's focus on the steps:





- **Auto time stepping**: If enabled, three fields must be defined: initial, minimum, and maximum substeps (otherwise, the software manages it autonomously). This feature is powerful because it minimizes computational time while slowing down the simulation (taking smaller steps) whenever convergence issues arise.

  The underlying concept is straightforward: if the solution converges easily, the software decreases the number of substeps (down to the minimum limit) to reduce iterations. If it struggles to converge, it increases the number of substeps to apply a smaller percentage of the load, thereby improving the chances of convergence.

  The software dynamically evaluates the model's condition and finds the optimal number of substeps within the defined range.

  NOTE: If Ansys *Mechanical* has trouble converging and needs to adjust the number of substeps, it will perform a *bisection*.

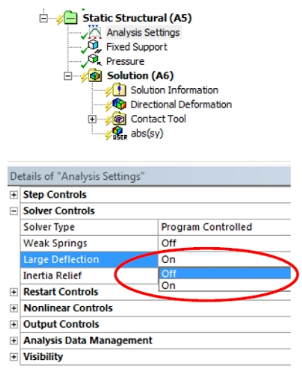
  It returns to the last converged substep and re-applies the load using a smaller increment.

Solver Controls

First, you must choose the type of solver; it is usually best to leave it as "Program Controlled". This setting determines how the software builds the stiffness matrix for each Newton-Raphson equilibrium iteration.

- **Direct (Sparse)**: More robust and recommended for nonlinear models, though it is computationally more demanding.
- **Iterative (PCG)**: More computationally efficient (recommended for large bulk solid models dominated by linear elastic behavior).

Large Deflection: This option enables most of the checks and controls required for nonlinear problems. It accounts not only for large deflections but also for: Large rotation,Large strain,Stress stiffening,Spin softening.



Enabling "Large Deflection" activates checks for large displacements, which is the most straightforward effect.

NOTE: Be aware that large deflection does not necessarily imply large strains. For example, a beam undergoing rigid body rotation or bending may experience significant displacement at the tip, altering its orientation, without generating high strains locally. This is a classic case of large deflection without large strain. This distinction is crucial when applying Euler-Bernoulli beam theory.

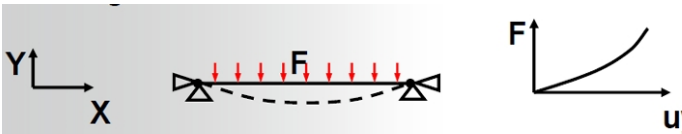
If "Large Deflection" is OFF, the solver assumes small deformations: it does not update the orientation of the tip as forces are applied, and cross-sections are assumed to remain perpendicular to the undeformed axis. For large displacements, this assumption yields physically impossible results (e.g., the beam will just deform vertically and artificially elongate). Turning ”Large Deflection” ON ensures the geometry is updated, making the beam behave realistically.

Setting Large Deflection = ON also accounts for local large deformations (large strains). The software changes how it computes mechanical strain: instead of the standard engineering strain $\varepsilon_l = \frac{l-l_0}{l_0}$, it uses the *nonlinear* (true/logarithmic) strain expression:

$$\varepsilon_l = \int_{l_0}^{l} \frac{dl}{l} = \ln\left(\frac{l}{l_0}\right)$$

This formulation is more accurate for high deformations, especially because it accounts for the fact that the elements may become highly distorted.

Another phenomenon accounted for when Large Deflection is ON is stress stiffening. Consider the following structure:



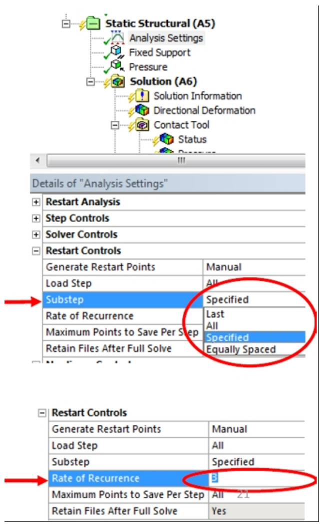
With both ends fixed, the transverse deformation induced by the applied force generates axial tension. This tension causes the material to behave more stiffly against further transverse loads. Thus, you get a stiffening effect induced by the applied stress state.

NOTE: The professor didn’t explicitly say it, but my understanding is that we should always turn this ON. He mentioned that one of the most common mistakes in the exam is forgetting to enable ”Large Deflection”. If you leave it OFF, you must be able to justify your choice. While leaving it OFF might be technically correct in some specific scenarios, those cases are rare for our purposes.

Restart Controls

This feature allows you to stop a simulation, make adjustments, and then resume it. It can also be used to resume a simulation that stopped due to convergence issues, potentially saving a significant amount of computational time.

For example, if your simulation consists of 10 load steps and convergence problems occur only during the final one, this feature allows you to restart directly from the 10th step. You are not forced to restart the entire simulation from zero, saving the progress of the previously converged steps and allowing you to focus on troubleshooting the last one.



You have several options available:

- **Pause & Stop**: After stopping the simulation, you can modify parameters in the ”Analysis Settings” or adjust the magnitudes of existing loads. However, you cannot introduce new loads or remove existing ones. Be aware that *restart points* are saved if the simulation is unsuccessful or if it is manually stopped.

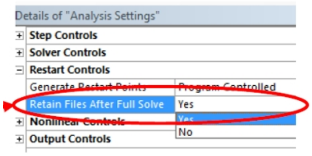
- **Generate Restart Points**: You can set this to ”Manual” to gain control over:
  - *Load Step*: Specify at which load steps to create restart points.
  - *Substep*: Specify how often restart points are created. You can choose between: Last, All, Specified, and Equally Spaced.

    For the ”Specified” option, you define the *rate of recurrence*, which dictates the maximum number of substep restart files retained per load step. For example, it can be set to only save the last 3 generated substeps.

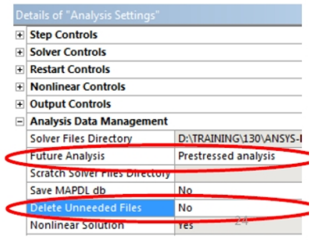
    For the ”Equally Spaced” option, the software selects substeps that span the load step as evenly as possible.

    Remember that you can choose to save only specific substeps based on these settings, or opt to save all of them.
    You can also choose to ”Retain Files After Full Solve”; otherwise, they are typically deleted once the simulation completes successfully.

Analysis Data Management:

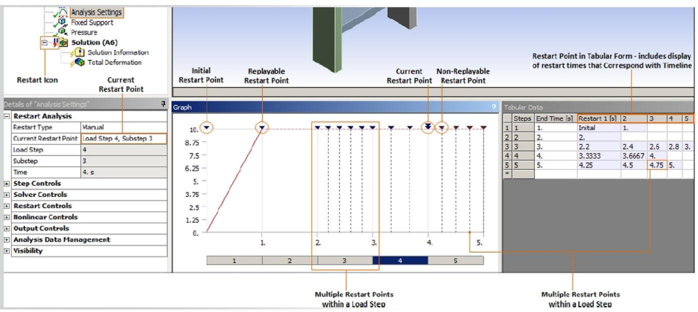


Here you can instruct the software to retain generated data (e.g., for a ”Prestressed Analysis”) if you intend to use this data as an input for a subsequent analysis. You can also modify the ”Delete Unneeded Files” setting to manage which files the software discards.



How can we use this data?

If restart options are configured and the simulation is halted (either manually or due to convergence issues), you will find the option to restart the simulation within the ”Analysis Settings”. By setting the restart type to ”Manual”, you can specify exactly from which point to resume.



Once you have selected the desired restart point, you simply press ”Solve” to resume the simulation from that exact state.

The restart points are color-coded in the interface:

- **RED**: These points will be overwritten and are not replayable. If you change a load or analysis setting for a given step and restart from an earlier point, the subsequent red points will be eliminated and will no longer be available.

- **BLUE**: These are the valid, replayable points.

These are the types of loads that you can modify:

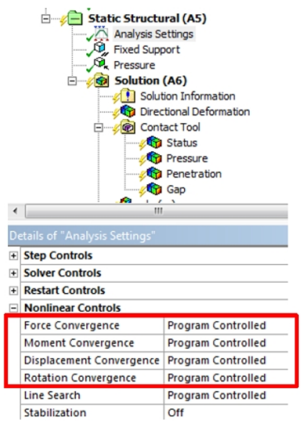
Load Type	Load Specified As...		
	Constant	Tabular	Function
Pressure	X	X	X
Line Pressure	X	X	X
Force	X	X	X
Remote Force	X	X	X
Moment	X	X	X
Displacement	X	X	N/A
Remote Displacement	X	X	N/A
Rotational Velocity	X	X	X
Bolt Pretension	X	X	N/A
Acceleration	X	X	X
Earth Gravity	N/A	N/A	N/A
Hydrostatic Pressure	X	N/A	N/A
Bearing Load	X	X	N/A
Joint Load	X	X	X
Pipe Temperature	X	X	X
Pipe Pressure	X	X	X
Thermal Condition	X	X	X
Imported Load	N/A	N//A	N/A
Nodal Force	X	X	X
Nodal Pressure	X	X	X
Nodal Displacement	X	X	N/A

Nonlinear Controls

We need to define convergence criteria, and typically we use ”Force Convergence”. We should only change this from ”Program Controlled” if strictly necessary and we know exactly what we are doing. Recalling the Newton-Raphson method, we use the following formula:

$$\left[K^T\right]\{\Delta u\} = \{F\} - \{F^{nr}\}$$

The method attempts to reduce the residual (force imbalance) until it becomes acceptably small.



Let’s mathematically define the **convergence criterion**:

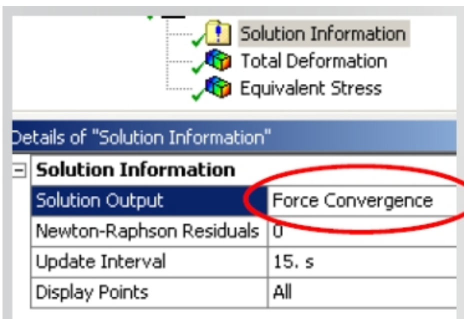
$$||\{R\}|| < (\varepsilon_R R_{ref})$$

Where:

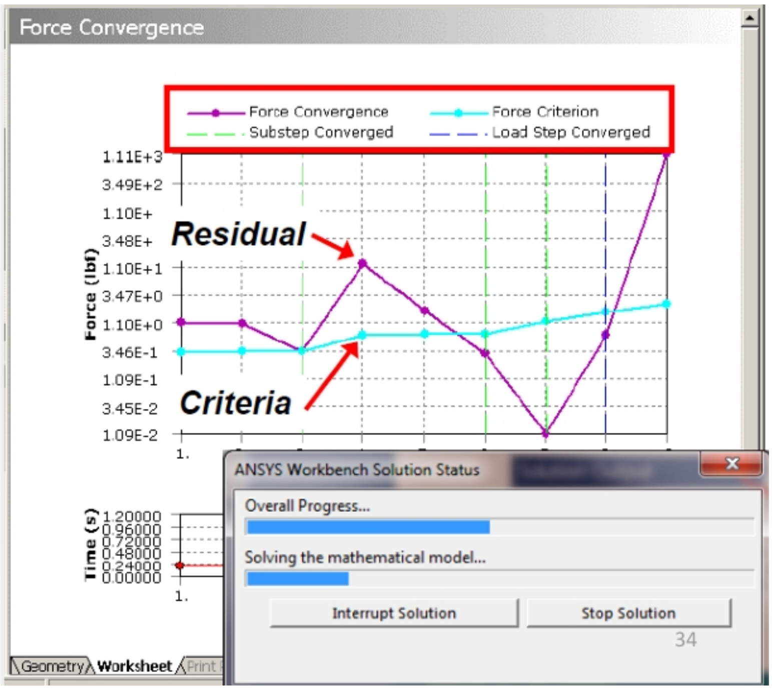
- $\{R\} = \{F^a\} - \{F^{nr}\}$ is the residual vector.
- $||\{R\}|| = \left(\sum R_i^2\right)^{1/2}$ is the vector norm of the residual (used to extract a scalar magnitude from the vector).
  - $\varepsilon_R R_{ref}$ is the force convergence criterion.
  - ε_R is a tolerance factor and R_{ref} is a reference force value.
  - R_{ref} is the norm of all applied forces and reactions, $||\{F\}||$ (which automatically scales the criterion relative to the magnitude of the load).

There is also a moment convergence criterion that applies to rotations.

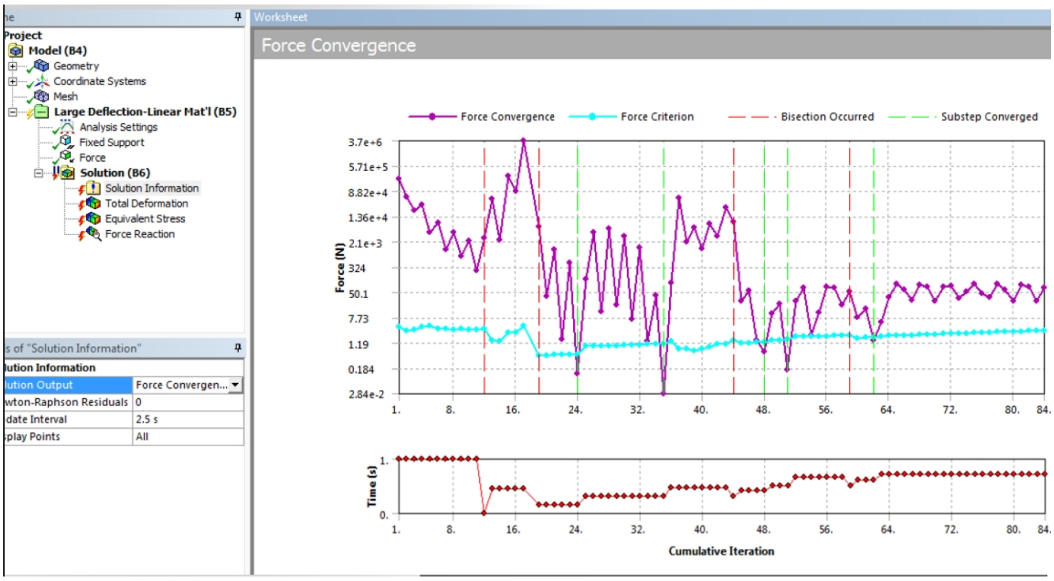
NOTE: A crucial practice is to monitor the ”Force Convergence” graph.



This allows you to observe how the simulation is progressing in real-time. Checking this plot during the simulation is important because it is generated point by point, where each point represents an iteration. Every iteration provides the value of the residual and the calculated convergence limit. If the residual is too high (i.e., it exceeds the convergence criterion), the current solution is not acceptable. Thus, the graph visually indicates when a substep has converged.



Let's focus on the next figure:



What may happen—particularly if "Auto Time Stepping" is enabled—is that the applied load increment (the size of the substep) is too large. Consequently, your initial guess falls outside the radius of convergence, and the iterations begin to diverge. The software is capable of detecting this.
Since the residual curve is not converging towards the force criterion line, the software automatically discards the current substep size and reduces it.
Recall that "time" in this context represents the percentage of the applied loading condition.
If you observe the smaller time plot, you can see that initially, the software attempts to apply 100% of the load. Because convergence is impossible, a bisection occurs (indicated by the red dotted line). It then tries applying 50% of the load. After several iterations without convergence, it bisects again to 25%, and so on. This process continues until the substep size is small enough to achieve convergence (indicated by the

green dotted lines). This represents an unsuccessful simulation, as the full load condition is ultimately not applied.

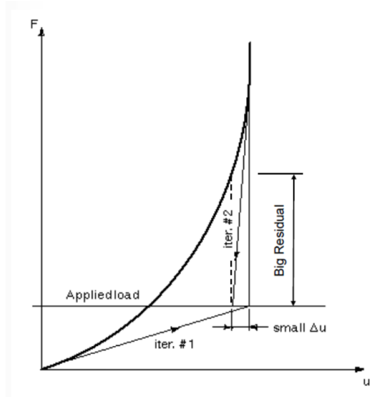
How can we change these convergence criteria?
When modifying these criteria, you must be aware of one important detail:

- If you add a custom convergence criterion, the program automatically deletes all default ones.

Therefore, you must explicitly re-establish the force convergence check.

Let's observe a force-displacement plot. Suppose we evaluate Newton-Raphson iterations based on a displacement criterion rather than a force criterion. As shown in the graph, the displacement residual might be small while the force residual remains very large. Relying solely on a displacement criterion can thus lead to an incorrect or unconverged solution.

- Displacement-based checking is a relative measure and should only be used as a supplementary check.
- Force-based convergence provides an absolute measure of convergence, as it directly verifies the equilibrium between internal and external forces.



What happens if no external force is applied?

The software employs a safety feature to prevent the solution from requiring a zero tolerance (which is mathematically impossible to achieve).

- **Minimum Reference Value (MINREF)**:
  - In free-body (unconstrained) systems or mechanisms with no external forces, the calculated force criterion ($\varepsilon_R \cdot ||\{F\}||$) would be zero. If the criterion is exactly zero, the solution will *never converge*.
  - In such cases, the program redefines the criterion to be $\varepsilon_R \cdot$ MINREF, where ε_R is the convergence tolerance factor.

Nonlinear Controls	
Force Convergence	On
--Value	Calculated by solver
--Tolerance	0.5%
--Minimum Reference	2.2481e-003 lbf

Notes about Convergence Criteria: You are allowed to modify the convergence criteria, particularly the force criterion, but this must be done with caution.

- Do not "loosen" a criterion excessively. Doing so widens the acceptable residual window; while this increases the likelihood of the simulation converging, it may converge to a physically incorrect state.
- Conversely, you can "tighten" the criteria to increase accuracy, but be aware that this will increase the number of iterations and the overall computational cost.

Another important consideration is **stabilization**:

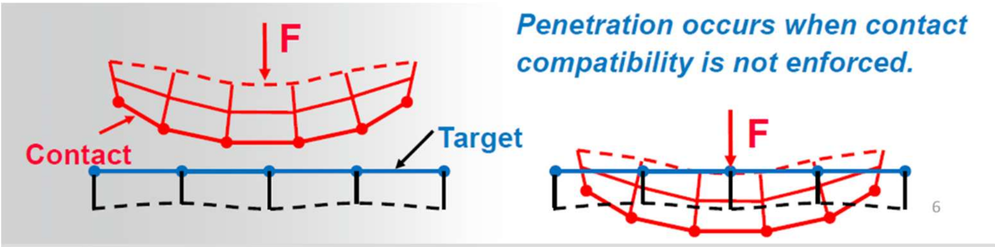
- See Chapter 5.

What about reviewing nonlinear results?
Generally, the main difference is that there is much more data to review.
Remember to always check the actual deformation scale and do not be misled by auto-scaling features. The last point relates to **materials**: since material behavior can also be nonlinear and may depend on the direction of the applied load, you might need to evaluate specifically user-defined directional parameters.

Contacts

In a physical sense, two objects that are in contact exchange compressive normal forces and tangential normal forces, but not side forces. The two bodies cannot interpenetrate (in the sense they cannot occupy the same volume at the same time). They do not transmit tensile normal force, so the surfaces are free to separate and move away from each other. In more technical way, when two separate surfaces touch each other such that they become mutually tangent, they are said to be in contact.

In Ansys, when defining the type of contact we have to take in consideration two elements: separation and sliding.



Types of contact

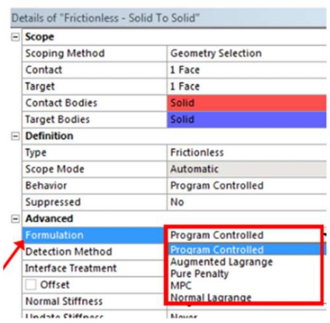
The main 5 types of contact behaviours are:

- Bounded: no separation and no sliding between faces or edges
- No Separation: similar to bounded, but frictionless sliding can occur along contacting faces
- Frictionless: surfaces are free to slide and separate without resistances
- Rough: similar to frictionless, but sliding is not allowed (is similar to have a friction coefficient equal to ∞).
- Frictional: allows separation without resistances and sliding with resistance proportional to user defined coefficient of friction

Table 1: Contact Names and Behavior

Name	Gap Open/Close?	Sliding Allowed?
Bonded	No	No
Rough	Yes	No, infinite μ
No Separation	No	Yes, μ = 0
Frictionless	Yes	Yes, μ = 0
Frictional	Yes	Yes, if $F_{sliding} > F_{friction}$

Contact Formulations



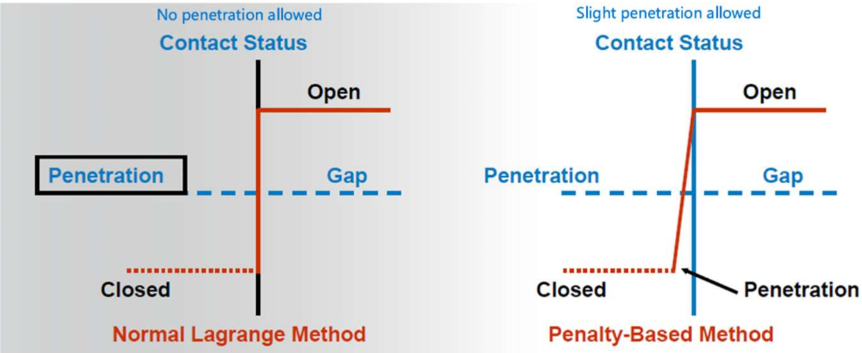
When you need to constraint physics in the software and mathematically describe the equations that do not allow the bodies to penetrate each other, and so to occupy the same physical space, you must enforce contact compatibility. 'Ensures contact compatibility' means that the program establishes a relationship between the two surfaces to prevent them from passing each other in the analysis. To do so, the compatibility must be defined mathematically through 'contact formulations' . There are several kinds of formulations: Programmed controlled, augmented Lagrange, pure penalty, normal Lagrange and MPC.

Pure penalty

You can imagine this method as placing a stiff spring between two parts that have come into contact during the simulation. The body is going to penetrate and in order to manage the overlapping area, we imagine having a stiff spring between the two surface and a generating of a normal force F_n equal to the multiplication between k_n and x_p. k_n is the contact normal stiffness that is the stiffness of the virtual spring. The degree of penetration depends on the stiffness of the spring, which must be selected by the user. Ideally, for an infinite k_{normal}, one would get zero penetration. This is not numerically possible with penalty-based methods, but as long as $x_{penetration}$ is small or negligible, the solution results will be accurate. The selection is fundamental, because a too high value minimize the penetration but increase the complexity of solving the problem (leads to convergence difficulties), while a small value is going to find an easy solution but at the cost of a remarkable penetration. If the penetration is on the same order of magnitude as the calculated displacements, the results will be quite suspect. Ansys automatically calculates the contact stiffness k_n, based on the underling solid element's size and material properties. However, the user should monitor the nonlinear solution. if convergence difficulties are met due to too high of a k_n value, ANSYS provides tools on determining which contact region's contact stiffness should be reduced. Likewise, postprocessing penetration x_p is available after a converged solution to ensure that k_n was not too low. The user can change it: in the Details view of a Contact Region, the "Advanced: Normal Stiffness" option should be changed to "Manual", and the user may input a "Normal Stiffness Factor".

Normal Lagrange

This method adds an extra degree of freedom (contact pressure) to satisfy contact compatibility. The satisfaction of this degree of freedom is explicit considered in the equations. The penetration is enforced to be 0 or nearly 0 with pressure DOF. The advantage of this method is that it said not require a normal contact stiffness (zero elastic slip), but the disadvantage is that it requires a direct solver, which can be more computationally expensive. Furthermore, it can only be used if the applied forces are normal to the contact surfaces. A limitation that often occurs with normal Lagrange is 'chattering': since the normal Lagrange method enforces zero or nearly-zero penetration, the contact status can only be open or closed (step function): this can make the convergence more difficult because contact points may oscillate between closed/opened status, which is called *chattering*. If some slight penetration is allowed, it can make it easier to converge since contact is no longer a step change (enlarge the window of acceptability of the solution).



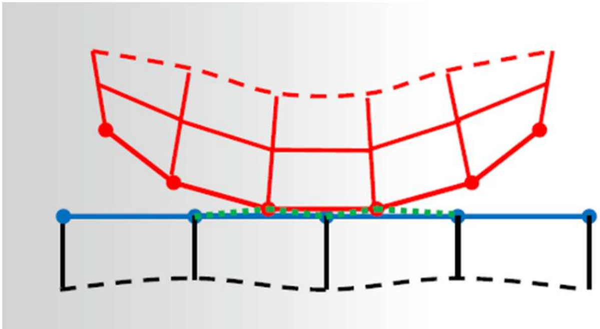
Augmented Lagrange

The augmented Lagrange method is a penalty-based formulation, that 'augment' the pure penalty formulation with an extra Lagrangian term, λ. This method is a lot used because it has all the advantages of pure penalty plus the Lagrangian term which makes the selection of the normal stiffness less impactful on achieving a sufficient level of accuracy. In pure method, the level of penetration can be modulated only by selecting different normal stiffness (in the penetration-force graph: changing the slope of the line), but a very small modification of k_n can lead to an important change into penetration elongation. In the Augmented Lagrange, λ acts as an offset. Varying λ has a direct impact, while varying the normal stiffness has an amplified effect on the evaluation of the penetration.

The problem of setting k_n, in the augmented Lagrange method is less impactful since it is less sensitive to the magnitude of k_n. For these reasons, the augmented Lagrange method is the default formulation used for 'Program Controlled' option.

Multi-point constraint (MPC) formulation

The method internally adds constraint equations to 'tie' the displacement between contacting surfaces. It is visible in the following image, where the green line represents connectors between nodes, which constraint the possibility of moving and penetrates. This approach is not penalty-based or Lagrange multiplier-based. It is a direct, efficient way of relating surfaces of contact regions which are bonded. Large deformation effects are also supported with MPC-based bonded contact. This method applied only to 'bounded' and 'no separation' types of contact.



Another option is 'beam', which is similar to MPC, but utilizes internally-created massless linear beam elements to connect the two sides of a contact interface together. This can be more efficient than the traditional formulations and can avoid the over constraints that can happen if multiple contact regions utilizing the MPC option end up generating constraint equations that tend to conflict with each other.

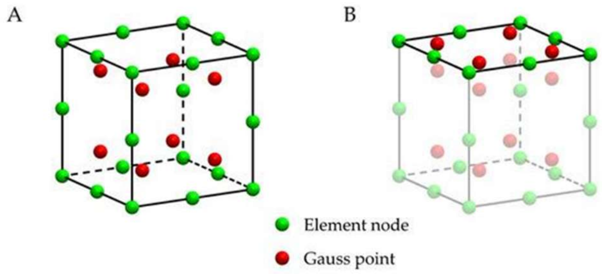
Both MPC and Normal Lagrange are applied on nodes, while pure penalty and augmented lagrange are applied on gauss points.

Focus: Nodes and Gauss Points

Nodes are corners or edges of an elements. They are the primary locations where the solver calculates primary unknowns such as displacement (in structural analysis), temperatures (in thermal analysis) and pressures (in fluid analysis). In the picture are the green dots.

Gauss points are mathematical locations inside the element where the actual integration of the governing equations happens. ANSYS uses "Gaussian Quadrature" to calculate stiffness matrices, strain and stresses (derived from nodal displacements) and heat flux. Red dots in the image.

The nodes are primary order elements, this means that we have a direct result. We obtain the solutions at the nodes and then linearizing we found other solutions in between the nodes. The equations are simpler but less accurate. The gauss points are second order elements, where solutions are found only in midpoints. They are more complicate but also accurate. The gauss points are called also integration points, which are the points where integration happens numerically. Every derived quantity is calculated by integration at the gauss points.



To increase the accuracy two strategies can be used: increase the number of first order elements (easier but more equations to solve) or introduce second order elements (less equations to solve but more difficult).

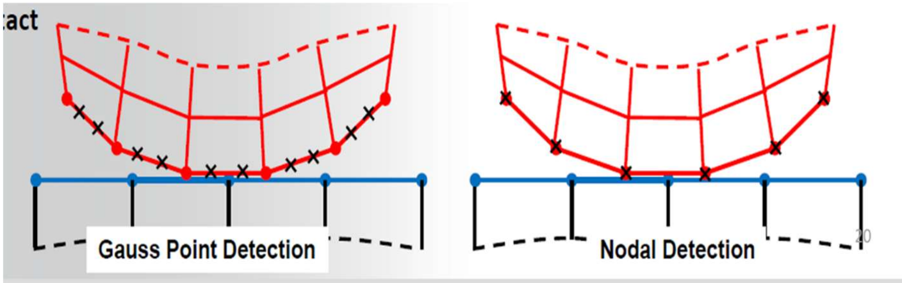
In every type of contact, there are a contact surface and a target surface. The user must define which is the contact surface and which is the target one. there are some automatic options that allow ansys to choose the contact and target surface. There are some guidelines:

- Shape: if a convex surface comes into contact with a flat surface or concave surface, the flat or concave surface should be the target
- Mesh size: the surface with the coarser mesh should be the target
- Stiffness: the stiffer surface should be the target
- Element order: the lower order surface should be the target
- Geometrical size: the larger surface should be the target

However, no object has all the characteristics to be the target and these guidelines have no order of importance.

Detection method

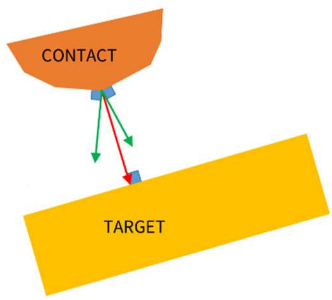
Detection method allows to choose the location of contact detection used in the analysis in order to obtain a good convergence. Pure penalty and augmented Lagrange formulations usually use gauss point detection by default, instead Normal Lagrange and MPC formulations use 'Nodal-Normal to Target' by default (this results in fewer detection points). There are 3 nodal based detection methods: nodal-normal to target, nodal-normal from contact, nodal-projected normal from contact.



In case of rounded object (as in the image) the detection of contact happens more or less at the same time using gauss point detection or nodal detection. In case of a sharp object: if you use gauss points the detection of contact started once the penetration has already happened, instead the nodal-based method will detect penetration immediately.

Nodal-normal to target and nodal-normal from contact dictates the direction of forces to be applied at the interface.

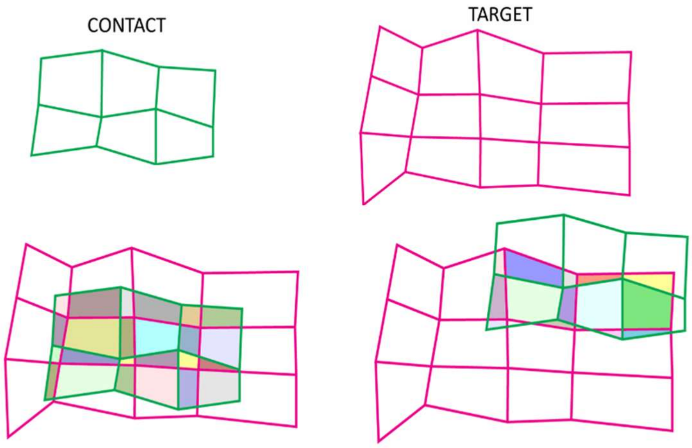
An example: When two bodies make contact, the solver must create an equal and opposite force to prevent penetration, but what direction should that force be applied? Consider a 2D case of a flat face on one side of the contact pair and two facets on the other side of the contact pair. A single node on one element is touching the face of the element on the other side.



Nodal – Normal to target: The contact detection location is on a nodal point where the contact normal is perpendicular to the target surface. (red arrow)

Nodal - Normal from contact: The contact detection location is on a nodal point where the contact normal is perpendicular to the contact surface (average). (green arrows)

Nodal - Projected Normal from contact: it is still based on nodes, but instead of be used in pairs, in this case the nodes are considered as neighbours, because we are considering an overlapping area between the two bodies that are going to be in contact. This method enforces a contact constraint on an overlapping region of contact and target surfaces. The contact penetration/gap is computed over the overlapping region in an average sense. It provides more accurate contact tractions and stresses of undelying elements compared with other settings. Results are less sensitive to the designation of the contact and target surface. It satisfies moment equilibrium when an offset exists between contact and target surfaces with friction. Calculated contact force distribution has smoother variation across multiple target elements. The contact detection location is at contact nodal points in an overlapping region of the contact and target surfaces (projection-based method). Instead of calculating penetration/gap at each integration point or node, it calculates 24 an average value for each overlapping area.

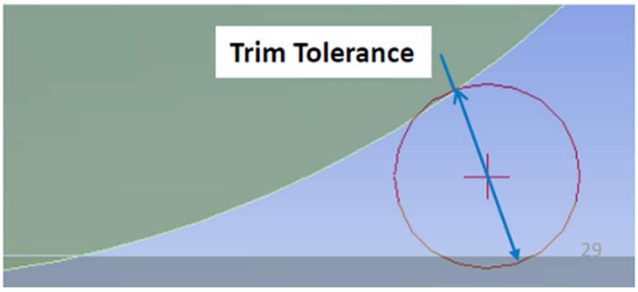


For example, if we have a contact (green) and a target (pink) surfaces, when they are in contact, the two surface overlaps and the intersection of meshes of the two surfaces ‘generate’ new nodes. We can estimate how large is the overlapping area. The advantage in considering the entire overlapping area is that when two bodies in contact slides one on the other, if you consider pairs one by one, when they lose contact is an abrupt change, because is on or off. But if we use an overlapping area, once the two area slides out, the overlapping area starts gradually reducing without abrupt changes.

Property	Description
Program Controlled	This is the default setting. The application uses Gauss integration points (On Gauss Point) when the formulation is set to Pure Penalty and Augmented Lagrange. It uses nodal point (Nodal-Normal to Target) for MPC and Normal Lagrange formulations.
On Gauss Point	The contact detection location is at the Gauss integration points. This option is not applicable to contacts with MPC or Normal Lagrange formulation.
Nodal - Normal From Contact	The contact detection location is on a nodal point where the contact normal is perpendicular to the contact surface.
Nodal - Normal To Target	The contact detection location is on a nodal point where the contact normal is perpendicular to the target surface.
Nodal - Projected Normal From Contact	The contact detection location is at contact nodal points in an overlapping region of the contact and target surfaces (projection-based method).

Trim contact and trim tolerance

Trim contact is in ‘Definition’ of control panel. It is useful when there are extended surfaces, but the contact/interaction area is very small. In fact, when you select contact regions, you are allowed to select only entire surfaces. However, if we a-priori know that only a limited region will be in contact, and do not define anything calculations will happen in between nodes that are very far away from the contact region. However, we can set the trim contact with numbers that already define an area which will be in contact, excluding all the rest. This reduced the number of contact elements generated within each pair, speeding up the processer time. In this case, an important parameter to set is the 'trim tolerance'. Trim tolerance defines the maximum distance in between the nodes, above which pairs must not be considered anymore. It defines the upper bounding box dimension used for trimming operation. For automatic contacts, this property displays the value that was used for contact detection, and it is a read-only field. For manual contact, user can enter any value greater than zero.



The trimming operation take place in the pre-processing step.

Penetration Tolerances

The penetration should be as low as possible, but it cannot be 0 otherwise k will be infinite and it is not possible. Contact compatibility is satisfied in normal direction if normal penetration (x_p) is within allowable tolerance. The penetration tolerance can be defined in two ways: by inserting directly a value or defining it as the product between a factor and the element depth. The factor can be changed by the user, its default value is 0.1. This is only valid for penalty-based formulations (pure penalty and Augmented Lagrange).

Contact stiffness

Normal stiffness is a multiplier or factor (FKN) on the code calculated stiffness. By default, in bounded and no-separation types of contact is 10, while for the other contact behaviours the default value is 1. Normal stiffness can also be set manually to any value. Of course, this is only valid for penalty-based formulations. The normal stiffness k_n is the most important parameter affecting both accuracy and convergence behaviour. A large value of stiffness gives better accuracy, but the problem may become more difficult to converge. If the contact stiffness is too large, the model may oscillate with contacting surfaces bouncing of each other. The normal stiffness can also be automatically adjusted during the simulation to enhances convergence. The 'update stiffness' option can be set to program controlled, never, each iteration or each iteration aggressive (which allows for a broader range of adjustment and adjusts also other sub-parameters). With the update stiffness option activated, the program will modify the stiffness (raise/reduce/leave unchanged) based on the physics of the modes. Stiffness is automatically determined in order to allows both convergence and minimal penetration.

Pinball Region

The pinball region is a contact element parameter that differentiates between far field open and near field open status. The pinball region allows us to exclude some elements to have possible contacts. It can be thought of as a spherical boundary surrounding each contact detection point.

When the two bodies are far away, the bodies are called to be 'far open', in this case there is further calculation possible. If the two bodies start moving closer and get close within a certain threshold, the contact is still open because the bodies are not in contact but this condition is called 'near open'. In this case ansys starts to evaluate more in details how much is the distance and how could be the contact. There are 3 conditions: not considered at all, considered but not evaluated because they are far open, considered and evaluated because are near open.

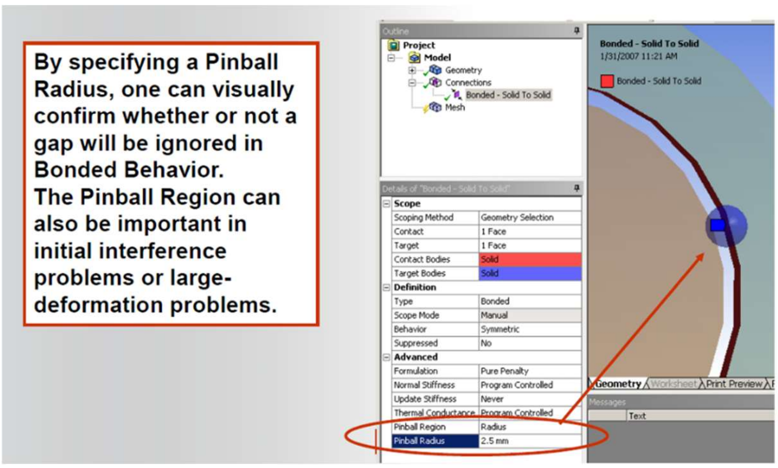
The threshold that separates the far open to the near open condition, is defined by the pinball region. The pinball region is defined by the pinball radius, which generate a sphere that is set at the boundary and define a volume around the surface, if the nodes of the other body are inside the pinball region, the bodies are considered to be near open, the nodes that are outside the pinball region are far open.

If you used bounded or no separation contact type, you must be very careful in having a too large pinpall region, because in this case, all the nodes that are inside the pinball region at the beginning of the simulation, will be considered to be already in contact and so the gap in between the nodes will not close during the simulation. However, this is a good trick to have two body in contact already at the beginning of the simulation.

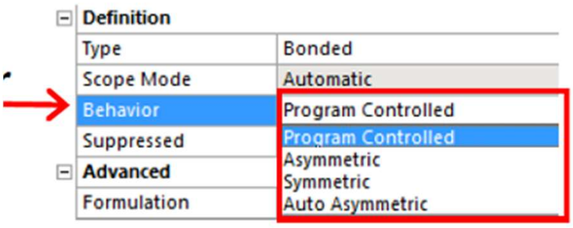
The size of the pinball region can be controlled with 3 different options:

- Program controlled: is the default option. The pinball region will be calculated by the program based on underlying element type and size.
- Auto detection value: the pinball region will be equal to the tolerance value as set on the global contact settings. This ensures that contact pairs created through the automatic contact detection have a pinball radius that envelops gap between target and contact. Is the recommended option for cases where the automatic contact region is larger than the program-controlled pinball value. In such cases, some contact pairs that were detected automatically a not be initially closed at start of solution.
- Radius: user manually specifies a value for a pinball region.

In the autodetection value and the radius options will appear a sphere on the contact region for easy verification be before the simulation allowing the user to visually check is the pinball region is covering entirely the gap, while program control option computes the radius and the sphere only during the simulation.

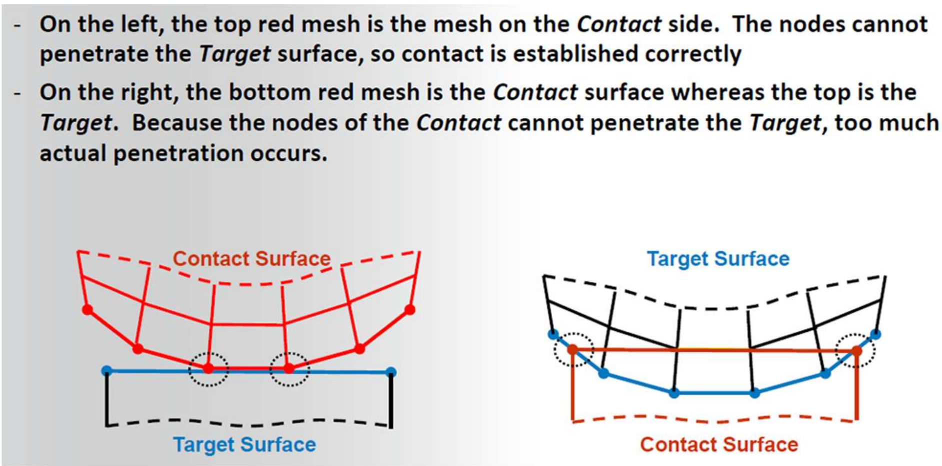


Symmetric/Asymmetric behaviour



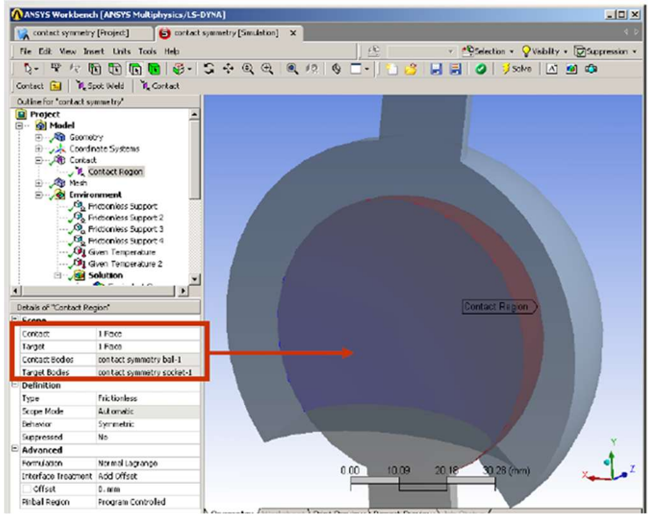
This behaviour is related to the selection and definition of what is the contact and what is the target surface in defining a contact feature. Through the symmetric/asymmetric behaviours we can manage the difficulty in defining which is the target and which is the contact. The concept of symmetric/asymmetric behaviour only applies to penalty-based methods.

Asymmetric behaviour only constraint nodes of the contact surface to penetrate the target surface. Internally, contact elements Are meshed onto the red surface and corresponding target elements are meshed onto the blue surface, constituting one contact 'pair'. For asymmetric behaviour, the nodes of the contact surface cannot penetrate the target surface. (one-direction)



The asymmetric behaviour may allow some penetration at edges because of the location of contact detection points. In asymmetric behaviour, integration points only residing on one side.

Symmetric behaviour both the contact surface is constrained from penetrating the target surface and the target surface is constrained from penetrating the contact surface. Internally, the program uses two contact pairs with contact and target elements residing on both red and blue surface. (bi-direction)



Auto-symmetric (default with program-controlled option) the program evaluates the contact region and chooses which surface should be meshed with contact elements and which should be meshed with target elements. Internally, this may or may not result in one contact pair, but the contact elements may end up on the blue surface and target elements on the red surface.

Only pure penalty and augmented Lagrange formulations actually support symmetric behaviour. Normal Lagrange and MCP require asymmetric behaviour because of the nature of the equations, symmetric behaviour would be over constraining the model mathematically, so auto-asymmetric behaviour is used even when symmetric behaviour is selected.

	Specified Option	Pure Penalty	Augmented Lagrange	Normal Lagrange	MPC
Behavior Internally Used	Symmetric Behavior	Symmetric	Symmetric	Auto-Asymmetric	Auto-Asymmetric
	Asymmetric Behavior	Asymmetric	Asymmetric	Asymmetric	Asymmetric
	Auto-Asymmetric Behavior	Auto-Asymmetric	Auto-Asymmetric	Auto-Asymmetric	Auto-Asymmetric
Reviewing Results	Symmetric Behavior	Results on Both	Results on Both	Results on Either	Results on Either
	Asymmetric Behavior	Results on Contact	Results on Contact	Results on Contact	Results on Contact
	Auto-Asymmetric Behavior	Results on Either	Results on Either	Results on Either	Results on Either
Notes	Symmetric Behavior	Easier to set up	Easier to set up	Let program designate	Let program designate
	Asymmetric Behavior	Efficiency and control	Efficiency and control	User has control	User has control
	Auto-Asymmetric Behavior	Let program designate	Let program designate	Let program designate	Let program designate

Summing up, the symmetric behaviour is easier to set up, but more computationally expensive. Interprets data can be more difficult. Results are reported on both sets of surfaces. Only for penalty-based formulations. The Asymmetric behaviour the proper selection of the target and contact surfaces is more relevant (an incorrect selection could affect results), reviewing results is easy and straightforward because all data are on the contact side.

Contact tool and contact results

For asymmetric behaviour results are only available on contact surface, while for symmetric behaviour results are both on contact and target surfaces. The contact tool allows to analyse the results in more details. When viewing the contact tool worksheet, the user may select contact or target surface to review results.

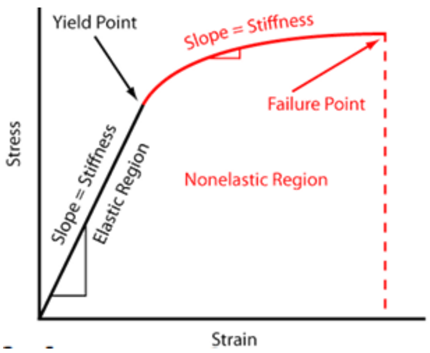
PLASTICITY

Chapter overview

- **A.** **Background Elasticity/Plasticity**
- **B.** **Yield Criteria**
- **C.** **Flow Rule**
- **D.** **Hardening Rules**
- **E.** **Material Data Input**
- **F.** **Analysis Settings**
- **G.** **Reviewing Results**
- **H.** **Workshop**

A. Metal plasticity overview

Elastic and Plastic Behavior



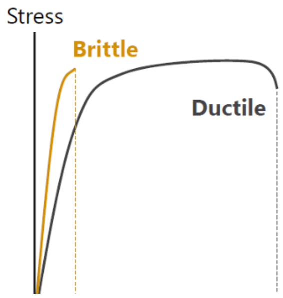
Elasticity refers to a material's ability to fully recover its original shape upon unloading, provided the induced stresses remain below the **yield strength**. At the atomic level, this is due to the stretching of chemical bonds without breaking them. Consequently, elastic strains are typically small and fully recoverable.
In metals, elastic behavior is described by **Hooke's Law**, which establishes a linear relationship between stress and stress. $\sigma = E\epsilon$
However, if the applied stress exceeds the energy holding the atoms in their original positions, the material enters the **plastic region**. This leads to:

- **Dislocation motion:** Atomic planes slip and rearrange, forming new neighboring bonds.
- **Permanent deformation:** The resulting strain is unrecoverable even after the load is removed.
- **Constant volume:** Unlike elasticity, plastic slipping generally does not result in volumetric strain.

Ductile and Brittle Materials

Materials react differently to plastic deformation:

- **Brittle Materials (e.g., glass, ceramics):** These fail abruptly with little to no plastic deformation.
- **Ductile Materials (e.g., aluminum, steel, copper):** These are safer for engineering applications (such as automotive structures) because they yield and undergo large permanent deformations before failure, effectively absorbing energy during impacts.

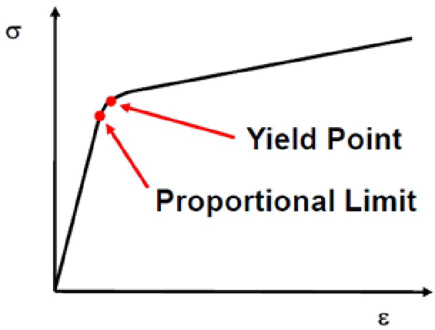


Key Transition Points

Within the material response, we distinguish two critical limits:

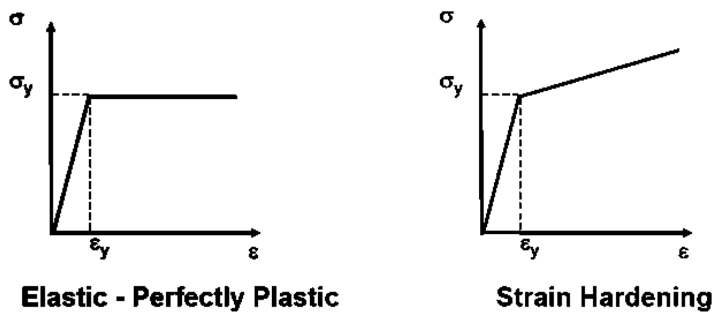
1. **Proportional Limit:** The maximum stress at which stress and strain are linearly related.
2. **Yield Point:** The point beyond which permanent plastic deformation occurs.

<u>Note</u>: In ANSYS the proportional limit and yield point are assumed to be <u>identical</u> due to their close proximity.



Types of Plastic Behavior
Plasticity can be categorized based on how the material responds to increasing loads:

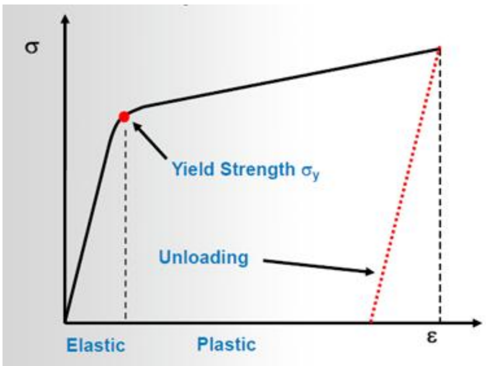
- **Elastic-Perfectly Plastic:** The material deforms indefinitely at a constant stress level once the yield point is reached.
- **Strain Hardening:** The yield stress increases as plastic strain accumulates.
- **Rate-Independent Plasticity:** The response is *independent* of the loading speed. Most metals exhibit this behaviour at low temperatures (<1/4 or 1/3 of the melting temperature) and low strain rates.



Incremental Plasticity Theory

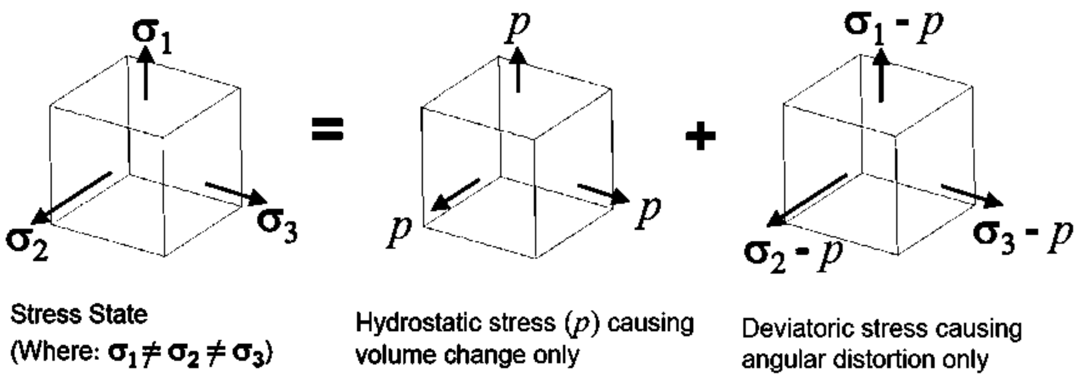
B. Yield criterion

To mathematically represent material behaviour in the plastic range, thus the relationship between $\Delta\varepsilon$ *and* $\Delta\sigma$, **Incremental Plasticity Theory** is used. It consists of three components: the **yield criterion**, the **flow rule**, and the **hardening rule**.



Yield criterion
While tensile tests provide **uniaxial** data (one-dimensional), real-world structures operate under **multiaxial** stress states. The **yield criterion** is used to *relate multiaxial stress state with the uniaxial case.*
In other words, the yield criteria serves as a tool to convert these complex stresses into a single **scalar invariant** that can be compared against uniaxial test results.
A stress state is composed of two parts:

- **Hydrostatic Stress:** Responsible for volume changes.
- **Deviatoric Stress:** Responsible for angular distortion and plastic flow.



Von Mises Yield Criterion
The **Von Mises theory** posits that yielding begins when the distortion energy in a material volume reaches the same energy level required to yield a uniaxial specimen.

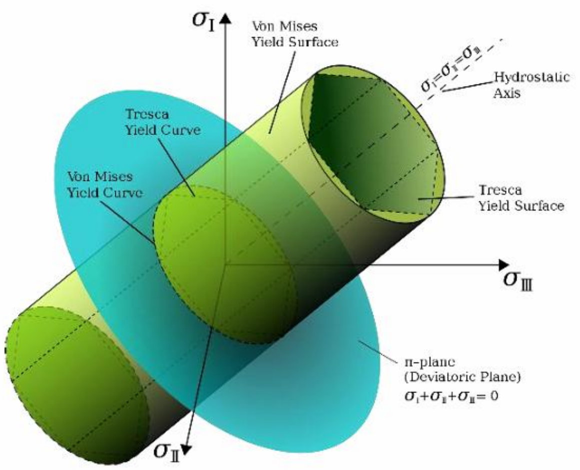
- **Von Mises Equivalent Stress:** A scalar value derived from this theory. If this value exceeds the uniaxial yield strength, yielding occurs.

$$\sigma_e = \sqrt{\frac{1}{2}\left[(\sigma_1 - \sigma_2)^2 + \quad (\sigma_2 - \sigma_3)^2 + \quad (\sigma_3 - \sigma_1)^2\right]}$$

Note sulla formula:

- σ_{eq} =0 (Stato Idrostatico o Nullo) → il materiale non subisce alcuna **distorsione** (variazione di forma). Ciò avviene in assenza di carico o quando tutte le tesnioni sono uguali. L'energia di Von Mises è zero, poiché le distorsioni sono nulle.
- $\sigma_{eq} > 0$ → se in campo elastico il materiale si deforma ma torna alla forma originale se rilasciato; se raggiunge la tensione di snervamento il materiale inizia a deformarsi plasticamente.

- **The Yield Surface:** In a 3D principal stress space, the Von Mises criterion is represented as a **cylinder** centered around the hydrostatic axis ($\sigma_1 = \sigma_2 = \sigma_3$).
- **Physical Significance:** Any stress state inside the cylinder remains elastic. Because the cylinder is aligned with the hydrostatic axis, pure hydrostatic pressure (no matter how high) will never cause yielding in metals; only deviatoric stresses (those deviating from the axis) contribute to plastic failure.



Note ulteriori: **La Forma Cilindrica:** poichè la "pressione" non causa snervamento, la superficie limite deve essere parallela a questo asse. Un cilindro con asse coincidente con la retta garantisce che, indipendentemente da quanto "lontano" si vada lungo l'asse (pressione infinita), il raggio del solido non cambi.

Il cerchio (Von Mises): Assume che il materiale sia perfettamente **isotropo**, ovvero che le sue proprietà siano identiche in ogni direzione. La sezione circolare del cilindro indica che la transizione tra elastico e plastico avviene in modo "morbido" e uniforme, indipendentemente dalla combinazione specifica di sforzi deviatorici. **L'esagono (Tresca):** È una versione più conservativa e semplificata, che si concentra solo sullo sforzo di taglio massimo e rappresenta solo un altro modo di vedere il confine tra elastic e plastic region.

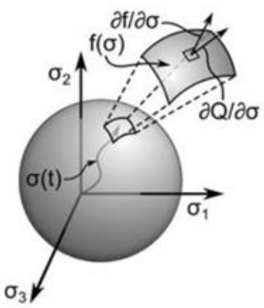
C. Flow rules

The flow rule determines the evolution of plastic strain.

The evolution of plastic strain is determined by the formula: $\boldsymbol{d\varepsilon^{pl} = d\lambda \frac{\delta Q}{\delta \sigma}}$

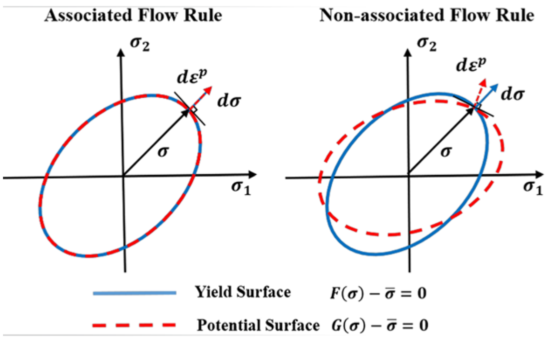
where

- $d\varepsilon^{pl}\ increment\ of\ the\ plastic\ deformation$
- $d\lambda$ is the magnitude of plastic strain increment
- $Q\ is\ the\ plastic\ potential$



It defines how individual plastic strain components ($\varepsilon_x^{pl}, \varepsilon_y^{pl}, etc$) develop once the material reaches the yielding point.

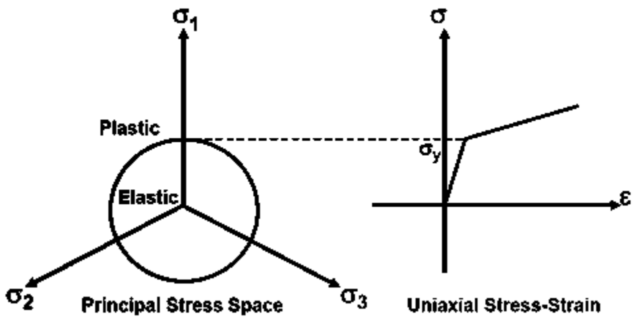
- **Associative Flow Rule:** The plastic strains develop in a <u>direction normal</u> to the yield surface. In other words, the flow rule is derived directly from the yield criterion. Most metal plasticity models in standard engineering software (like ANSYS) use associative flow rules.
- **Non-Associative Flow Rule**: The plastic potential is defined by a function different from the yield criterion. This is typically used in more advanced models for specific materials (e.g., soils or polymers).



D. Hardening Rules

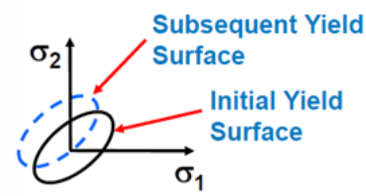
While elastic-perfectly-plastic materials have a fixed yield surface, most metals exhibit **hardening**, where the <u>yield surface changes size, center, or shape due to plastic deformation</u>.

Note on Geometry: If we visualize the yield surface as a cylinder (or circle in 2D), no stress state can exist outside of it. When the stress state reaches the "edge" of the cylinder, yielding occurs. Instead, hardening rules will describe how this cylinder evolves and changes with respect to yielding.

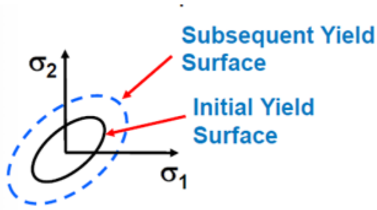


There are two primary types of hardening:

1. Kinematic Hardening
The yield surface **remains constant in size but translates** (shifts) in the direction of the yielding. This model is essential for simulating cyclic loading and captures the **Bauschinger Effect**.

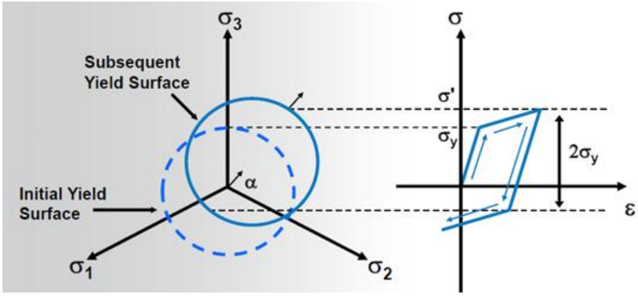


2. Isotropic Hardening
The yield surface **expands uniformly** in all directions. The center of the surface remains fixed at the origin, but its diameter increases. This model **is best suited for large-strain simulations** where cyclic effects are not the primary concern.



Kinematic Hardening
Most metals exhibhit kinematic hardening behaviour for small strain cyclic loading. The stress-strain behavior for linear kinematic hardening is illustrated below:

1. **Initial Loading:** The material behaves elastically until it hits the initial yield stress (σy). If you continue loading, it enters the plastic region.
2. **Surface Translation:** In kinematic hardening, as you push into the plastic region, you are "dragging" the yield surface along with the stress state. The diameter ($2\sigma y$) stays the same, but the center moves.
3. **Unloading:** When you release the load, the material recovers elastically. Because of the permanent plastic deformation, the strain does not return to zero.

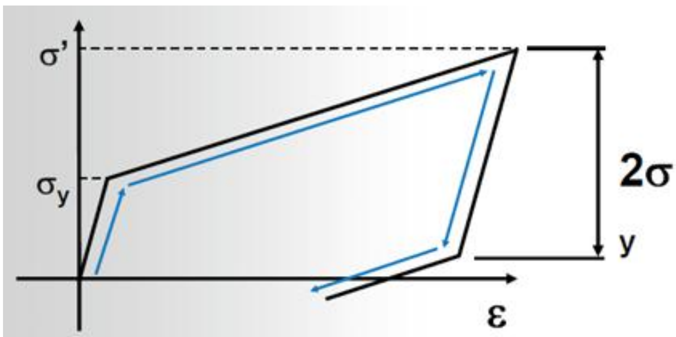


4. **Reverse Loading (The Bauschinger Effect):** Since the yield surface shifted forward, the "bottom" edge of the circle is now closer to the origin than it was originally. When you load the material in the opposite direction (compression), it reaches the new, shifted yield surface **much sooner**. (Nota: dall'immagine si vede bene che con la trazione il limite di servamento è dato da σ'; durante la compressione il nuovo limite di snervamento dalla intersezione tra la linea tratteggiata orizzontale in basso, che è molto più vicina all'asse delle ascisse rispetto a quella originale. In pratica, "sacrifichi" la resistenza a compressione per aver aumentato quella a trazione.)

   Rule: A constant range of $2\sigma y$ is maintained. If the yield stress increased in tension, it decreases by the same amount in compression.

<u>Limitation of this method:</u>

- **Small Strains:** Kinematic hardening is generally used for small-strain cyclic loading.
- **Large Strains:** In very high plastic deformation, the model can become mathematically "inappropriate." As we can see in the figure, if the surface shifts too far without changing size, the model might predict (erroneously) that the material yields during what should be a simple unloading phase. Indeed, if $\sigma' > 2\sigma y$, from thee moment you remove the load, the model thinks you are out from the circle even before this may happen, and it interprets this as a new yield point. (In pratica, il software direbbe che il materiale sta già snervando in compressione mentre tu stai ancora solo cercando di togliere il carico di trazione. È un paradosso fisico.). For such cases, isotropic hardening is preferred.

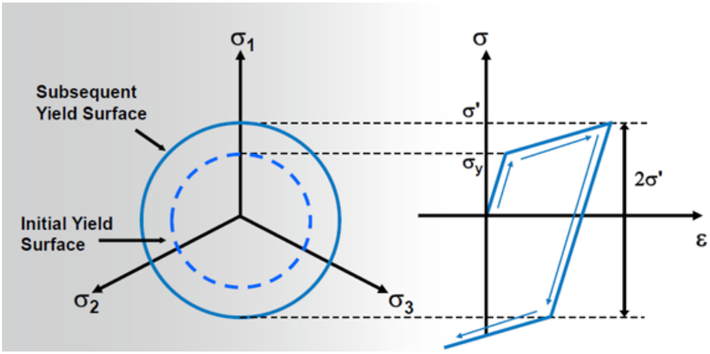


- **Anisotropy:** A material that starts as isotropic (the same properties in all directions) becomes **anisotropic** after kinematic hardening, because its yield strength is now different in tension vs. compression.

Isotropic Hardening

Isotropic hardening specifies that the yield surface expands uniformly in all directions during plastic flow.

- **Uniform Dilatation**: The term "isotropic" in this context refers to the uniform expansion of the yield surface's diameter, rather than the orientation of the material itself.
- **Yield Surface Evolution**: Unlike kinematic hardening (where the surface shifts), here the center of the yield surface remains fixed at the origin of the stress space, while the radius increases.



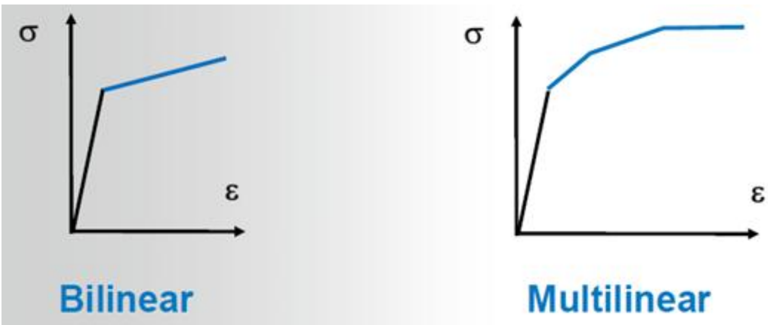
The stress-strain curve illustrates the material's behavior during a complete loading and reverse loading cycle:

1. **Tensile Phase:** When the material is loaded beyond the initial yield stress σ, it undergoes plastic deformation. This increases the "size" of the yield surface. The highest stress reached during this phase is denoted as $\boldsymbol{\sigma'}$.
2. **Elastic Recovery:** When the load is released, the material follows an elastic unloading path (parallel to the initial elastic slope).
3. **Reverse Loading:** In this model, the new yield limit in compression is equal to the maximum stress attained during the tensile phase. This means the elastic range has grown from $2\sigma y$ to $2\sigma'$. Because the yield surface expanded in all directions, the material is now *"stronger"* (has a higher yield point) in both tension and compression.

E. Material Data Input

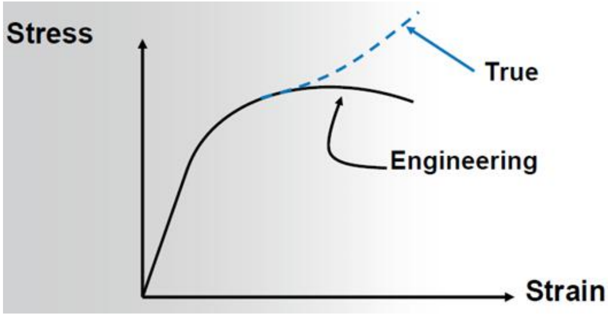
Stress-Strain Representations
There are two primary ways to represent the plastic domain in a material model:

- **Bilinear**: The plastic region is simplified into a single straight line defined by the Tangent Modulus.
- **Multilinear:** A more precise representation where the plastic curve is defined by multiple points (stress vs. plastic strain), providing a smoother transition and better fit for experimental data.



Engineering vs. True Stress-Strain
For accurate plasticity simulations, True Stress and True Strain must be used.

- Engineering Stress/Strain: Based on the original dimensions of the specimen. It is only suitable for small-strain analysis.
- True Stress/Strain: Accounts for the instantaneous changes in cross-sectional area and length during deformation.



Conversion and Limitations

The conversion from engineering to true values is valid only until necking occurs.

The formulas used are:

$$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_{\mathbf{eng}} \text{ and } \boldsymbol{\sigma} = \boldsymbol{\sigma}_{\mathbf{eng}}$$

$$\boldsymbol{\varepsilon} = \mathbf{ln}\left(\mathbf{1} + \boldsymbol{\varepsilon}_{\boldsymbol{eng}}\right) \text{ and } \boldsymbol{\sigma} = \boldsymbol{\sigma}_{\boldsymbol{eng}}\ (\mathbf{1} + \boldsymbol{\varepsilon}_{\mathbf{eng}})$$

These conversions assume that the material is **incompressible** (volume remains constant, V = A×L; acceptable approximation for large strain) and that the stress distribution across the cross-section remains uniform.

Once necking begins, the stress state becomes complex and non-uniform, making these simple **conversions invalid.** The instantaneous cross-section must be measured.

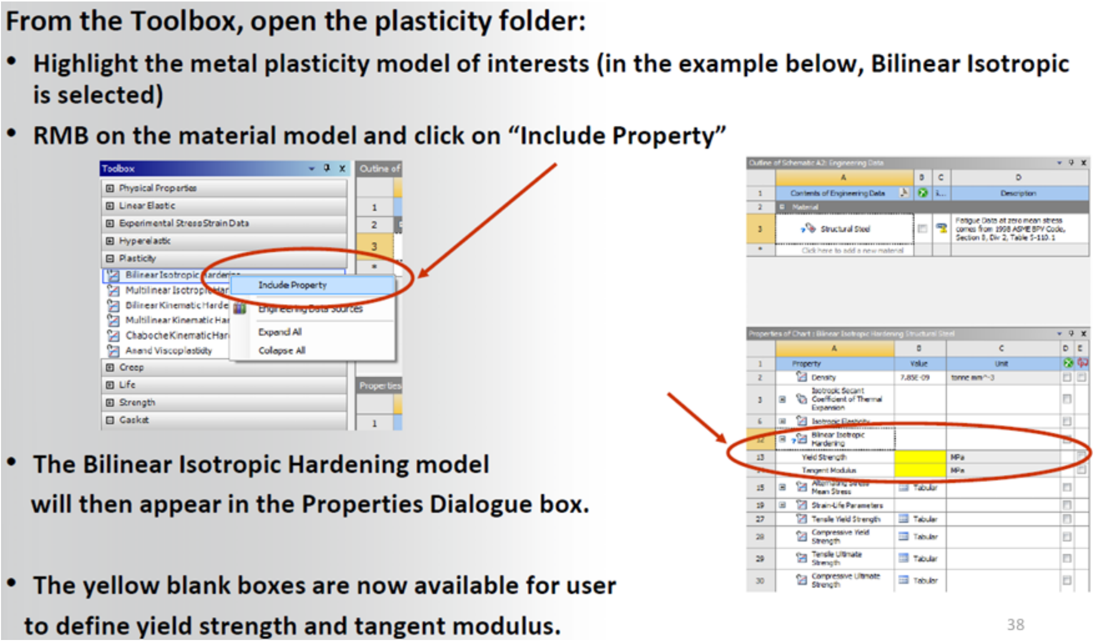
Implementing Plasticity in ANSYS (Workbench)

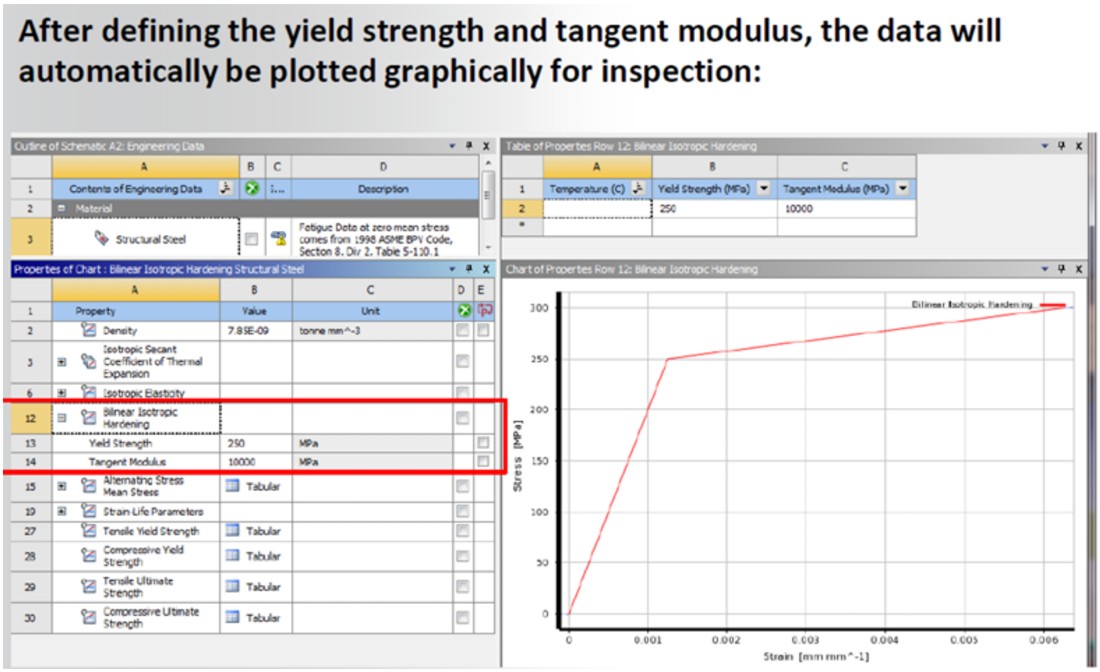
There are 2 ways to introduce metal plasticity:

1. From the project schematic, highlight the engineering data branch, doble click or RMB and click on Edit.
2. From the mechanical GUI, within the details window of the body to be modified, highlight the current material assignment and RMB to choose one of 3 options: new material→ import →edit.

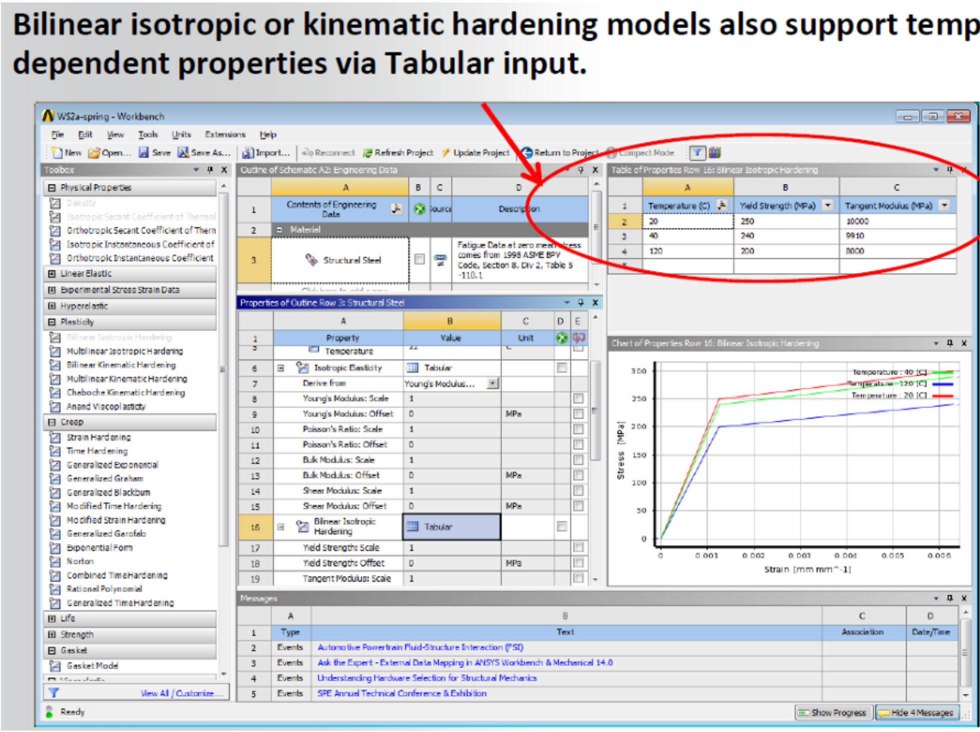
When defining material properties in the Engineering Data cell:

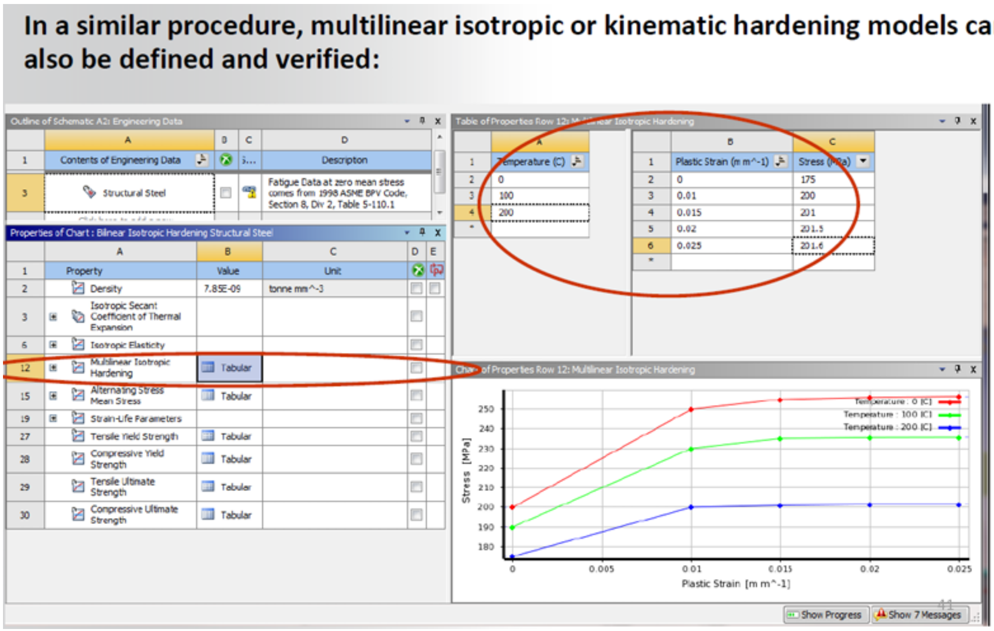
- **Yield Strength**: Defines the stress level where plastic deformation begins.
- **Tangent Modulus**: For bilinear models, this defines the slope of the curve after yielding.





- **Elastic Modulus**: Defines the initial slope (stiffness) before yielding.
- **Temperature Dependence**: Both bilinear and multilinear models support temperature-dependent properties. Generally, as temperature increases, both the Yield Strength and the Tangent Modulus decrease.



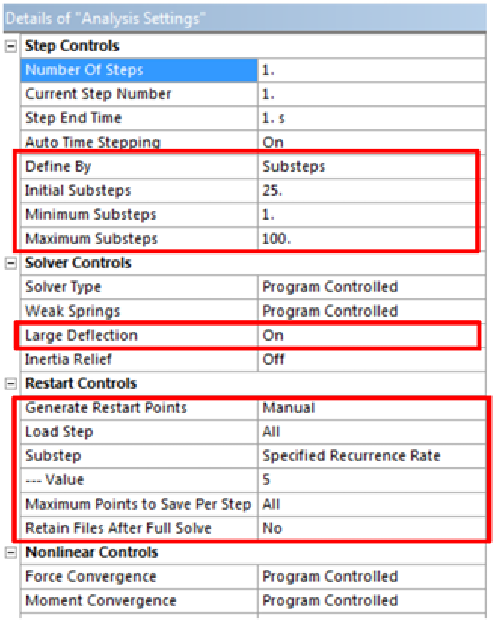


- **Multilinear Input:** When entering data for multilinear models, the first point in the plastic strain table must be 0 (corresponding to the initial Yield Stress).

F. Analysis Settings and Solver Controls

To ensure convergence in a plasticity model, the following settings are recommended:

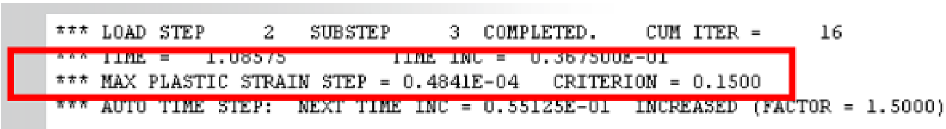
- **Large Deflection** = ON: Essential for capturing the geometric nonlinearities associated with plasticity.
- **Substep Size:** Must be adequate to capture path-dependent behavior. If the plastic strain in a single substep exceeds 15%, the solver will automatically trigger a bisection (splitting the time step to find a solution).
- **Restart Control:** Recommended for long simulations. This allows you to resume the analysis from a specific point if it fails to converge, rather than starting over.



G. Reviewing Results

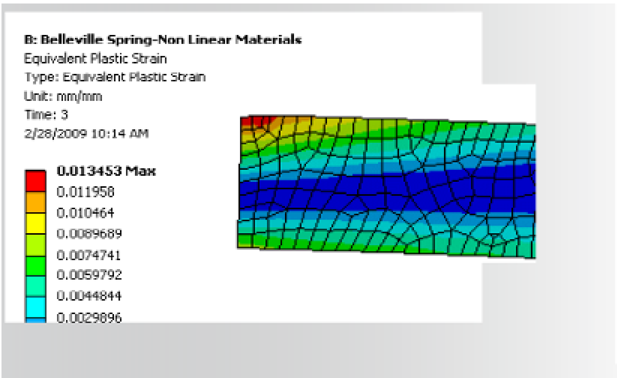
Reviewing a plasticity model is similar to linear analysis, but with the addition of path-dependent results:

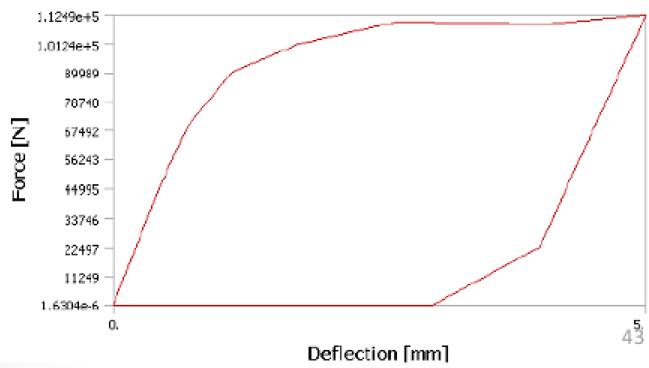
- **Plastic Strain**: Unlike elastic strain, this is permanent and depends on the loading history.



- **Nonlinear Force-Deflection Curve:** Examining this curve helps visualize how plastic deformation contributes to the overall structural nonlinearity.

- **Unloading**: Remember that upon unloading, the material always recovers its elastic strain, but the plastic strain remains as a permanent set.





H. Workshop: W4A-Metal plasticity

L21 14/05/2026

VIRTUALIZING SOFT MATERIALS

Before diving into the details, it is useful to briefly review the production process of **elastomers.** The **manufacturing process** involves pouring silicone into a mold to define its shape. Raw experimental data is then collected to describe the material within a simulation environment. This lecture focuses on the final stage of this process.

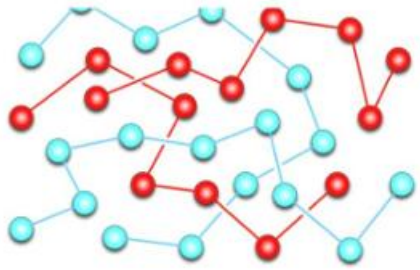
In ANSYS, this procedure often feels like "magic": you input the raw data and parameters, and everything seems to work perfectly without providing a clear understanding of the underlying mechanics. To gain deeper insight, we will use MATLAB. This allows us not only to input data but also to analyze the underlying phenomena by evaluating and comparing different hyperelastic models.

Background on elastomers

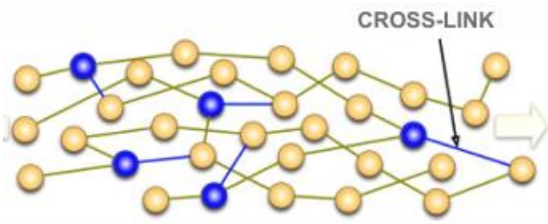
Classification of Polymers

Polymers are generally classified into two main categories based on their thermal behavior:

1. **Thermoplastics (Thermosoftening plastics):** These materials soften when heated due to their molecular structure. They consist of tangled polymer chains with no cross-links and exhibit weak intermolecular forces of attraction. Consequently, they can be repeatedly melted and reshaped. *(Examples: Polyethylene (PE), Polyvinyl Chloride (PVC), Polystyrene)*.

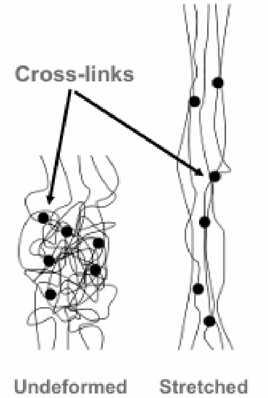


2. **Thermosets (Thermosetting plastics):** These materials do not soften upon heating. Their structure is stabilized by strong, covalent cross-links holding the polymer chains together. These bonds do not break when heat is applied, meaning the material remains hard and cannot be remolded. *(Examples: Epoxy resin, Polyurethane, Vulcanized rubber)*.



Physical Properties of Elastomers

Elastomers are a specific class of polymers that exhibit rubber-like elasticity, meaning they can undergo very high reversible deformations. They include both natural and synthetic rubbers, which are typically amorphous and composed of long molecular chains. Their behavior is characterized by:



- **Undeformed state:** Chains are highly twisted, coiled, and randomly oriented.

- **Under tensile load:** Chains partially straighten and untwist, reverting to their original configuration once the load is removed.

The Thermodynamic Origin of Elasticity
Unlike metals, where elasticity arises from the stretching of crystalline atomic bonds, rubber elasticity is **entropic**. It is driven by:

- Chain flexibility
- Chain-chain entanglement
- Changes in chain-chain interactions

When an elastomer is stretched the polymer chains align, leading to a **reduction in entropy** (an increase in structural order). Because the system naturally tends to maximize its entropy, a restoring force is generated, pulling the material back to its original, disordered configuration.

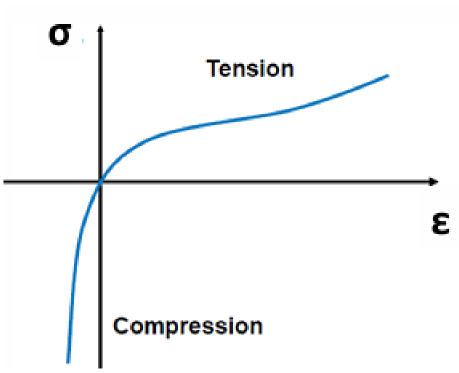
Peculiar Mechanical Properties

- **Large Elastic Deformations:** They can undergo large, recoverable deformations, easily exceeding 200% and sometimes reaching up to 1000%.

- **Near Incompressibility:** There is negligible volume change under applied stress because the deformation is primarily related to chain straightening rather than volumetric expansion.

- **High Nonlinearity:** Their mechanical behavior is highly nonlinear, meaning **Hooke's Law** ($\sigma = E \times \epsilon$) **does not apply.**

- **Typical Tensile/Compressive Response:** In tension, the material initially softens and then stiffens at higher strains (remember that the rigidity is represented by the inclination of the curve: the more is steep, the stiffer it is.) . In compression, the response becomes very stiff almost immediately, we can see it since the blue line ends immediately.



Looking at the graph we can see 3 main regions related to:

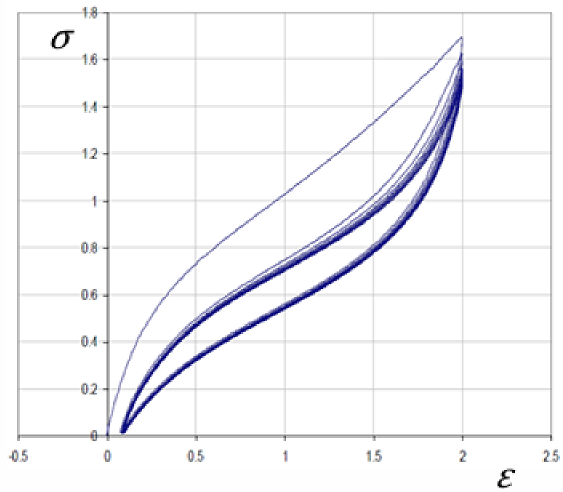
1. **Linear Elasticity (Small deformations):** Associated with the initial rearrangement of polymer chains and the initial decrease in entropy.

2. **Softening (Intermediate deformations):** A geometric softening effect related to the incompressibility of the material.

3. **Stiffening (Large deformations):** Occurs because the polymer chains become fully realigned and straightened. From this point onward, further deformation requires stretching the covalent bonds of the polymer backbone, which possesses a much higher Young's modulus.

Stress-Strain Relationship in Elastomers

Let's see in detail the stress-strain relationship, especially analyzing tensile test data for elastomers.



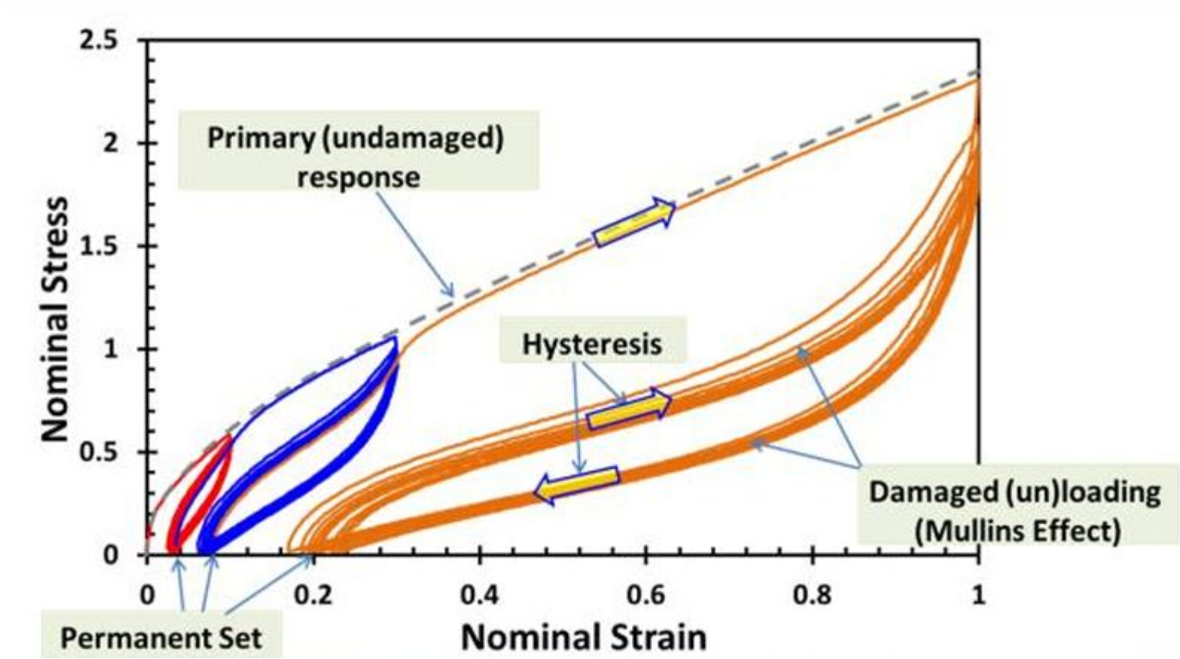
During a loading-unloading cycle, a distinct gap appears between the loading and unloading pathways. The area enclosed by this loop represents the **dissipated energy** (calculated as the integral of the loading curve minus the integral of the unloading curve).

This behavior occurs because **elastomers** are not perfectly elastic; they **are viscoelastic**. Viscoelasticity is an intrinsic property that allows the material to dissipate energy, meaning less stress is required to maintain the same strain during the unloading phase than during the loading phase.

During the very first loading cycles, the material does not return to zero strain when the load is removed. Instead, it exhibits a **permanent set** (residual plastic deformation). This initial gap is caused by the breakage of certain microscopic cross-links during the first deformation cycle.

- **Stabilization:** If the material is repeatedly cycled within a specific strain range without exceeding it, no further cross-links break, and the stress-strain loop stabilizes into a repeatable pathway.

- **The Mullins Effect:** This is the phenomenon where the mechanical properties depend on the maximum strain previously encountered. If the material experiences a **load that exceeds its historical maximum**, more internal bonds break. When unloaded, it follows a new pathway and settles into a new, permanent set (as shown by the transition from the red loop to the blue and orange loops in the graph).



Note: The Mullins effect can occur even before a component is put into service. For instance, if a component is designed to operate only up to 30% strain (the blue loop), but it is subjected to a much higher tensile force when being pulled out of its manufacturing mold, its original mechanical state is permanently lost. It will now follow a damaged pathway (like the orange curve). Therefore, it is crucial to align the strain ranges used during mechanical testing with the actual manufacturing and operational history of the product.

Factors Influencing the Stress-Strain Relationship

- **Manufacturing Method:** For cast silicones, the components (usually a two-part system) are mixed and poured into a mold. Curing can occur at room temperature or in an oven. Curing at room temperature is often preferred to allow stable chemical bonds to form uniformly.
- **Cure Time:** The total time allowed for the cross-linking reaction to complete.
- **Environmental Conditions:** Ambient temperature and humidity during storage and testing can significantly alter the final mechanical stiffness.
- **Aging and Oxidation:** Testing a fresh sample yields different results compared to testing an older one. Over time, polymer chains react with oxygen in the air (**oxidation**), incorporating oxygen atoms into the backbone. This process makes the molecular network more **brittle,**

reducing the product's lifespan. Identifying these changes is critical for predicting the service life of the product.

Introduction to hyperelastic theory

Linear elastic models (such as Hooke's Law) cannot accurately describe the mechanical behavior of elastomers under large deformations. **Hyperelasticity** provides a mathematical framework to model the highly **nonlinear stress-strain behavior** of these materials.

To apply standard hyperelastic models, the material is typically assumed to be:

- **Isotropic:** Its mechanical properties are identical in all directions.
- **Isothermal:** The deformation process occurs at a constant temperature.
- **Elastic:** Deformations are fully recoverable (neglecting the viscoelastic effects discussed previously).
- **Fully or Nearly Incompressible:** The material undergoes negligible volume change during deformation.

In hyperelasticity, stresses and strains are not directly related through a constant Young's modulus (E) and Poisson's ratio (ν). Instead, constitutive hyperelastic models are defined using a **strain energy potential function** (**W**).

Basic background

1.STRESS and STRAIN

	STRESS	STRAIN
Engineering Original dimensions of specimen	$\sigma = \frac{F}{A_0}$	$\varepsilon = \frac{L - L_0}{L_0}$
True Actual dimensions of specimen	$\sigma' = \frac{F}{A}$	$\varepsilon' = \int_{L_0}^{L} \frac{dL}{L} = \ln\left(\frac{L}{L_0}\right)$
Equivalence (by constancy of volume)	$\sigma' = \frac{F}{A_0}(1+\varepsilon) = \sigma(1+\varepsilon)$	$\varepsilon' = \ln(1+\varepsilon)$

2. The Stretch Ratio (λ)

To properly describe large deformations, standard strain measures are often replaced or supplemented by the **stretch ratio** (or simply "stretch"), denoted by λ. It is defined as the ratio between the deformed length (L) and the original undeformed length (L0) of a specimen:

$$\lambda = \frac{L}{L_0} = \frac{L_0 + \Delta u}{L_0} = 1 + \epsilon$$

(example of stretch ratio as defined for uniaxial tension of rubber specimen)

For a general three-dimensional deformation, there are three **principal stretch ratios** λ1, λ2 and λ3 which provide a directional, tensorial measure of the deformation.

3. Strain Invariants

While principal stretches provide a directional description, we often need **scalar equivalents** that characterize the deformation state independently of the chosen coordinate system. This is crucial for satisfying our assumption of material isotropy. These scalar quantities are known as **strain invariants** (I1, I2, I3) and are defined as functions of the principal stretch ratios:

$$I1 = \lambda1^2 + \lambda2^2 + \lambda3^2$$

$$I2 = \lambda1^2 \lambda2^2 + \lambda2^2\lambda3^2 + \lambda3^2\lambda_1^2$$

$$I3 = \lambda1^2\lambda2^2\lambda3^2$$

4. Volume Ratio (J)

The **volume ratio** (J) represents the ratio of the deformed volume (V) to the initial, undeformed volume (V0). It is calculated as the product of the three principal stretches:

$$J = \lambda1\lambda2\lambda3 = \frac{V}{V_0}$$

Note on Incompressibility: Mathematically, for a perfectly incompressible material, the volume ratio should be exactly equal to 1 (J = 1, which also means I3 = 1). However, in finite element simulation environments, enforcing absolute incompressibility can generate mathematical singularities and numerical instability. Therefore, we treat the material as *nearly incompressible*, allowing J to deviate *slightly* from 1 to ensure numerical convergence.

The Strain Energy Potential Function (W)

The **strain energy potential function**, typically denoted as **W**, represents the elastic energy stored per unit volume. Depending on the formulation of the hyperelastic model, W can be expressed either in terms of the principal stretch ratios or the strain invariants:

$W($ λ1, λ2, λ3) Or $W($ I1, I2, I3)

Deriving Stress from Energy

Once the specific form of the strain energy potential function W is defined, the components of the stress tensor (Sij) can be derived analytically by taking the derivative of W with respect to the components of the strain tensor (Eij}):

$$Sij = \frac{dW}{dE_{ij}}$$

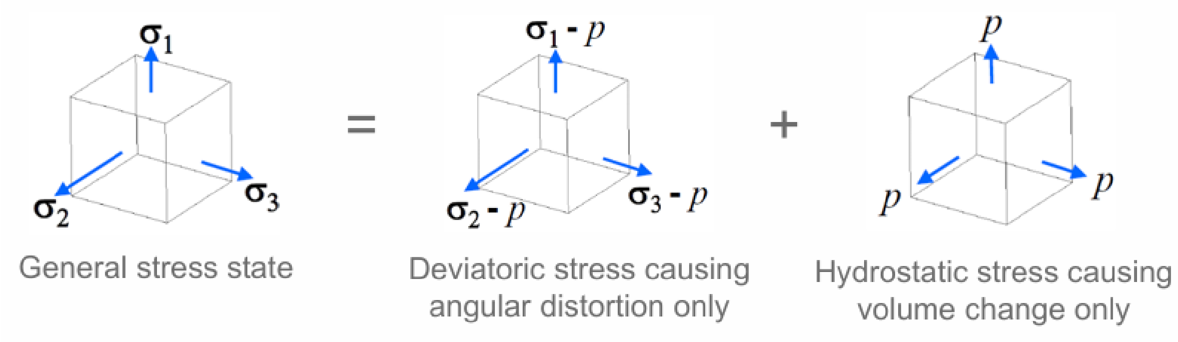
where Eij are components of strain tensor.

To handle near-incompressibility computationally, the strain energy function is typically split into two decoupled parts: a **deviatoric (or isochoric)** term and a **volumetric** term:

$$W = W_d(\overline{I_1}, \overline{I_2}) + W_b(J)$$

$$W = W_d(\overline{\lambda_1}, \overline{\lambda_2}, \overline{\lambda_3}) + W_b(J)$$

- **Wd:** Describes the energy associated with shape change (distortion) at a constant volume.
- **Wb:** Describes the energy associated with pure volume change (bulk compression/expansion) and is formulated purely as a function of the volume ratio J.



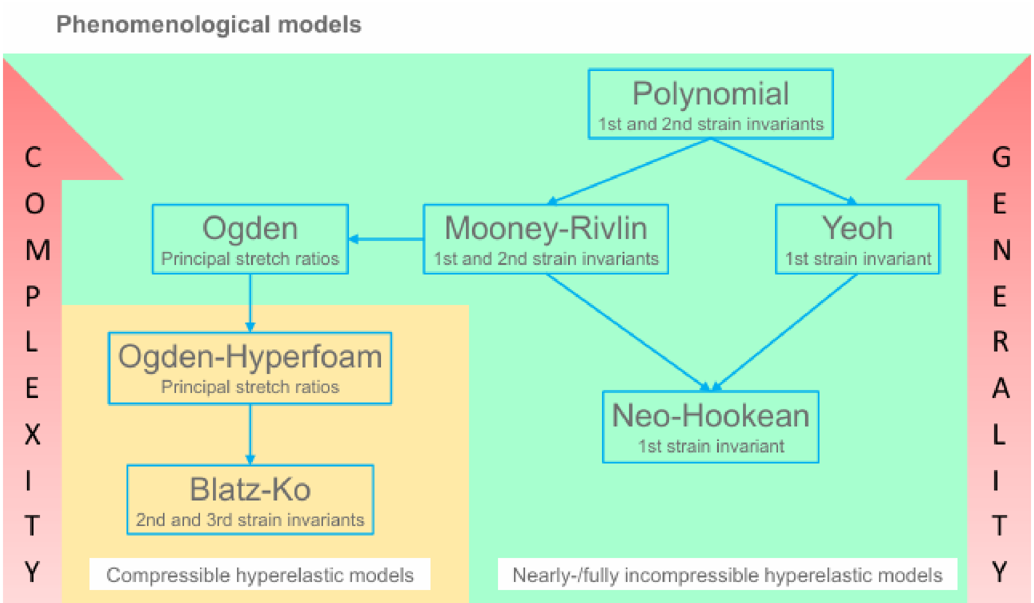
Forms of Strain Energy Potential (W)

There are several mathematical formulations for the strain energy potential function (W). The choice of the model depends on multiple factors:

- **Material Constants:** Every model requires a specific set of input parameters (material constants), and the number of these constants varies between models.
- **Type of Elastomer:** Different formulations excel at describing different types of rubber or silicone.
- **Loading Conditions:** The maximum expected deformation (% strain) influences which model remains stable.
- **Availability of Experimental Data:** The number of independent material tests (e.g., uniaxial tension, biaxial tension, pure shear) you can perform dictates how many parameters you can reliably fit.

In general, **the best strain energy function is the one that produces the closest curve fit to the experimental stress-strain data.** While some functions yield similar responses, highly challenging applications with complex loading states may require one specific formulation.

General guidelines can assist in selecting W, but they do not cover 100% of scenarios. Hyperelastic models typically balance generality against mathematical complexity.



Polynomial form

The polynomial form is a **phenomenological model** based on the first and second strain invariants (I1 and I2). Its general mathematical formulation is expressed as:

$$W = \sum_{i+j=1}^{N} C_{ij}(\bar{I}_1 - 3)^i\,(\bar{I}_2 - 3)^j + \sum_{k=1}^{N} \frac{1}{d_k}(J-1)^{2k}$$

Where:

- Cij are the material constants governing the deviatoric (shear) behavior.
- dk are the material constants governing the volumetric (compressibility) behavior.
- N represents the order of the polynomial.

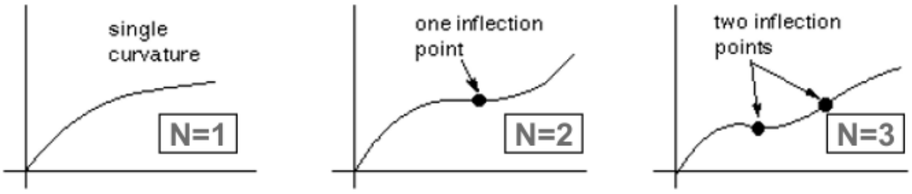
From this function, the initial shear and bulk moduli are

$$\mu_0 = 2(c_{10} + c_{01}) \qquad \kappa_0 = \frac{2}{d_1}$$

<u>Technical Comments and Guidelines</u>

- **Generality:** It is a highly versatile form capable of producing excellent curve fits for complex material data.

- **Inflection Points:** Higher-order terms (higher N) are required to capture inflection points (changes in curvature) in the stress-strain curve. However, using higher-order terms requires a larger and more comprehensive set of experimental data to avoid numerical instability.



- **Strain Ranges:** A polynomial form with N = 1 (which corresponds to the *Mooney-Rivlin* model) or N = 3 can typically be used for strains up to 100%–300%. Orders higher than N = 3 are rarely used in practice.

- **Data Requirements:** Multi-axial experimental data (e.g., uniaxial, biaxial, and shear tests) are strictly required when curve-fitting higher-order polynomial models. If your data is limited (e.g., only uniaxial tensile test data is available), you should consider alternative, simpler models to prevent unphysical behavior outside the tested range.

To use these models in a simulation, we must determine the numerical values of the material constants (Cij and dk)

Why can we not differentiate the raw data directly?
To find the stress, we need the derivative of the strain energy function. We cannot simply differentiate the experimental raw data because it consists of discrete, noisy data points rather than a continuous analytical equation. You cannot take a mathematical derivative of a discrete set of points.

<u>The MATLAB Approach</u>
Instead, the parameter identification is performed numerically within an environment like MATLAB using optimization algorithms. The workflow is as follows:

1. **Define the Order (N):** The user selects the order of the polynomial equation based on the complexity of the curve and the available data.
2. **Analytical Derivation:** The hyperelastic model equation (W) is mathematically differentiated to obtain an analytical expression for stress.
3. **Optimization (Curve Fitting):** MATLAB runs an optimization routine to find the values of the constants (Cij) that minimize the error between the analytical stress formula and the discrete raw data points.

Ultimately, you are choosing the specific set of parameters that forces the derivative of the hyperelastic model to match your experimental curve as accurately as possible.

The Mooney-Rivlin Model

The **Mooney-Rivlin** model is a specific and widely used variation of the general polynomial form. Depending on the complexity of the material behavior and the amount of available data, it can be formulated with different numbers of parameters—most commonly using two, three, five, or nine terms.

Two-term Mooney-Rivlin model: The two-term version is the most basic form and is mathematically equivalent to the general polynomial form when the polynomial order is set to N=1

$$W = c_{10}(\bar{I}_1 - 3) + c_{01}(\bar{I}_2 - 3) + \frac{1}{d}(J - 1)^2$$

Three-term Mooney-Rivlin model: The three-term version introduces a higher-order effect. Equivalent to the polynomial when N=2 and C20=C02=0:

$$W = c_{10}(\bar{I}_1 - 3) + c_{01}(\bar{I}_2 - 3) + c_{11}(\bar{I}_1 - 3)(\bar{I}_2 - 3) + \frac{1}{d}(J - 1)^2$$

Five-term Mooney-Rivlin model: The five-term version provides more flexibility to capture nonlinearities. It is fully equivalent to the general polynomial form when N = 2, meaning all first and second-order terms of the invariants are considered.

Nine-term Mooney-Rivlin model: The nine-term version is a highly complex formulation, equivalent to the general polynomial form when N = 3. It allows for highly accurate curve fitting but requires extensive experimental data.

For all variations of the Mooney-Rivlin model, the initial shear modulus (μ_0) and the initial bulk modulus (κ_0) are defined consistently as:

$$\mu_0 = 2(c_{10} + c_{01}) \qquad\qquad \kappa_0 = \frac{2}{d}$$

Technical Comments and Guidelines

Because Mooney-Rivlin models are directly derived from the polynomial framework, they share the same general limitations and operational guidelines as the polynomial forms (N=1,2,3).

While the **two-term Mooney-Rivlin model** is the most frequently implemented model in engineering applications due to its simplicity, its mathematical structure imposes specific physical limits:

- **Tensile Strain Limit:** It is generally valid only up to **90%–100%** tensile strains. Because it lacks higher-order terms, it cannot capture the upturn or **stiffening effect** that elastomers typically exhibit at larger deformations (when polymer chains become fully straightened).

- **Shear Strain Limit:** It accurately characterizes pure shear behavior only up to **70%–90%** strain. This is because the model inherently predicts a **constant shear modulus**, which fails to mimic real rubber behavior under severe shear.
- **Compressive Strain Limit:** While it can capture moderate compression reasonably well (up to **30%** strain), it cannot accurately describe significant or severe compressive responses. To capture steep compressive stiffening, a higher number of terms (such as the five- or nine-term models) is mandatory.

The Yeoh Model

The **Yeoh model** is one of the most widely used formulations for describing the elastic behavior of rubber-like materials. It is structurally similar to the general polynomial form and is frequently referred to as the **"reduced polynomial form"** because it depends exclusively on the first strain invariant (I1)

Its mathematical expression is:

$$W = \sum_{i=1}^{N} C_{i0}(\bar{I}_1 - 3)^i + \sum_{i=1}^{N} \frac{1}{d_i}(J - 1)^{2i}$$

Although the model can be configured with any polynomial order N, it is most commonly implemented with N = 3, which is known as the **cubic form**.

The initial shear modulus and initial bulk modulus are defined as:

$$\mu_0 = 2c_{10} \qquad \kappa_0 = \frac{2}{d_1}$$

<u>Comments and Guidelines</u>

- **Omission of I2:** The second strain invariant I2 is intentionally omitted based on the physical observation that the strain energy potential is far less sensitive to changes in I2 than in I1 (i.e., $\frac{dW}{dI_1} \gg \frac{dW}{dI_2}$ especially under large strains.
- **Predictive Capability with Limited Data:** If experimental data is scarce (e.g., only uniaxial tensile test data is available), ignoring the second invariant prevents mathematical instability. It has been proven that the Yeoh model yields much better predictions of general, un-tested deformation states under these conditions than a standard polynomial model.
- **Modeling the S-Shaped Curve:** As tensile strain increases, the shear modulus (the slope of the stress-strain curve) initially decreases slightly (softening) and then increases sharply (stiffening). To capture this unique behavior, the cubic form (N=3) requires specific parameter relationships:

- C10: Must be positive, as it represents half of the initial shear modulus value.
- C20: Must be negative to capture the intermediate softening effect at small-to-moderate strains ≈ 0.1 to $0.01*c_{10}$
- C30: Must be positive to capture the sharp upturn and stiffening effect at large strains $\approx$ 1e-2 to 1e-4*c_{10}

The Neo-Hookean Model

If you need to simulate a complex assembly using nonlinear material properties rather than relying on an overly complex material definition from the start, the **Neo-Hookean** model is the ideal baseline. It represents the simplest transition from linear elastic properties to hyperelastic mechanics.

The Neo-Hookean formulation is a strict subset of the polynomial form where N = 1, C01=0, and C10=$\mu/2$

$$W = \frac{\mu}{2}(\bar{I}_1 - 3) + \frac{1}{d}(J - 1)^2$$

Where the initial bulk modulus is defined as $\kappa_0 = \frac{2}{d}$

<u>Technical Comments and Guidelines</u>

- **Simplicity:** This is the most basic hyperelastic model available. It serves as an excellent starting point for establishing a working, converging simulation.
- **Strain Limits:** It does not predict moderate-to-large strains accurately and is strictly suitable for small strain applications.
- **Comparison to Mooney-Rivlin:** The operational limits of the two-term Mooney-Rivlin apply here, but are more restrictive:
  - **Tension:** Valid only up to **30%–40%** tensile strain. It cannot capture any material stiffening.
  - **Shear:** Valid for pure shear behavior up to **70%–90%** strain due to its mathematically constant shear modulus.
  - **Compression:** Captures moderate compression up to **30%**, but fails completely at higher compression levels.

The Ogden Model (Incompressible)

Unlike the previous models, which are invariant-based, the **Ogden model** is formulated directly as a function of the principal stretch ratios:

The initial shear and bulk moduli are defined as:

$$W = \sum_{i=1}^{N} \frac{\mu_i}{\alpha_i}\left(\bar{\lambda}_1^{\alpha_i} + \bar{\lambda}_2^{\alpha_i} + \bar{\lambda}_3^{\alpha_i} - 3\right) + \sum_{i=1}^{N} \frac{1}{d_i}(J-1)^{2i}$$

Model Equivalence

- **To Mooney-Rivlin (N=2):** The Ogden model simplifies exactly to the two-term Mooney-Rivlin form if N=2, $\mu_1 = 2C_{10}, \alpha_1 = 2, \mu_2 = -2C_{01}, and\ \alpha_2 = -2$.
- **To Neo-Hookean (N=1):** It degenerates to the Neo-Hookean form when N=1, $\mu_1 = \mu, and\ \alpha_1 = 2$.

<u>Technical Comments and Guidelines</u>

- **Accuracy:** Because it uses principal stretch ratios directly, it provides exceptional mathematical flexibility, leading to superior curve-fitting accuracy. However, this makes it slightly more computationally expensive during FEA solving.
- **Data Sensitivity:** If multiple deformation modes are expected in your physical application, curve-fitting the Ogden model using *only* uniaxial data can result in highly unrealistic, unphysical behavior in shear or compression.
- **The Three-Term Rule:** Ogden noted that a minimum of three terms (N=3) should be used to capture full elastomer mechanics correctly:
    1. **First term ($\boldsymbol{1.0 < \alpha_1 < 2.0\ and\ \mu_1 > 0}$):** Governs the small strain values.
    2. **Second term** ($\boldsymbol{\alpha_2 > 2.0\ and\ \mu_2 > 0, with\ \mu_2 \ll \mu_1}$): Captures the stiffening behavior at large tensile strains.
    3. **Third term** ($\boldsymbol{\alpha_3 < -0.5\ and\ \mu_3 < 0, with\ \mu_3 \ll \mu_1}$): Governs the material behavior under compression.

    Note: The product $\boldsymbol{\mu_i\ \alpha_i}$ must always be positive to ensure a physically valid, positive shear modulus.

Compressible Models (Foam Rubbers)

When dealing with highly porous or foam-like rubber materials, the assumption of incompressibility (J $\approx$ 1) no longer holds. Specialized compressible formulations must be used.

1. The Ogden Compressible Foam Model (Hyperfoam)

The **Ogden Compressible Foam model** handles highly compressible behaviors. Unlike the standard Ogden model, its volumetric and deviatoric components are tightly coupled within the main summation:

$$W = \sum_{i=1}^{N} \frac{\mu_i}{\alpha_i}\left(J^{\alpha_i/3}\left(\bar{\lambda}_1^{\alpha_i} + \bar{\lambda}_2^{\alpha_i} + \bar{\lambda}_3^{\alpha_i}\right) - 3\right) + \sum_{i=1}^{N} \frac{\mu_i}{\alpha_i \beta_i}\left(J^{-\alpha_i \beta_i} - 1\right)$$

The initial shear modulus (μ_0) and bulk modulus (κ_0) are given by:

$$\mu_0 = \frac{\sum_{i=1}^{N} \mu_i \alpha_i}{2} \qquad \kappa_0 = \sum_{i=1}^{N} \mu_i \alpha_i \left(\frac{1}{3} + \beta_i \right)$$

Here, the parameter β_i is directly related to the material's compressibility and determines the effective Poisson's ratio.

2. The Blatz-Ko Model

The **Blatz-Ko model** is a specialized formulation developed specifically for highly compressible **polyurethane foam rubbers** (originally validated on a 47% volume percent polyurethane foam). Its strain energy potential is defined as:

$$W = \frac{\mu}{2} \left(\frac{I_2}{I_3} + 2\sqrt{I_3} - 5 \right)$$

Where:

- μ is the initial shear modulus.
- I_2 *and* I_3 are the standard, **total (not deviatoric)** strain invariants.

The Blatz-Ko model can be viewed as a restricted subset of the Ogden Compressible Foam model where N=1, $\mu_1 = -\mu, \alpha_1 = -2, and\ \beta_1 = 0.5$.

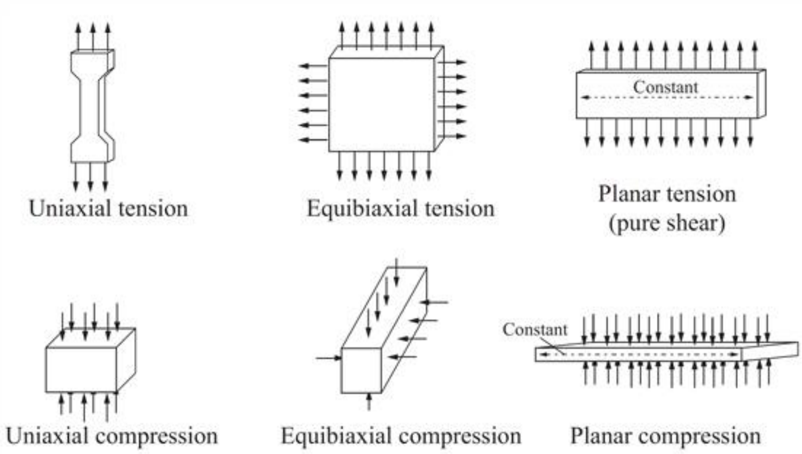
This specific formulation mathematically forces the relationship between the initial bulk modulus (k) and shear modulus (μ) to be: $\kappa = \frac{2(1+\nu)}{3(1-2\nu)} \mu \rightarrow \kappa = \frac{5}{3} \mu$

This constant ratio dictates that the Blatz-Ko model inherently assumes a fixed effective Poisson's ratio of $\boldsymbol{\nu = 0.25}$.

Experimental Data Collection

To fully characterize a hyperelastic material and provide a complete dataset for constitutive modeling, experimental test data can be collected from up to **six different deformation modes**:

- **Uniaxial Tension**
- **Equibiaxial Tension**
- **Planar Tension** (also known as Pure Shear)
- **Uniaxial Compression**
- **Equibiaxial Compression**
- **Planar Compression**



Experimental testing machines typically record raw data as **engineering stress** and **engineering strain**. This raw data must be converted into true stress and true strain or directly into stretch ratios before being implemented into numerical solvers like MATLAB or ANSYS.

Reducing the Test Matrix via Incompressibility

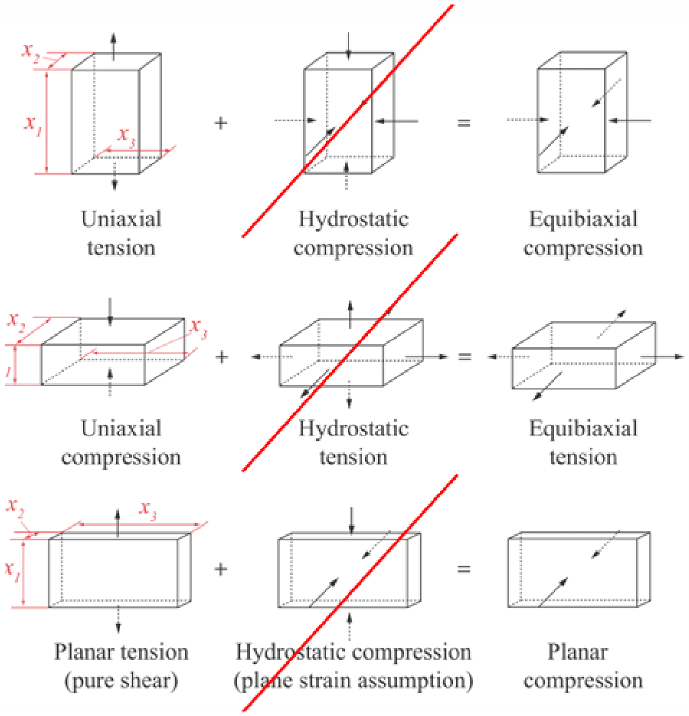
While performing all six tests provides the most comprehensive data, we can significantly reduce the testing matrix by leveraging the assumption of **perfect material incompressibility** (Poisson's ratio $\nu = 0.5$).

Under this assumption, adding a pure hydrostatic pressure state to the material changes the stress field but does not cause any physical deformation. Consequently, tension and compression modes are kinematically linked, allowing us to reduce the six experimental setups down to **three independent modes of deformation**.

The mathematically identical modes are paired as follows:

1. Uniaxial Tension ⟷ Equibiaxial Compression

Pulling a specimen in one direction (uniaxial tension) creates a deformation field identical to compressing a specimen uniformly from all sides along the perpendicular plane (equibiaxial compression).



2. Uniaxial Compression⟷Equibiaxial Tension

Compressing a specimen along one axis (uniaxial compression) forces the material to expand equally in the other two orthogonal directions, which is mechanically identical to stretching it uniformly in a plane (equibiaxial tension).

3. Planar Tension ⟷Planar Compression

Planar tension occurs when a specimen is stretched along one axis while being constrained from contracting along the transverse axis (typically using a wide, short specimen). Because the material cannot contract laterally, it thins out in the thickness direction. This kinematics is perfectly identical to planar compression.

By conducting just three of these tests (typically **Uniaxial Tension**, **Equibiaxial Tension**, and **Planar Tension**), you obtain all the necessary mechanical data required to fully calibrate higher-order hyperelastic models like the Ogden or polynomial formulations.

CURVE FITTING

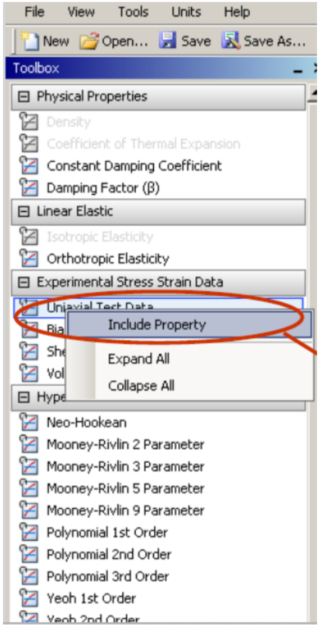
Data Usage

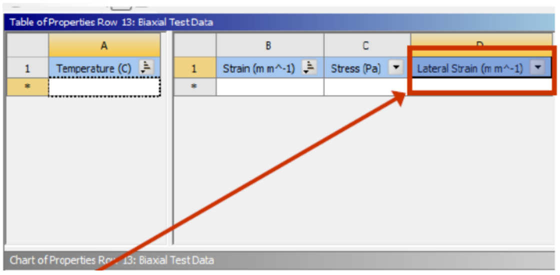
Raw data obtained from testing machines is used as input to define the material properties of custom materials. Whether using a default or custom material, properties are defined within the Engineering Data cell by importing laboratory measurements.
NOTE: The label "Experimental Stress Strain Data" might be outdated.
Specifically, data can be added to the following toolboxes in a tabular format (via copy-and-paste):

- **Uniaxial test data**
- **Biaxial test data**
- **Shear test data**

NOTE: Always ensure measurement units are consistent between your laboratory tests and the software settings.

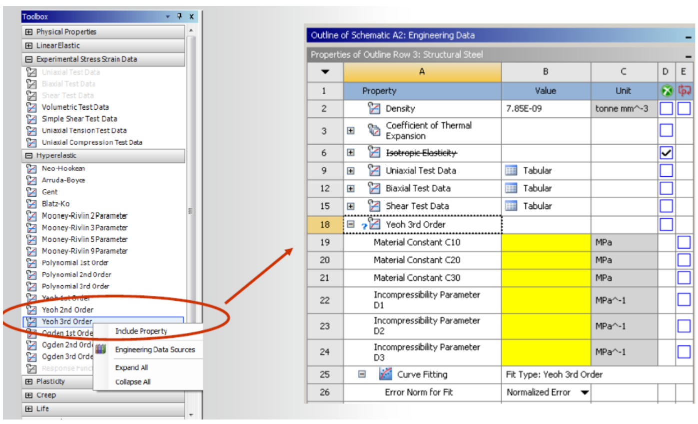
NOTE 2.0: If the material is compressible, the *Lateral strain* must also be defined.





A mathematical relationship between the increase in length and the decrease in cross-sectional area cannot be assumed. The software requires the specific lateral strain corresponding to the applied stress.
If lateral strain is disabled, this input is not required because the isovolumetric constraint applies.
Additionally, a *temperature* field is available; leaving it blank implies that the data is temperature-independent.
REMEMBER: Always use *engineering strain* and *engineering stress*.

Curve-Fitting

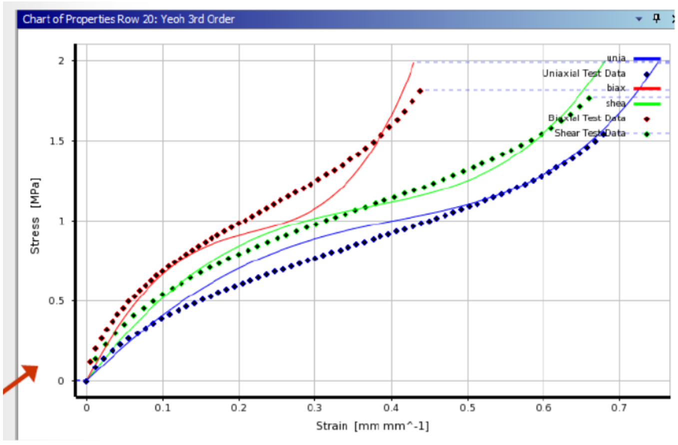


Raw data cannot be used directly as an analytical model for simulations. Therefore, a *Hyperelastic model* must be selected, which initially contains unassigned constants.
The **GOAL** is to determine the constants that best fit the raw data. These can either be entered manually or calculated using the built-in curve-fitting tool.
It is crucial to follow this precise sequence:

1. **Input** and **Load** the raw data.

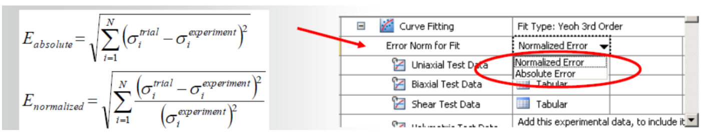
2. **Choose** a Hyperelastic model.

NOTE: If this order is reversed, the software assumes the curve-fitting tool is unnecessary, and the option will not appear.



After evaluation, the software displays a comparison between the analytical fit and the raw experimental data. The graph above demonstrates an excellent fit.
Generally, achieving a perfect fit simultaneously across all three testing modes is challenging. The example above represents a well-balanced trade-off where all modes fit reasonably well. Optimizing the fit for a single mode often requires sacrificing the accuracy of the remaining two.
To improve the accuracy for a specific mode, the other modes can be suppressed. Conversely, increasing the number of active modes introduces more constraints, forcing the curve-fitting tool to balance multiple datasets rather than focusing on just one, which may reduce the quality of the individual fits.

Error Handling



Curve-fitting errors can be formulated in two ways:

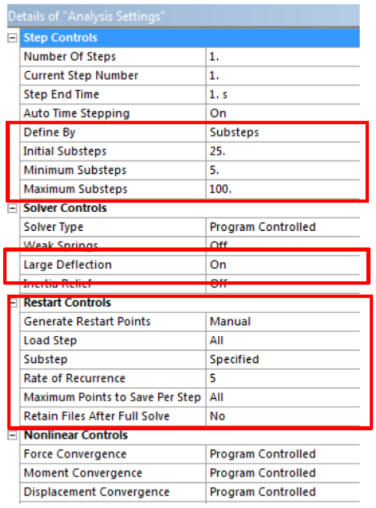
- **Absolute**
- **Normalized**

You can easily switch between them. The default option is *Normalized*, which performs very well in most cases. The *Absolute* formulation is typically preferred when dealing with high strains, as it tends to better capture and manage the upper values of the strain range.
Choosing between them is generally a matter of trial-and-error, guided by an understanding of your expected error and strain ranges.

Analysis Settings

When using a *Hyperelastic model*, large deformations are anticipated. This introduces significant geometric non-linearities and adds substantial computational complexity to the simulation.
Key **recommendations** include:

- **Adjust substep values**: Ensure the simulation proceeds incrementally and at a controlled pace to avoid running into convergence or numerical stability issues.
- **Large Deflection**: This setting must always be turned **ON**. Given the inherent non-linearities of hyperelastic materials, it is mandatory.
- **Restart Controls**: Enabling restart points is highly recommended, especially for multi-load step simulations. If a specific step encounters convergence issues, you can troubleshoot or resume without restarting the entire computationally expensive simulation.



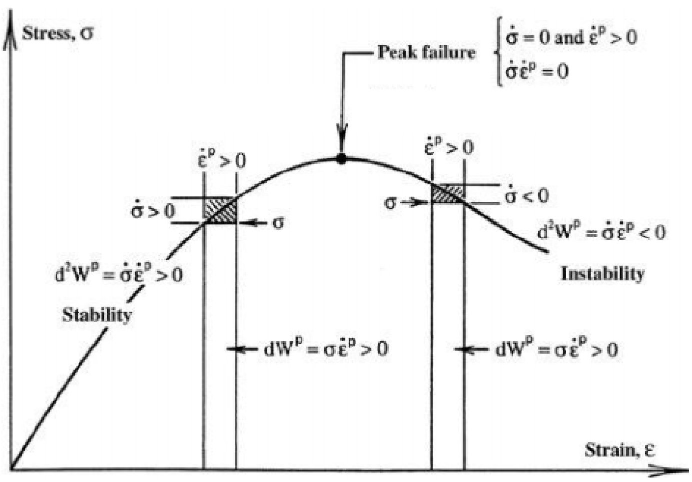
Running the Solution

Two types of instabilities can occur during a simulation:

- **Physical instabilities**: Such as *buckling*, where a specific loading condition suddenly triggers rapid, large deformations.
- **Numerical instabilities**: In hyperelastic models, we do not define the stress-strain relation directly. Instead, it is mathematically derived from a strain-energy density function.

Because of this mathematical derivation, the resulting analytical stress-strain relationship may sometimes exhibit physically unrealistic behavior.

What constitutes realistic behavior?
Under normal physical conditions, there must be a positive relationship between stress and strain; increasing the strain should always result in an increase in stress.



Therefore, the material slope (stiffness) must remain positive:

$$d\boldsymbol{\sigma} : d\boldsymbol{\varepsilon} > 0$$

Both the user and the software must verify that the derived relationship maintains a positive slope. For instance, in the curve shown above, the material behavior is realistic only in the first region, whereas the second region exhibits an unstable negative slope.

To evaluate this, the software utilizes the **Drucker stability criterion**, which checks for a positive definite tangent material stiffness matrix within a specified interval. By default, the solver performs a preliminary check across a stretch ratio range of:

$$0.1 \leq \lambda \leq 10$$

What does this mean in terms of stretch?
When the material is completely unloaded, the stretch ratio is $\lambda = 1$. Checking the range from 0.1 to 10 means the software evaluates the material stability under severe compression ($\lambda < 1$) and high tension ($\lambda > 1$). This extensive check is necessary because while your experimental data ensures validity within a specific testing window, the material behavior during extrapolation remains unknown.

If the software detects an instability (i.e., a negative slope) during this extrapolation check, it will not abort the simulation immediately; instead, it displays a warning.

```
*** WARNING ***                         CP=      0.219   TIME= 12:50:52
Hyper-elastic material may become unstable, material number 1 at
 temperature 0.
The nominal-strain limits where the material becomes unstable are:

    UNIAXIAL TENSION                     0.110E+01
    EQUIBIAXIAL COMPRESSION             -0.309E+00
    PLANAR TENSION                       0.118E+01
    PLANAR COMPRESSION                  -0.540E+00
```

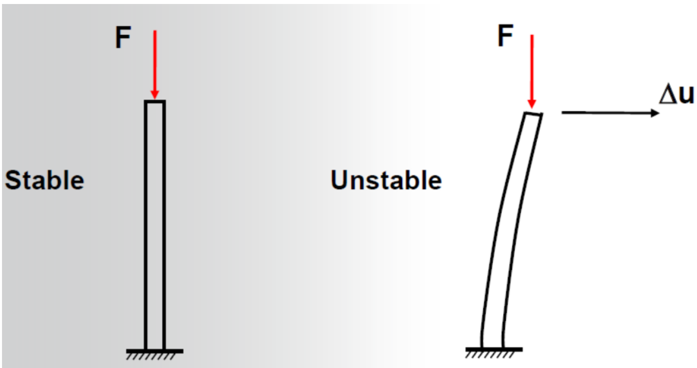
The user must then evaluate the warning based on the specific application:

- The negative slope region might occur at a strain level well beyond what the model will actually experience during the simulation, making it negligible.
- The simulation operating range may never enter the unstable strain domain, meaning the instability will not affect the results.

BUCKLING

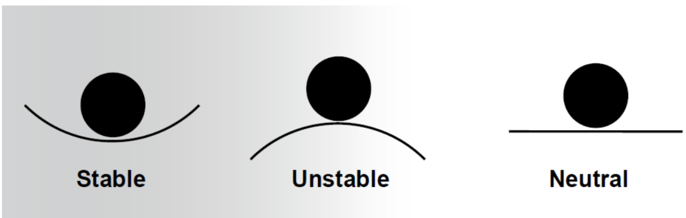
Background on Structural Stability

Buckling is a specific loading condition that causes sudden and dramatic structural deformation. The core concept of mechanical instability is that an infinitesimal increase in load can cause the deformation to escalate rapidly.
A classic example of this is global instability, such as a thin membrane abruptly flipping from one curved state to another. To visualize this, consider a flat sheet of paper curved upward by bringing its opposite edges closer together. If a small downward force is applied to the apex of the curve, it will suddenly snap and invert completely to the opposite side—a phenomenon known as **snap-through**.
This rapid evolution of deformation is precisely what introduces extreme complexity when analyzing the behavior and progression of the system.
Let us consider a simple example:



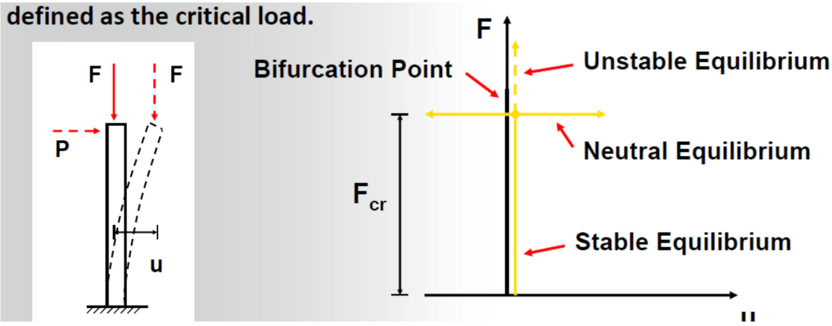
On the left, the system remains perfectly vertical and symmetric; since the load aligns perfectly with the structural axis, stability is maintained.
The emergence of **instability** occurs when this symmetry is broken—either due to a slight misalignment of the force or an asymmetry in the interaction between the load and the rod—resulting in a lateral (orthogonal) displacement, as shown on the right.
In the stable configuration (left), the axial force propagates vertically and is fully counteracted by the physical support. On the right, once the force loses alignment, it induces a lateral displacement that alters the equilibrium state entirely. While this provides a simplified physical intuition, the formal definition of structural stability is more rigorous.

States of Equilibrium: Stable, Unstable, and Neutral



- **Stable**: If the structure is subjected to a minor perturbation, it temporarily displaces from its initial state but subsequently returns to its original equilibrium position.
- **Unstable**: A minor perturbation triggers a continuous evolution toward an entirely new configuration. From a structural standpoint, this new state is often unacceptable as it compromises the intended load-bearing function of the component.

- **Neutral**: A minor perturbation shifts the structure into a new position without causing catastrophic failure or a return to the origin. Each subsequent perturbation simply establishes a new, slightly altered stable equilibrium configuration.

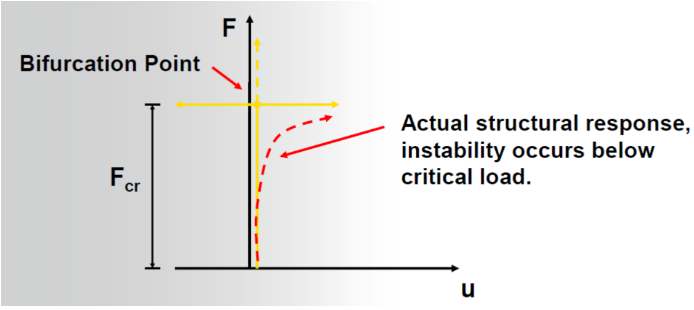


An idealized, fixed-end column exhibits the behavior shown above under an increasing axial load F.

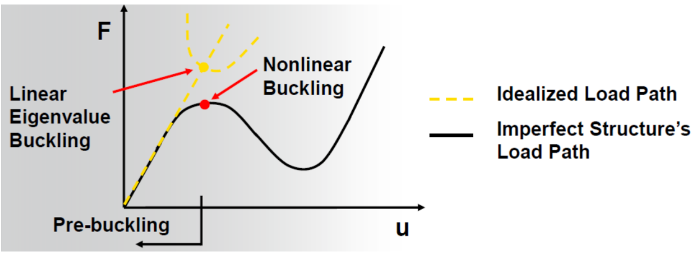
The **bifurcation point** defines the transition from a stable equilibrium path to an unstable one (a region of the plot that is physically inaccessible). This point corresponds to the **critical force** F_{cr}. At this threshold, multiple equilibrium configurations become possible (for instance, the structure can buckle symmetrically to either the left or the right).

- $F < F_{cr}$ (**Stable**): A small perturbing force causes a temporary displacement, but the structure safely returns to its original configuration over time.
- $F = F_{cr}$ (**Neutral**): The system reaches the *critical load* threshold. Any small perturbation leads to a new, adjacent stable equilibrium state.
- $F > F_{cr}$ (**Unstable**): The current configuration is completely unstable, and even an infinitesimal perturbation triggers structural failure or immediate collapse.

NOTE: In real-world structures, unavoidable geometric imperfections and material non-linearities cause instability to occur at loads lower than the theoretical critical load.



An even more realistic representation of this behavior is illustrated below:



The system initially exhibits linear behavior until it reaches a threshold where the idealized and real paths diverge. Structurally, this response is divided into two distinct phases: the **pre-buckling phase**, during which the structure remains stable, and the **post-buckling phase**, which occurs after crossing the critical load limit.

Linear Buckling Procedure

This procedure is used to evaluate pre-buckling behavior and is formulated as a classic **eigenvalue** problem. Prior to buckling, a structure subjected to a baseline load behaves according to Hooke's law. This initial loading state produces an elastic deformation directly related to the structural stiffness. Specifically, the applied reference load vector $\{P_0\}$ induces an initial displacement vector $\{u_0\}$, expressed as:

$$\{P_0\} = [K_e]\{u_0\}$$

From this initial step, we obtain:

- $\{u_0\}$ = the displacement vector resulting from the applied reference load $\{P_0\}$.
- $\{\sigma\}$ = the internal stress state resulting from $\{u_0\}$.

However, the development of internal stresses alters the structural stiffness of the component. Assuming the pre-buckling displacements remain small, the incremental equilibrium at an *arbitrary state* $(\{P\}, \{u\}, \{\sigma\})$ under an incremental load $\{\Delta P\}$ is given by:

$$\{\Delta P\} = [[K_e] + [K_\sigma(\sigma)]]\,\{\Delta u\}$$

where:

- $[K_e]$ = standard elastic stiffness matrix.
- $[K_\sigma(\sigma)]$ = *initial stress matrix* (or geometric stiffness matrix) evaluated at the current stress state $\{\sigma\}$.

Consequently, the total stiffness of the system depends on two terms: the baseline elastic stiffness (governed solely by geometry and material properties) and the stress stiffness (induced by internal loading).
What occurs at the onset of buckling?
At the *onset of instability*, an infinitesimal increase in the external load triggers a rapid, disproportionate escalation in deformation. Under these conditions, the incremental external load vector can be considered negligible:

$$\{\Delta P\} \approx 0$$

Despite the absence of a load increment, a non-zero incremental displacement $\{\Delta u\}$ occurs, yielding:

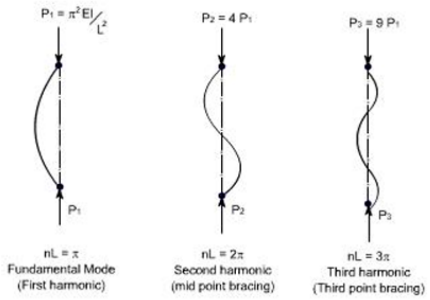
$$[[K_e] + [K_\sigma(\sigma)]]\,\{\Delta u\} = \{0\}$$

This expression matches the exact form of a *classic eigenvalue problem*.
To standardize notation for *linear buckling analysis*, the eigenvalue problem is typically rewritten to solve for the buckling load multipliers λ_i and their corresponding buckling modes Ψ_i:

$$([K] + \lambda_i\,[S])\,\{\Psi_i\} = \{0\}$$

where:

- $[K]$ = conventional elastic stiffness matrix (held constant).
- $[S]$ = stress stiffness matrix derived from the reference load (held constant).
- $\lambda_i = i^{\text{th}}$ buckling load factor (multiplier).
- $\Psi_i = i^{\text{th}}$ buckling mode shape vector.
  - Linear elastic material behavior is assumed throughout.
  - Small deflection theory is assumed during the baseline step.
  - Non-linear material properties, if defined, are ignored.



Physically, the load factor λ_i represents the scaling factor by which the reference load must be multiplied to induce buckling, while the mode shape Ψ_i represents the expected deformed geometry at that threshold.
As shown in the classic Euler columns above, the first eigenvalue corresponds to the fundamental buckling mode (left), while higher eigenvalues correspond to subsequent harmonic modes. Each distinct mode shape is uniquely linked to its respective critical load factor.
NOTE: Eigenvalues and mode shapes are calculated in pairs. Because this formulation is strictly **linear**, any post-buckling behavior or geometric and material non-linearities are completely disregarded.

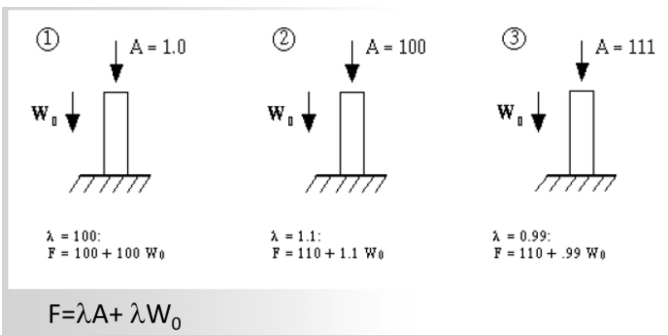
Contacts

Contact Type	Linear Buckling Analysis		
	Initially Touching	Inside Pinball Region	Outside Pinball Region
Bonded	Bonded	Bonded	Free
No Separation	No Separation	No Separation	Free
Rough	Bonded	Free	Free
Frictionless	No Separation	Free	Free

NOTE: In a linear buckling analysis, non-linear contact behaviors cannot be modeled directly. The solver automatically simplifies the setup by approximating non-linear contacts with their closest equivalent linear contact types (e.g., treating friction or frictionless settings as bonded or no-separation states during the eigenvalue extraction).

Evaluating the Load Multiplier (λ)

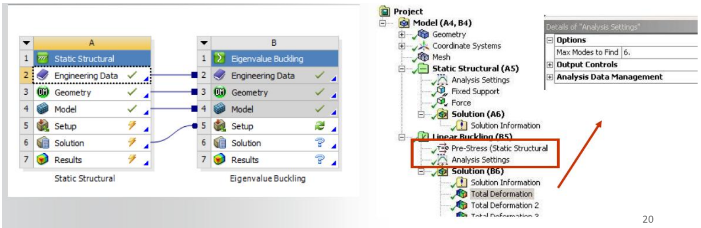
As established, λ is a global scaling factor applied to the reference loads to identify the conditions that trigger structural instability.



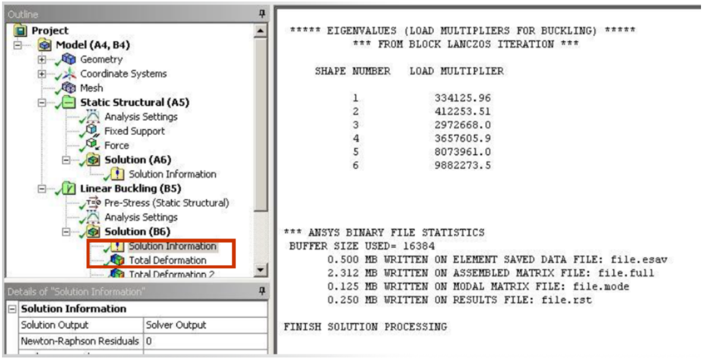
However, the solver applies this multiplier uniformly to all forces active in the load step. As illustrated above, if a model contains both an external operating force and a constant dead weight (such as gravity), the software will scale the structural weight along with the force. Physically, scaling gravitational constants or invariant loads is incorrect and unrealistic.
To resolve problems involving a combination of fixed (constant) and variable loads, an alternative iterative approach must be used. Instead of relying on a random multiplier, the user systematically adjusts the magnitude

of the variable load (e.g., scaling factor A) across successive iterations until the calculated eigenvalue λ converges close to 1.0. By reversing the methodology into an iterative process, the true critical value of the variable load can be precisely isolated while keeping constant loads physically accurate.

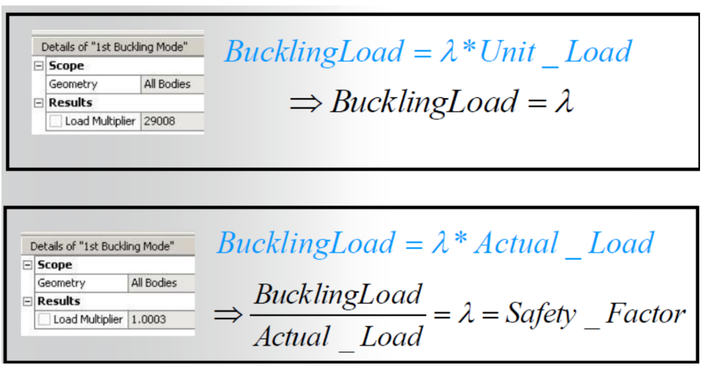
Linear Buckling interface



In the linear buckling interface, there are options related to the pre-stress state imported from the static structural block, alongside additional "Analysis Settings" specific to linear buckling. Buckling analysis is significantly more computationally expensive than static structural analysis. Specific fields are dedicated to linear buckling, particularly the one highlighted in the figure below:



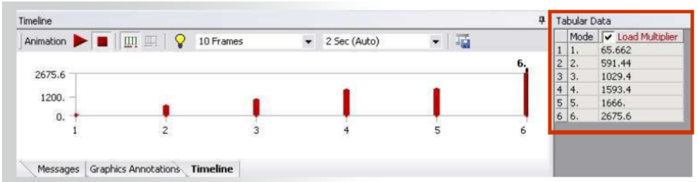
This field defines the number of mode shapes (and corresponding load multipliers) the simulation should calculate. In the figure above, the user is analyzing the first six modes. These represent the fundamental mode plus five higher-order harmonics, each associated with progressively higher loads and different deformation shapes. The first mode corresponds to the lowest load multiplier, the second is higher, and so on. Let's focus on *why it is useful to analyze more than one buckling mode*, considering two specific scenarios:



- First, we want to determine the exact buckling load required to trigger instability in the structure.
- Second, we know the applied load and want to determine our safety margin before buckling occurs.

If the goal is to find the absolute buckling load, the best approach is to subject the model to a unit load (e.g., 1 N). The calculated load multiplier (λ) will scale this imposed load. With a unit load, the resulting multiplier directly represents the true buckling load. Conversely, if you apply the actual expected load, the multiplier λ acts as a **safety factor** (Buckling Load / Actual Load), indicating how close the system is to instability.

Important suggestion Studying multiple deformation modes is highly recommended.

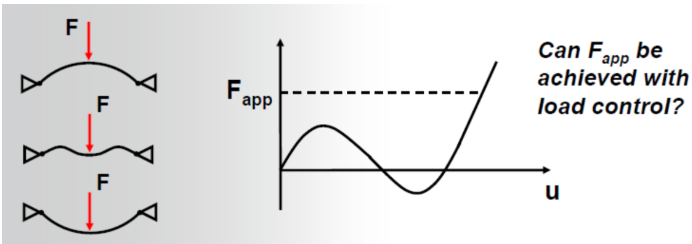


While there is a natural sequence—the first mode occurs earlier due to its lower load limit—the structural implications of different buckling modes can vary significantly.
NOTE: A mode associated with a very low load multiplier might indicate **local buckling**. This instability is confined to a specific part of the system and does not necessarily cause complete global failure of the structure. For instance, if the multiplier is ¡ 1000 ¡1000, it may just be local yielding; as shown in the picture above, complete global failure might only occur after the third mode.
NOTE 2.0: Pay close attention when load multipliers are very close in value (e.g., the fourth and fifth modes in this example). At lower load levels, this proximity can be problematic. Any slight defect or imprecision in the system could cause these modes to swap. Therefore, analyzing the gap between these values and their associated modes is crucial.

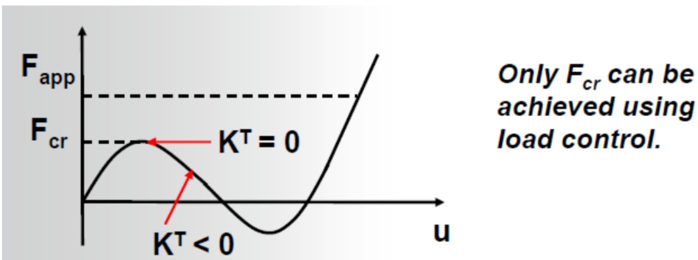
Non-linear buckling techniques

Once the buckling limit is identified, standard linear analysis is no longer valid. To analyze the post-buckling behavior after the stability limit is exceeded, non-linear techniques are required.

Load control

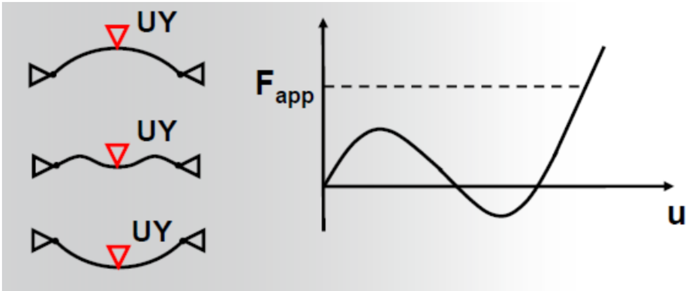


This illustrates a **snap-through** phenomenon. When a force is applied, the body eventually snaps through to the other side, causing a rapid evolution of deformation. The fundamental issue is that tracing this post-buckling path using pure load control is practically impossible. Once the maximum force (the buckling point) is reached, the structure loses stability. Beyond this point, the necessary force actually decreases before increasing again. This reversing post-buckling region is inaccessible via load control. Mathematically, at the point of instability, the stiffness matrix becomes singular (the tangent is 0). Traditional force-controlled Newton-Raphson approaches cannot solve this.

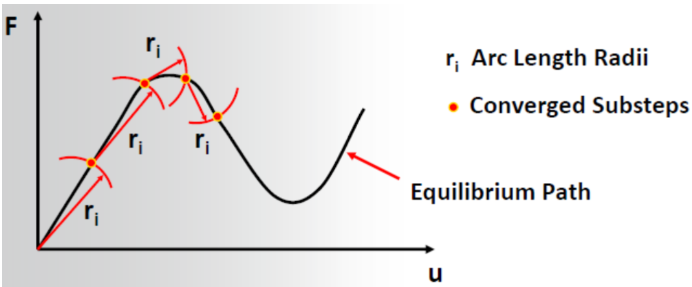


Displacement control

This approach works well because it allows you to trace the beam's behavior even after the buckling load is exceeded. From a displacement perspective, the force-displacement relationship remains single-valued: every specific deformation state corresponds to a measurable reaction force. However, this method is only viable when the force is applied locally, as shown below.



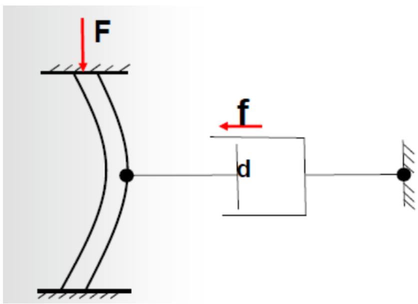
Conversely, if a distributed pressure is applied, tracing the equivalent force is much harder. You cannot impose a unique displacement control for a homogeneously applied surface pressure. When displacement control cannot be used, and the standard Newton-Raphson method fails due to a singular stiffness matrix, an alternative numerical method must be employed. You can use the **Arc-length method**:



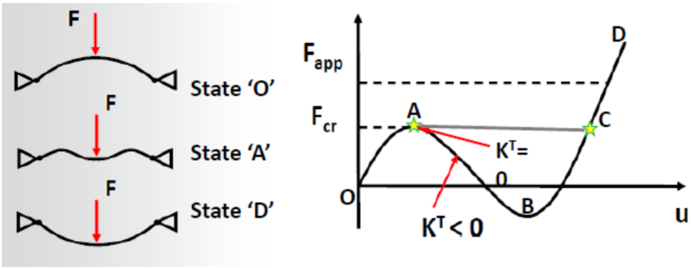
The Arc-length method can track the equilibrium path even when the required force decreases, making it ideal for studying post-buckling behavior. While not directly exposed in the standard ANSYS interface, the solver supports it, though it is quite complex to configure.

Nonlinear stabilization

This is a mathematical technique used to stabilize and slow down rapid deformations during simulation.



The main difficulty in analyzing buckling is the velocity of the orthogonal deformation, which occurs almost instantaneously. To counter this, a virtual damper is added to slow down the deformation. The damping force increases proportionally with the deformation velocity, thereby resisting the system's unnatural acceleration. Essentially, an artificial dashpot is added to each structural node. This process is time-dependent. The dashpot element operates on a pseudo-velocity derived from the simulation substeps (change in deformation over a time step, which represents velocity). Starting from each substep, the solver calculates a pseudo-velocity to determine the appropriate reactive damping force. Before buckling occurs, the deformation velocity is very low, meaning the damper has almost no influence on the results. Once the buckling limit is exceeded and deformation accelerates, the damping influence increases significantly. **Mathematically speaking**, the stiffness matrix now includes not only the elastic constants and internal stress terms but also a stiffness component associated with this virtual damper, which prevents the matrix from becoming singular (reaching 0).



As seen in the graph above, this method stabilizes the unstable region. The virtual damper allows the solver to "jump" across the instability to find the next stable equilibrium point corresponding to the applied force. **NOTE**: This artificial damper has no physical meaning, but it introduces numerical implications that must be monitored. As explained, if the artificial damping force is too high—approaching the magnitude of the actual applied load—it will heavily skew the simulation results. We can control this stabilization force in two ways:

- **Energy**
  - Damping is calculated automatically.
  - It varies element by element, rather than applying globally across the entire volume.
  - This is the best approach for local instability, as the damping factor adapts to local element rules.
  - It is more computationally demanding due to this localized approach.
- **Damping**
  - Manually input by the user.
  - Applied uniformly to every element.

Stabilization	Constant
--Method	Energy
--Energy Dissipation Ratio	1.e-004
--Activation For First Substep	No
--Stabilization Force Limit	0.2
Output Controls	
Analysis Data Management	
Visibility	

Stabilization	Constant
--Method	Damping
--Damping Factor	0.1
--Activation For First Substep	No
--Stabilization Force Limit	0.2
Output Controls	
Analysis Data Management	
Visibility	

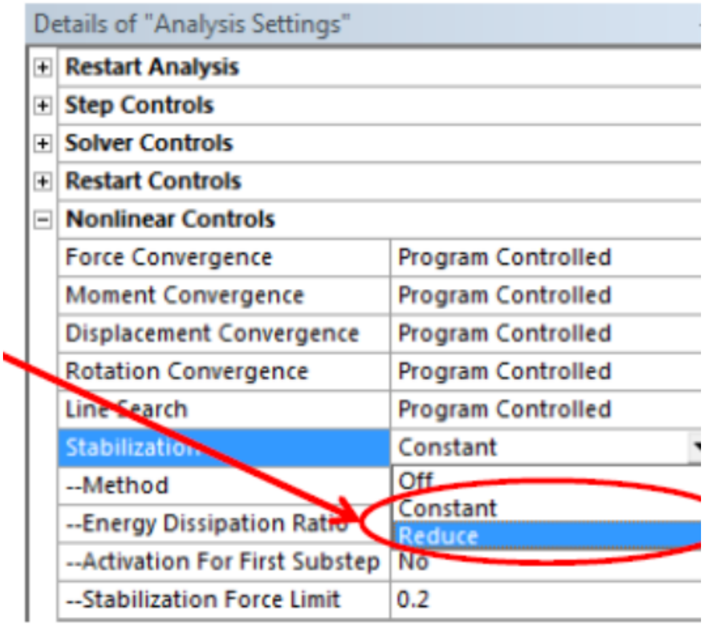
Sometimes it is useful to combine approaches, particularly by starting with the Energy method.
In the Energy method, you can also set the **Energy dissipation ratio**:

- This is the ratio of the work done by the artificial stabilization forces to the element's actual potential energy.
- It should be large enough to prevent divergence, but small enough to avoid introducing excessive artificial stiffness (typically between 0 and 1).

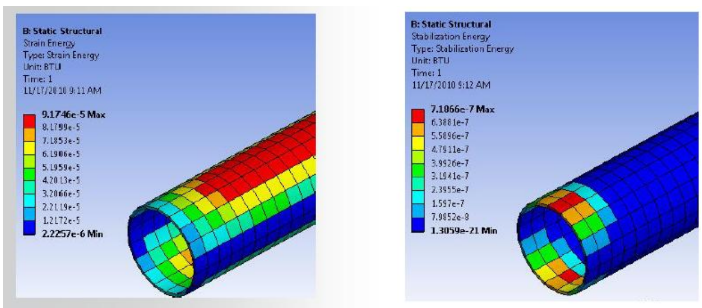
It is good practice to monitor how much artificial energy has been introduced.

Regardless of the chosen method, the user can modify how stabilization is applied over time:

- **Constant**: Maintains a constant damping factor throughout the analysis without varying it.
- **Reduced**: The solver calculates the necessary stabilization values initially but gradually reduces the factor to zero once the highly unstable portion of the simulation is resolved. The final result represents a true force equilibrium without artificial stabilization forces, as they are no longer needed. If the 'Reduced' option is not used—especially when stabilization is activated for a single load step—abrupt changes in boundary conditions can cause convergence issues. Therefore, using the Reduced option is almost mandatory.



Additionally, a **stabilization force limit** can be set. The default limit is 20 20It is highly recommended to check reaction forces and moments to understand the magnitude of the introduced stabilization forces. You can also visually inspect the ”stabilization energy” distribution across the model.



The general rule of thumb when comparing ”Strain energy” to ”Stabilization energy” is that the artificial stabilization energy should remain a very small fraction (typically ¡ 5 % ¡5%) of the total physical strain energy.